

Math 210 Introductory Applied Mathematics

Numbered Transcript

MATH 210 · Dr. Robert Snellman

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Lecture 1: A Competitive Edge in Math

At a glance — Snellman introduces applied mathematics and the competitive edge it provides.

- 1.1 All right, welcome to my introductory course in Applied Mathematics. I want to give you a feel for the expectations you should have in the course. What are you going to learn? Why? How is it applicable? And things like that.
- 1.2 Ultimately, a course like this, there's lots of different ways to go about it. If you take an off-the-shelf course like this, it's going to teach you a lot of really important mathematics, and it's interesting, and things like that. But at the end of the day, why are you here? Why do you want to take a class like this? Why should you take a class like this?
- 1.3 That's really what I want to get through, because I think we could all agree that we've maybe had this experience. Okay, maybe we haven't, but I would argue that most of us have probably had an experience like this, where you go in, you take a class, and you learn a lot of really interesting things, but at the end of the class, you think, "Now what? What did I actually get out of it? Did I just memorize a lot of facts—or something along those lines—that are interesting in their own right and that are valuable in their own right? Or did my motivation for taking the class actually get reached?"
- 1.4 Does that make sense? "I want to take the class for a reason. I want to develop a skill set. I want to develop a knowledge. I want to develop marketable skills or things like that at the end of the day. Did I actually achieve those goals? Or did I just learn a bunch of interesting facts for the sake of learning a bunch of interesting facts?"
- 1.5 What I want to do is I want to, rather than taking these off-the-shelf type courses, what I want to do is ultimately look at what is a motivating problem in the world that we want to solve or a skill set that we want to develop. Then I want to cut through all of the material that otherwise wouldn't necessarily be necessary in a certain type of course.
- 1.6 It's not going to help me understand how to solve these problems. It's not going to help me understand the skill set that I need to be able to solve these problems or make a difference. It's learning a topic for the sake of learning a topic. It's not necessarily something I can apply or I can then take out into the real world or mark it as a skill set. Does that make sense overall?
- 1.7 I want to cut through that and give you everything that you need to be successful on that front so that you can come out of the course and you can say, "Look, I learned all of this interesting theory. I learned all of these interesting techniques, but now I have a marketable skill that I can actually say, I can do this and apply that to whatever your trajectory is on that front."

- 1.8** Let's think about this. What is applied mathematics? Applied mathematics, it's hard to explain. I mean, ultimately, you're trying to take mathematical principles and look at a real-world problem, whatever that may be, whether it's a business analytics problem, whether it's a supply chain problem, whether it's a workflow problem, whether it's anything you could possibly imagine, you want to apply mathematical principles to try and model that situation. After you've been able to model that situation, you want to say, "What does that give me or what does it not give me?"
- 1.9** For example, if you're running a business and you are selling a particular product, it's in your best interest to sell more of that product, and you want to be able to do that as efficiently as possible. There's a lot of variables that come into play here. If you think about it, if you're selling a product x , whatever that is, whether it's books or phones or watches or whatever your product is that you're selling, you want to optimize your profits. What does that mean? That means you want to sell as many things as you possibly can, ultimately, at the optimal price, whatever that is, with minimal cost. You want to maximize your margins, your profit margins at the end of the day.
- 1.10** Think about all the variables involved with that. You have your business. Where is it located? You might need to rent out a space to be able to do that. What are rental rates for that? What are property taxes and things like that? What are licensing fees? Then you're going to have your vendors that you're potentially going to be partnering with to get your product from A to B. What are the costs associated with that? What are the costs of trying to manufacture the product that I'm looking at? What are the costs of hiring employees and covering insurance and covering all of these different things?
- 1.11** Suddenly, you sit there and you're like, "Oh, my goodness. This is a monster of a problem and a monster of a situation that I'm trying to optimize." But at the end of the day, that's what you're trying to do. You're trying to optimize it. You want the minimal cost to be able to produce your goods. You want to maximize your revenue so that you can have the thickest profit margins possible. But at the end of the day, you're looking at an optimization problem: "How do I optimize something along those lines?"
- 1.12** Now, you're not going to be able to sit down with pen and paper and say, "Okay, here's my function that's telling me what my profit is, and here's all the variables, and here's all of this, and so on," and try and solve it by hand. You need a skill set where you can plug this into a machine, code it up, and then be able to find the optimal path forward with your business. Whatever the machine churns out, it says, "Hey, you should be doing this." Fantastic. And we're going to get there. We have to start somewhere, but that's our motivation.
- 1.13** Now, the world is changing at such a rapid pace, and a lot of it has to do with the amount of data that we're consuming and the data that's being produced. There's all these buzzwords with data science and machine learning and artificial intelligence and

all these other different things, which are amazing, but they're changing the world at such a rapid pace.

- 1.14** What I would argue is that I've had students in the past who tell me, "I want to do data science. I want to do machine learning. I want to do all this." That's fantastic. They learn all the techniques, and they learn all of the software, and they learn all of the programs that they need to use to actually run these particular algorithms to learn data science and make actionable decisions based on that data. Or machine learning and be able to say, "Oh, this is actually going to predict this particular thing," or artificial intelligence, and that's fine.
- 1.15** But if you look at where the world is going, this artificial intelligence side of things, you can literally just go online, type in something, "I want code for this particular thing," and suddenly, boom, there it is. My question is: Is there value in simply being able to run these algorithms? Absolutely. Absolutely, there's value in that. But I would argue that there's more value in understanding what's actually going on under the hood. Do you actually understand the underlying mathematical principles that are going into these particular algorithms?
- 1.16** It's one thing to be able to say, "Hey, here's a tool. Can you use it?" And pull that off the shelf tool and say, "I can run it." That's fine. That's a valuable skill set to have. That is very valuable. It's another entirely to be able to say, "I understand how that tool was made, and I understand how I can modify that tool to create new tools that can potentially be revolutionary." That's what understanding the underlying mathematics is going to give you.
- 1.17** It's going to allow you to see what's under the hood, to see, "Oh, why does my algorithm work the way that it does?" That's an important thing to know. We need to understand why these algorithms work the way they do, because we don't know, for example, if there's inherent bias built into these things. Somebody codes it up and they think that's fantastic. What if their personal biases are built into the algorithm? That could have detrimental effects all over the world. You need to understand how to be able to look underneath the hood. That's what I'm going to give you.
- 1.18** What are some of the areas where applied mathematics is being revolutionized or is revolutionizing the world? One of the biggest ones for me is in the medical sector. We see this all the time. Medicine is changing at such a rapid pace. So, we need to understand how do we optimize the situations that we're in.
- 1.19** Let's say, for example, you have an accident of some sort and you go to the hospital and they do imaging. They do an X-ray, an MRI, or a CT scan or whatever the case may be. They say, "Oh, you have a broken bone somewhere." Or it looks like somebody goes in, they didn't have an accident, somebody goes in and they're like, "I just had pain in this particular area," something like that. "My hip is hurting, my knee is hurting, my shoulder," whatever the case may be. They look at it and they're like, "Whoa, you need

a knee replacement, or you need a hip replacement, or you need something along those lines, you're bone on bone in that area, that's where your pain is coming from. We need to replace that joint."

- 1.20 Think about that. We're all unique, and that's what makes us amazing. We all bring different things to the table. But physiologically, yes, we have similar components, but they might look differently depending on who you are. My hip joint looks different than your hip joints. My shoulder joint might look different from your shoulder joints, things like that. So, why should I expect, if I need to go in for a knee replacement or something like that, that some off-the-shelf knee replacement prosthetic would somehow just be the right one for me? Does that make sense?
- 1.21 One of the things that's happening is, with machine learning, what's being able to happen is they take images of patients, and they take very detailed MRIs, X-rays, things like that, and they develop—they feed all of that into these machine learning algorithms, which can then customize a prosthetic for your new knee joint, or your new ankle, or your new shoulder, or your new hip, or whatever the case, whatever joint is being replaced, it's customized to your physiology. That's a massive improvement.
- 1.22 What's another one in medicine? Another one in medicine would be drug discovery. You have pharmaceutical drugs, for example. A new drug is trying to come to market, whatever that may be. It's shown to be effective to decrease symptoms for disease x or cure disease y or whatever the case may be, but it's got to go through these phased trials, which usually take a lot of time for a good reason. How can I optimize that process? How can I get it through the trials in a way which is ethical, safe, but cuts away the time to actually get it to the marketplace so that it can actually help individuals? It's an optimization problem. There are a lot of companies that are working on this.
- 1.23 Another one in the medical sector would be increased diagnostic accuracy for images. Okay, so go back to the analogy that we had where you had an accident, you go to the hospital, they take some images, things like that. I'll share a personal story about my experience with that. But let's say you go in, you have an image. How do you know that whatever the image was is actually accurate?
- 1.24 You put a lot of faith in a radiologist who's going to read the film and tell you, "Oh, it looks like you have a compound fracture in your femur." My bone looks like this. It's like, "Oh, okay, something's wrong." Something along those lines. Some of them are glaringly obvious. You have a bone that suddenly used to look like this but looks like this. It's like, okay, that's a serious problem.
- 1.25 But what about smaller little details? Let's say you go in and you're like, "I'm having a really hard time breathing. I'm not quite sure why. I'm coughing a lot more. I'm not sure why." They take a CT scan of your lungs, and the radiologist comes back and they're like, "There's inflammation. You might have some pleurisy or something, some inflammation in the pleural lining of your lungs, but there's not much else I can see."

You hope that that's correct. You hope that that's actually accurate at the end of the day.

- 1.26** Nine times out of ten, it is. I mean, these are professionals that we're talking about. They've got years of experience, things like that. But what if I could have an algorithm running in the background to assist the radiologist, that I've trained that algorithm. It'd be a machine learning algorithm which is doing image recognition at the end of the day, where I've fed it thousands of CT scans of lungs with the corresponding diagnostics associated to it. This one has a malignant tumor here. This one has a tumor here, but it's benign. This shadow over here means this.
- 1.27** I can have that running in the background in a software program so that the radiologist takes their years of experience and they look at it and they say, "This little thing over here, that's concerning. Everything else seems a little clear. We need to investigate that spot a little bit." But what if I had a software program running in the background that said, "I agree with what the radiologist said, However, there's a little tiny dot in the bottom left-hand corner that I think is concerning that maybe the radiologist missed." It would increase medical diagnostic imaging significantly. Significantly. But at the end of the day, we need to understand the underlying optimization.
- 1.28** Let me share a couple of stories. I had a student not too long ago. They were interning at a big company over the summer. This is one of my applied mathematics students. They came to me at the end of their internship and I said, "How did it go? How was your experience? Do you want to work for them long term? Tell me everything." They said, "It was actually pretty good. They started telling us about these investment models that they were using."
- 1.29** They said, "Here's the code that we're using, but it's slow. It's slow." My student looked at them and they said, "Can I look under the hood? Can I actually look at the hard coding of this and see what's going on?" They said, "Yeah, sure. That's fine." So they let them take a look at it, in toy examples. This wasn't what they were actually—they're not going to give an intern their proprietary software, things like that. They gave them a little toy example to play with.
- 1.30** My student looked at it and they said, "You're doing a computation in this way. Why aren't you doing it like this?" The manager looked at the student and they were like, "What do you mean?" He said, "Let me code up a toy example to show you what I mean." He codes up a toy example a week later and presents it to his manager. He says, "Look, the stuff that you guys are using, it takes about an hour or two to get through this computation." It's a massive computation that they're going through. It takes about an hour to run. He said, "If you just tweak this little thing, I can get that thing run in an hour to run in about 30 seconds." The manager looked at him and said, "Are you serious?" He said, "Yeah."

- 1.31** So he started working with their development team, started implementing that in larger scale models, and they realized things that were taking them a day and a half to compute, suddenly, it was taking them an hour to compute. So my student's telling me this, and they get this big smile on their face, and they look at me at the end, and I looked at them and I said, "So what was their offer?" And he said it was pretty big because they realized the value that was being brought to the table. The only reason that that offer was made is because the student understood the underlying applied mathematics. They understood the applied optimization necessary. A very small tweak suddenly just changed their life like that.
- 1.32** The second example that I have was another student of mine. They were interviewing big company out of their undergrad. They show up to the interview and there's a technical interview question. Technical interview question goes along these lines. Typically, these interview questions, they're asking you, "How do you think?" Not necessarily, "Can you solve the problem that we give you?" They want to know how you process information, how you attack problems, how you solve problems, things like that, even if you can't necessarily get to a solution to the actual problem.
- 1.33** My student walks into the interview and they said, "How would you solve this particular problem?" They outline everything. My student looks at it and goes— asks some clarifying questions first, looks at it and they say, "I would first do this, and then I would use this particular mathematical object to put it in this framework. Once it's in that framework, then I would be able to run this particular procedure on it to optimize that. Then once I have that, I'd be able to apply this technique to extract the information that I'm looking at. Then once I have that information, I've got actionable information that I can actually make a decision on."
- 1.34** The one conducting the interview looked at my student and goes, "What was that technique that you said you used right here? Can you teach me about that?" That's the answer you want because immediately, they knew the value that that person was bringing to the table. They came back from that interview, and they had an email offer immediately out of the gate with an outrageous salary attached to it. Just outrageous. Just an amazing opportunity. But you need to understand what's going on under the hood, and this course is a beginning to that.
- 1.35** Now, I'm a huge cyclist. I love cycling. I love riding on the roads. I love doing all that. One of the biggest things right now going on in applied mathematics is autonomous vehicles, vehicles that can drive themselves. I want you to get your head wrapped around that and how difficult of a problem this actually is to try and solve.
- 1.36** From a cyclist, I would be more than happy to have autonomous vehicles that were 100% accurate, that takes the guesswork out of anything. If somebody's on their phone and they're driving, I don't have to worry as a cyclist that I'm going to be hit by a vehicle, things like that. But it's a very difficult problem. You're trying to teach a

machine to be able to recognize its surroundings in such a way that it can differentiate between things that are safe to bypass versus things that are not.

- 1.37** How would you teach a vehicle, for example, that if it is in the middle of fall and leaves are falling all over the place, how do you train a machine to actually say, "It's okay to run through that pile of leaves because it's harmless. It's just a pile of leaves in the middle of the road. Not a big deal. You can go through that." Versus if a toddler runs into the road, "You need to stop immediately!" How do you train a machine to differentiate between those things in real time. That's what humans are good at. But how do you teach a machine to do that? That's an incredibly difficult problem.
- 1.38** Getting back to the cycling analogy. In 2019, I was involved in a bike crash that was horrific. Blew out my jaw completely. My left jaw blew out, fractured my cheekbone, blew through my jaw, all of that. I had to be wired shut for eight weeks. My mouth was completely cinched down for eight weeks. When I met with my medical team and all that, they said, "Look, we could do this procedure, we could do that procedure, we could do this, this, this, and this." And they said, "But at the end of the day, you're going to have residual damage from that. It was a very severe accident."
- 1.39** What I would have loved more than anything is to be able to look at the medical team and say, "Look, how did the fracture occur? Look at the MRI imaging, look at the CT scans, look at all of that. Where are the fractures? Specifically. I don't want an off-the-shelf, out-of-the-box surgical procedure simply saying, 'I'm just going to do this, and then you're done.' One size fits all mentality. No. I want it custom. I want it—my fracture occurred this way. I want those images to be as precise as possible. I want the best surgeon to be able to put together the bones, put the pins in wherever they need to go, put the wires in, all that stuff." That's where the world is headed, but it's not going that way without applied mathematics, without optimization, without understanding what's going on under the hood.
- 1.40** Now, one of the biggest things that is challenging with math, I would say, arguably, math is one of the most common disciplines where it's okay to express your disgust about the discipline to somebody's face, for example. It's okay. When I was studying, I would have friends who were going to be doctors and lawyers and dentists and all that stuff. Whenever you meet somebody new, they would say something to the effect of, "Look, what are you going into?" It's like, "I'm going into medicine." It's like, "Oh, my mom's a doctor or my uncle is a doctor" or something. "That's so cool. What medicine are you going to study?"
- 1.41** But then when it came down to me and they said, "What are you studying?" It's like, "Mathematics." The answer was nine times out of ten, "I hate math. I don't like it. I don't understand it. It doesn't make sense." Things along those lines. I would always joke saying that math was the only socially acceptable discipline where you could express your disgust about the discipline to somebody openly.

- 1.42 But the fact of the matter is this: Every single person is capable of understanding mathematics. Every single person can learn these deep, hard principles of algorithms associated to optimization and machine learning. You can understand gradient descent. You can understand Newton's method. You can understand how to find zeros of functions through things like the bisection method.
- 1.43 Those are all things that we're going to cover in this class. You don't need an advanced degree to go through it. You just need somebody who cares enough to believe in you that you can actually achieve these things, and you have that person in me. Welcome to the course. Let's get started.

Lecture 2: Derivatives in Action

At a glance — Derivatives in action — one of mathematics' key concepts and its uses.

- 2.1 The notion of a derivative is one of the key concepts in mathematics overall. But it's one of these things that is elusive and seems very challenging and very intimidating to some extent. But ultimately, it governs a lot of the questions and a lot of the ideas in modern mathematics and data science in particular.
- 2.2 One of the things that's really interesting is, in today's world, everything is about data. Everybody's screaming data, data, data, data. It's coming up all the time. There's too much of it for us to be able to actually make a lot of sense about how to make decisions, business analytics, things like that.
- 2.3 But at the end of the day, all of this boils down to these big key ideas of artificial intelligence or machine learning or anything along those lines boils down effectively to, "Can I optimize a particular thing whatever that may be?" I want to consider, "What is the next step I should take in my business?" I collect all of this data, and then based on that information, I want to come up with, "What is my next step? How do I optimize my business's trajectory, my revenue?" Whatever the case may be. But it all boils down to a derivative.
- 2.4 What I want to do is go through what is a derivative, how do you compute it, and starting from basic principles, ultimately. How do I generate these big ideas simplistically to some extent?
- 2.5 The world around us is all about optimization. When you finally get that into your head, it ruins your day-to-day to some extent because you start viewing everything in an optimization standpoint. But if you think about it, it's very true. If I have 45 minutes to grab lunch, I have constraints upon me. I've got 45 minutes. I've got a geographical location that I could potentially go to. Maybe I only have 20 bucks in my pocket. I have time restraints. I've got financial constraints. I can't go too far. I might get stuck in traffic, things like that. All of these boil down to an optimization problem. At the heart of optimization is the derivative.
- 2.6 What I want to do is I want to cut through everything and get to these key ideas, such as in machine learning, you'll hear the concept of gradient descent. You'll hear the concept of supervised learning versus unsupervised learning and all of this.
- 2.7 At the heart of it, what we're doing is we model a situation from the data that we acquire, and then we want to minimize our error to some extent, which is encapsulated in what's called the loss function. Now, that loss function, we can take its derivative, we can optimize it, we can minimize it, we can figure out, "How do I actually cut my losses, so to speak, from a business standpoint?" But it all comes from the derivative. What I want to do is I want to start from basic principles and say, "Look, even in a single

variable calculus class, we can still get at these topics of machine learning, gradient descent, things along those lines."

- 2.8** Let's start with the derivative. What is it? The derivative is a lot of different types of things, ultimately. There's a lot of synonyms that you can say. What is the derivative? Number one, you can think about it as a rate of change. If I have a particular quantity and I want to know how is that quantity changing with respect to whatever the other variables are.
- 2.9** A natural thing would be, let's say I have position, and position is measured in meters. Let's say it's measured in meters. If I want to say, what is the rate of change of meters with respect to time, for example? I want to say something—what would the rate of change of this be? A rate would be something, like the units would be, let's say, meters per second. Then what's the quantity that I'm considering at that point? If I'm looking at how far am I going per unit of time, I'm now looking at a speed. I'm now looking at a velocity of some sorts. I can notice that if I have something that's modeling position, then taking its derivative, which would give me its rate of change with respect to time, would give me a speed or things like that.
- 2.10** You could think of all sorts of other examples. If I had something that was measuring, let's say, the velocity of a vehicle or something along those lines, and I want to take the rate of change of velocity, what would that actually be? If I have, let's say, velocity here, and its units are going to be meters per second, and if I look at its rate of change with respect to time, then what am I looking at? I'm going to be looking at the rate, so the derivative is going to be meters per second per second. I would look at this as meters per second squared, and I look at that as an acceleration associated to it. The rate of change in my velocity would be my acceleration.
- 2.11** Now, that's all fine and great, interpreting it as a rate of change, but at the end of the day, how do you compute this? What I want to do today is ultimately explain, if I give you a particular function, what are the rules that govern how I can actually compute the derivative? What can I actually do to compute these things without having to build up an entire machinery associated to it? I'm just going to give you simple rules for computing derivatives, and then we're going to see what we can actually do with them at that stage. Let's go through that.
- 2.12** How do I compute derivatives? The first thing we need to understand is we need a notation for this. The greatest thing about mathematics, and I think this is a fairly misunderstood part of mathematics at the end of the day, is that mathematics is a language. As with any language, if you don't practice the language, you're going to lose it. It's the so-called "use it or lose it" type idea. But mathematics is ultimately a language.
- 2.13** From a notation standpoint, what do we have? There's a lot of different notations for denoting a derivative. There's a lot of historical context for some of these things, but

one of the notations is this prime notation. If I give you a function—a function I denote by the letter f , and I have some input to it. f is my function, x is the input. You can think about whatever that may be. You could say that x is maybe a time variable associated to it. Maybe it'd be better to denote it by t . But the function $f(t)$ is telling me what is my position in time t . I have a particle that's moving in space, and at time t , it tells me where that is particularly located. But the prime notation is very simple. Whatever my function is and whatever its variable is, I simply put a prime there to indicate that that is my derivative. That's one notation for it.

- 2.14** That notation is very common, especially if I have a single variable function. However, it's useless if I now consider functions of multiple variables. You can think about what that would mean. What would a function of multiple variables mean? Well, that function can model, in one variable, it's modeling, let's say, the position of whatever the particle is. In another one, it has a particular velocity associated to it. In another one, it may be telling me how large the object is or things like that. So you could have functions of multiple variables, in which case saying f' , it ultimately means nothing because it's not telling me what variable is changing. Am I changing with respect to time? Am I changing with respect to speed? Am I changing with respect to size? What is actually changing?
- 2.15** There's another notation. This is Leibniz notation, just for Leibniz who created this. This is this df/dx notation. This is shorthand for saying the following thing. You have this operator, which we call d/dx , and I'm applying that to the function $f(x)$. Now, this is very confusing when you see this for the first time, but I want to make a connection between something that we all know.
- 2.16** When you're reading a book or anything, and suddenly you see a quotation mark pop up, your brain naturally goes, "Oh, somebody is speaking," as you're reading a novel or whatever the case may be. If I see something and suddenly I have dot, dot, dot quotes, dot, dot, dot, the end of quote, I can know, "Oh, in here, that's indicating that somebody is speaking and somebody is talking." They're quoting a particular person.
- 2.17** That's exactly what this is, mathematically speaking. This notation d/dx , rigorously, it's called an operator, ultimately. That's beside the point. It doesn't really matter what it's called. It's simply an indicator, like a quotation mark is when we're reading, saying, whatever comes to the right, I'm going to say, "Whatever the variable is down here, so x in this case, this is simply indicating whatever function you put here, take its derivative." That's it. That's literally all it's saying. Just like a quotation is saying, "Once I run up on this, oh, whatever comes next, that's a quote."
- 2.18** Okay, fantastic. So whenever I see this d/dx , I'm simply saying, that's cuing me to say, "Take the derivative of f "—with respect to what?—"with respect to the variable x ." Now, this being a single variable type idea, the x is irrelevant. Everything that we do is going to be with respect to the variable that we know. In this case, that's all I want you to think about when looking at this notation. This d/dx is really just indicating,

"Whatever comes to the right, take the derivative of that thing with respect to whatever the variable is." That's it.

- 2.19** What are the rules? Some of them are very, I would say, intuitive and what we would hope to actually be true when it comes to computing derivatives, but others are somewhat confusing. I want to list out the rules so that we know what moves we're allowed to do in the game of computing derivatives. These are the things that tell me what I am allowed to do and what I am not allowed to do. Then I want to just basically do a whole bunch of examples so we can see how those rules actually apply. Then I'll motivate how the derivative is relevant in the real world and why we care about it, ultimately. So that's the trajectory of what we're going for.
- 2.20** The first thing is that if I take the derivative of a function and I add it or subtract it, I add or subtract another function from it. I'm giving you two rules here, ultimately. I take a function $f(x)$, and I add or subtract a function $g(x)$ to it, and I say, "What is its derivative?" Its derivative is going to be the following. It's exactly what you hope it would be: it would be, hopefully it's just the derivative of $f(x)$ plus or minus the derivative of $g(x)$, and that's exactly what it is.
- 2.21** I'm going to mix notations a little bit so we get a feel for how this works. Again, if I add or subtract functions, the derivative of the sum or difference is the sum or difference of the derivatives, ultimately. That's what we're getting at. That's a nice rule to some extent.
- 2.22** The second one is, what happens if I have some constant, it doesn't matter what it is—I'm just going to call it c —and I multiply that by a particular function? I have a function $f(x)$, and I want to multiply it by 5. Then I want to say, "How do I take the derivative of $5 * f(x)$?" What's beautiful about this is that as long as it's multiplication, the constant can just come out front. So this is equal to the constant times $f'(x)$. That's a nice property. If I know the derivative of f , then I know the derivative of any scalar multiple or any constant multiple of f in that case.
- 2.23** Those are two of my rules. Unfortunately, those are the nice ones. All the other ones are relatively not intuitive, and they break our intuition. I'll show you how we can check that these rules are actually valid and that our intuition is not really great in these cases.
- 2.24** The next one is the arithmetic operations I'm allowed to do on functions—right now, I add or subtract them. Fantastic. We have other arithmetic operations. I can multiply things, I can divide things, so a natural question would be to say, "How would I take the derivative of the product of two functions?" I give you $f(x)$ and $g(x)$, and I say, "What is their derivative?"
- 2.25** If I go solely based off of my intuition here, I would say, "It's probably just $f'(x) * g'(x)$. That's what naturally I would go for. I think that's a natural thing overall. This is $f(x) * g(x)$. Okay, derivative of the sum was the sum of the derivatives, so the derivative of the

product, it might be the product of the derivatives. It turns out that that's completely false. That that's completely false; it doesn't work that way.

- 2.26** You have to do—what's a little strange is you have to take this symmetric sum between the two. That's not intuitive at all. The fact that this is actually the right answer is like, "What do you mean? That doesn't jive with my intuition whatsoever." But that's okay. But you have to do this symmetric thing where you say, "I'm going to take one of the functions. I'm going to take $f(x)$, and I'm going to multiply by the $g'(x)$. Then after that, I'm going to take the $g(x)$ and multiply by the $f'(x)$. So it's this symmetric thing where it's like one of them gets the derivative, and you just bring the other one along for the ride and multiply. Then you have to add in, the other one gets the derivative, and you multiply by that. I'll motivate this in a second once we actually have some functions that we can differentiate to see how that works.
- 2.27** The next one, the next operation we have is taking the derivative of quotients. I can divide things, and I can ask, "What's the derivative of that?" Ultimately, multiplication and division are the same operation to some extent, if you think about it. If I ask you to divide something by 2, if I say take 5 and divide it by 2, that's equivalent to saying take 5 and multiply it by $\frac{1}{2}$. Those operations are one in the same to some extent.
- 2.28** So if my intuition turned out to be wrong for products that $f(x) * g(x)$ is not—the derivative of that is not $f'(x) * g(x)$. Then it stands to reason that the derivative of the quotient is not going to be $f'(x) / g'(x)$. It's probably going to be something else.
- 2.29** Now, what is it? It's horrible. It's horrible. You take the bottom function, $g(x)$, you multiply it by the derivative of the top. It's like, all right, that's looking like this up here so far. I chose one of the functions and multiply it by the derivative, so that seems okay. But then my first deviation from this is that this is no longer plus, this becomes a minus. "All right, great. Why?" I don't know why. It falls out in the proof when you actually do this, but we're not worried about the proof.
- 2.30** We're worried about, "How do I get to the key mathematics necessary in understanding our world today?" I'm cutting through a lot of the nuances and the mathematical proof and theory and all that stuff. I want to get my hands dirty and be able to get to a point where I can just go straight to understanding what is going on in the world today with data science and machine learning and all of that.
- 2.31** I'm going to take the bottom function, $g(x)$, times the derivative of the top, minus the top function, $f(x)$, times the derivative of the bottom. Okay, I can get on board with that a little bit. I can get on board with that. It's very similar to the top one, except that rather than a plus, I have a minus. But then now I have to throw the worst part in it is that this whole thing is divided by the derivative of the bottom squared. That's the actual right answer. The derivative of the quotient, $f(x) / g(x)$, is $(g(x)f'(x) - f(x)g'(x)) / g(x)^2$.

- 2.32** This is terrible. I mean, this is absolutely horrible. I took something relatively benign, $f(x) / g(x)$, and now it's this monster of a thing that suddenly is the right answer, which is not intuitive whatsoever. But that is the correct answer.
- 2.33** These things have names. This one is I'm taking the derivative of a product—mathematicians, we're not very good at naming things. I'm taking the derivative of a product, "Oh, what should I call it? Probably the product rule. That seems like a reasonable thing." That turns out to be what we do. The next one is, "If I'm taking the quotient of things, then the rule for taking its derivative, I should probably call it the quotient rule. That's fine." That's what it's called.
- 2.34** These ones don't really have names. This is just the derivative of a sum and difference or the derivative a scale or multiple, and that's fine. But these are four of the five rules that we have for our five arithmetic operations, ultimately. We have addition, subtraction, multiplication, division. Then the last one, which is unfortunately the most complicated, but also arguably the most important, and it shows up every single time you take a derivative, is what's called the chain rule. That's when you take functions and you can compose them.
- 2.35** Now, function composition is confusing. I want to motivate how to think about function composition. What do I mean by taking the composition of functions? Composition of functions. The notation is strange, ultimately. The composition is $f(g(x))$, and the way that this is considered is $f(g(x))$ like this. That's ultimately what it is. Now, to understand this, you have to go back to understand what is a function in the first place? What does a function do? What is its purpose? What is this notation? Why do we use it that way?
- 2.36** A function is nothing more than a machine to some extent, where by that, I mean, I'm going to feed an input into the machine, and whatever that input is, if it's valid, then the function is simply just going to output something. That's all it is. That's ultimately what a function is. You give me the number 5, my machine says, "Hey, whatever I do to 5, here is what I do to 5, and that is my output," and that is what $f(x)$ is. You give me x is equal to 5, $f(5)$ will be whatever the machine is telling me $f(5)$ should be.
- 2.37** Now, at the end of the day, we're talking about functions where we're inputting numbers or things like that. But we could do something very simple, such as these recommendation engines. These recommendation engines, which are based on machine learning, artificial intelligence, things like that. At the end of the day, if I purchase a whole bunch of things from a particular retailer, and then suddenly I start getting emails saying, "Hey, I think you would like these as well." I like to buy a whole bunch of textbooks about a particular thing, and then suddenly I'm getting a bunch of emails saying, "Hey, here's a whole slew of textbooks that you may like to buy." Why? Because it's using my user history to try and generate a profile for what I might like to purchase. It's a recommendation engine.

- 2.38** It's the same thing as if I'm watching a certain type of show. Let's say I like a certain type of genre. I like dramas or something like that, and I watch a whole bunch of dramas. Then suddenly, the program from which I'm watching dramas says, "Hey, here's a whole slew of other dramas that you might be interested in." That's what a function is. It's taking as an input my user history, and then the function is saying, "Oh, this is your user history. These are the things that you've watched. These are the things that you've bought. Here's a recommendation for something that might interest you." At the end of the day, it's a function. That's all it is.
- 2.39** Function composition is saying the following. Rather than simply putting in a single input to it, I'm going to run that input through another function. I'm going to take as an input x , I'm going to feed it into another function g . That $g(x)$ will be a certain output, whatever that is, and then I'm going to take that and I'm going to feed it into another function. That's the idea.
- 2.40** My initial input is x . I feed that into a function g . It gives me an output, whatever that is, but that output becomes the input for the next function f . If you go back to this notion—I don't know what this would look like, but you could think of my input is my user history for things that I purchased from some retailer, and g is going to give me a recommendation based on that user history. It's going to say $g(x)$ is, "Hey, I think you should buy these things as well." Then I'm going to take that recommendation and I'm going to feed it into the function f , and whatever f is going to be in that case is going to give me an output associated to that. You can see how you could iterate this over and over and over again.
- 2.41** Now, from a mathematical standpoint, you could think about it like this. You could think about it as, let's say x is a particular point, and then $g(x)$ is going to take that point and it's going to move it to a certain place. Now, based on its new position of wherever it is, f is going to take that point and move it someplace else. Then I can iterate this over and over again as long as these inputs are valid.
- 2.42** I could say, for example, "What is x ?" I could do $f(x)$ and then $f(f(x))$ and then $f(f(f(x)))$ and I could just keep doing this. And maybe f moves x here, and then f moves that point to here, and that moves that to here. I can ask the question, "Where does it end up if I do this infinitely many times? Does it end up someplace? Does it go off to infinity? Does it go to the 0.7? Where does it go? How does it iterate? Where does it end up?"
- 2.43** Function composition is actually very useful. However, its derivative rule—if I say, "What is the derivative with respect to x of $f(g(x))$?" How do I handle this? The way I like to think about it, which I like to think about things very simplistically because Because I think mathematics is complicated enough. It's best to have very simple ideas for understanding these principles.
- 2.44** I like to think about this as an onion, which is embarrassing to admit, but I like to think about it as an onion. There is an outermost layer here. There is an outermost function,

and that outermost function in this case is the function f . I differentiate f . I take its derivative. We say "differentiate" to mean take the derivative. I differentiate the outermost layer first. That outermost layer is going to be an f' . Now, where am I going to evaluate f , in this case f' ? Well, f is being evaluated at g . I'm going to say, "Take the derivative of f , evaluate it at $g(x)$," and now I've worked my way through that particular layer. That's my outermost layer.

- 2.45** I start working my way back into the innermost layers. The next layer is going to be g , and I just multiply by $g'(x)$, and I keep going until I'm done. There's only finitely many layers, so there's going to be finitely many derivatives that I have to compute. In this case, I'm going to multiply by $g'(x)$, and then I'm done. That's how that works. If you think about it as, "What is the outermost function, and then what's the next outermost function, and then what's the next outermost function?" and things like that, it'll help tremendously in terms of computing these things.
- 2.46** This thing is called the chain rule, the chain rule, because you're chaining the functions together. You have this $f(g(x))$, and you can do this for multiple compositions. It doesn't just have to be $f(g(x))$. I could have 20 functions composed together and try and figure that out. That's actually what's happening from a machine learning standpoint.
- 2.47** There's a lot of these buzzwords that people like to throw around, and that's fine. That's completely fine. But at the end of the day, do you actually understand what these buzzwords mean? People like to throw around machine learning or neural networks or things along those lines. Do you actually understand what a neural network is? What is it doing? How are you computing with it? What are its implications and things like that? If we don't understand what goes on under the hood, that's dangerous. If I don't understand what my algorithm is actually doing, then maybe I've actually programmed some bias into it unknowingly. If I don't actually understand what's going on under the hood, if I don't understand the mathematics underlying everything, that's concerning. That's what we're getting at.
- 2.48** A neural network, at the end of the day, is literally a whole bunch of functions composed together. Depending on what are called the layers of your neural network, it depends on the number of compositions that I have. But at the end of the day, I'm trying to find a function and I'm trying to minimize my error. If I'm in minimizing my error, I'm looking at a composition of a ton of functions, and I'm trying to say, "How do I make that as small as possible? How do I minimize the error in this case?" You're boiling down to the chain rule. That's all it is.
- 2.49** Let's get a list of functions, how we can compute their derivatives, and actually just cut through straight through to say, "Look, we can just compute derivatives now based on simple principles and basic functions that we know." Let's start from there.
- 2.50** Typically, the derivative is built differently, and I personally think that we can streamline the process a little bit, and we don't have to build it up from this basic

principle. I'll tell you what that basic principle is later. However, right now, I just want to know, "How do I compute derivatives? How do I turn the crank and actually just going through the motions of computing derivatives without necessarily having to do it from basic principles?"

- 2.51** Here are some examples. Some examples. If I give you the function which is x^n —raised to any power, whatever that is, it doesn't matter what the power is; n is just a number. That's all it is; n is a number. I want to say, again, the d/dx —this is like our quotation mark, right? This is saying, "Take the derivative of whatever is to the right of it." If I look at x^n , whatever that is, x^2 , x^3 , x^4 , $x^{1/2}$, $x^{1/\sqrt{2}}$, $x^{1/\sqrt{7}}$, whatever it is, and I want to say, "What's its derivative?" Its derivative is very simple. It simply says take that exponent, drop it down in front, multiply it, and then take the variable and raise it to one less of the exponent. That's it.
- 2.52** As an example, I could say, what is the derivative of x^2 in this case? If I follow my rules, the rule says, okay, the 2 should come down, and then I just decrease the 2 by 1. That's it. I simply get $2 * x^{(2-1)}$, which is $2 * x^1$, which we would write as $2x$. Now we know how to compute the derivative of x^2 . Simply just $2x$. That's great.
- 2.53** Let's do a different exponent that's a little more difficult. If I take the derivative with respect to $x^{(-3/2)}$. Again, I said n could be anything. Let's just pick whatever number that comes to mind. How would I do this? Again, I follow the principle. It says: whatever is right here— I'm going to drop that in front, and I'm going to drop this n by one. The principle says: this comes out front, so I get a $-3/2$. I rewrite x , whatever that is, and then I have to decrease negative $-3/2$ by 1. I'm going to get $-3/2 - 1$.
- 2.54** Now I have to do the dreaded "simplify your fractions," which is nobody's favorite thing ever. But it's a thing. I get $-3/2 - 1$. And -1 , I could really write as $-2/2$, if I'd like. Then what do I get? I get $-3/2 * x^{(-5/2)}$. And I'm done. The rule is that simple, ultimately. We're what—25, 30 minutes in or something like that—and we're computing derivatives already. That's fine. That's my goal, to cut through everything and get to the point where you can actually use this. You can actually use this. This has a name. It's called the power rule. Every one of these things has a name associated to it. But at the end of the day, it doesn't really matter. If you roll up on a function and say, "Hey, what's its derivative?" You just need to know how to do it at the end of the day. That's it. That's how you differentiate x^n .
- 2.55** One that's really nice is that if I have a constant here, a constant. So think of a constant as a number, whatever that is. And I say the derivative of this. Now, I want you to think back to the beginning when we were talking about the derivative as a rate of change. If I have a position and I say, "What is its rate of change with respect to time?" We said, "That's a speed." Miles per hour, for example. What is your speedometer reading in your car? It's reading 65 miles per hour. That's your rate of change of position with respect to time, ultimately.

- 2.56** If I say, "If I'm constant, if I'm not moving, if my position is right here and I'm not moving, then what should my rate of change be?" It should be nothing. I'm not moving. There is no rate of change. If my position is I'm standing still, I'm not moving in any particular direction. My speed is zero. That's a general fact of derivatives, is that this will always be equal to zero. The derivative of a constant is zero. That's how we can see why it's zero is that it's a rate of change. At the end of the day.
- 2.57** All right, what are some other ones? Some other ones are we have all of these functions that are— you ask most people and they're going to be like ugh; they bristle a little bit when you start talking about these things, like trigonometric functions, sines and cosines and tangents and all these things associated to geometry. These functions are great, and they help us model oscillations and things of that nature. The sound that that we produce, we can model it through $\sin(x)$ and $\cos(x)$, and we can model manufacturing processes via $\sin(x)$ and $\cos(x)$, and we can create machines that will actually trace outlines of various things using $\sin(x)$ and $\cos(x)$. But they're intimidating, and they're a little frustrating to deal with. But at the end of the day, they're functions, and we need to know what are their derivatives.
- 2.58** The derivatives are the following. We have the six trigonometric functions. First one is $\sin(x)$. The derivative of $\sin(x)$ is $\cos(x)$. That's nice. The derivative of $\cos(x)$ is equal to— it would be ideal if it were $\sin(x)$. That'd be nice. They'd just play off of each other—the derivative of $\sin(x)$ is $\cos(x)$, the derivative of $\cos(x)$ is $\sin(x)$. Turns out that's not the case. You get $-\sin(x)$, so you get $-\sin(x)$, the pun intended, okay? S-I-G-N and S-I-N-E, okay? You get the negative sine associated to that.
- 2.59** Now, what do I get for the derivative of $\tan(x)$? The derivative of $\tan(x)$ in that case— this one is annoying. I get a $\sec^2(x)$ associated to that. Okay, I don't really like these functions all that much. The derivative of $\tan(x)$ is $\sec^2(x)$. That's starting to get a little out of control. But those are the first three, the $\sin(x)$, $\cos(x)$, and $\tan(x)$.
- 2.60** Now, what I'll show you later is that ultimately the only ones that you need to understand are these two, the derivative of $\sin(x)$ and $\cos(x)$ because if you have the quotient rule and you understand the definition of the other trigonometric functions, you can exploit that to get the other definition. And I'll show you how to do that.
- 2.61** So what are the other ones? We have the derivative with respect to x of $\sec(x)$. This one is $\sec(x) * \tan(x)$. The other one we have is $\csc(x)$, so the derivative of $\csc(x)$. This is very similar to $\sec(x)$. $\sec(x)$ was $\sec(x) * \tan(x)$. $\csc(x)$ is going to be $\csc(x) * \cot(x)$, but we're going to have to augment it with a negative sign in front. So this will be $-\csc(x) * \cot(x)$.
- 2.62** The very last one we have is the derivative of $\cot(x)$. Those are the six trigonometric functions: $\sin(x)$, $\cos(x)$, $\tan(x)$, $\sec(x)$, $\csc(x)$, $\cot(x)$. The derivative of $\cot(x)$ is very similar to $\tan(x)$. So $\cot(x)$, but this is $-\csc^2(x)$. You can see that there's a lot of similarities between these things. $\tan(x)$ —the derivative was $\sec^2(x)$. $\cot(x)$ was -

$\csc^2(x)$. So there's some similarities there. We had that $\sec(x)$ was $\sec(x) * \tan(x)$. $\csc(x)$ was $-\csc(x) * \cot(x)$. So there's a lot of similarities that we can play off with all of this.

- 2.63** But I want to show you using the quotient rule how you can get, for example, the derivative of $\tan(x)$, which is why I say you really only need to know what are the derivatives of $\sin(x)$ and $\cos(x)$. As long as you know the the rules, the quotient rule and things like that, you can then find the other four. If you're anything like me, I don't necessarily remember things very well associated to this, but I remember processes very well. I may not be able to just recall all the definitions of the derivatives, but I can say, "I know how to get them." That's the key thing.
- 2.64** I want you to think. I want you to be able to think through the problems. I don't want it to just necessarily be, "I want to regurgitate whatever the rules are for whatever the game is." I want you to be able to say, "I don't remember what it is, but I know I can think my way through to actually solve the problem," as opposed to just regurgitating a solution. Does that make sense how that mentality works?
- 2.65** So let me show you how we can find the derivative of $\tan(x)$ using the quotient rule. Now, the quotient rule has a silly, I guess you could say, mnemonic to remember it. Let's go back to this. If I have $d/dx [f(x) / g(x)]$ — remember? It was this nasty monster of a formula associated to it. How in the world am I ever going to remember that? It's the bottom times the derivative of the top minus the top times the derivative of the bottom all over the bottom squared. It's a ultimately very silly way to remember this.
- 2.66** The way you think about this is you designate $g(x)$ as being "low," and you designate the top as being "high." So then you get the following thing. Let's say it's this. What is this formula? It's really $(\text{low} * d(\text{high}) - \text{high} * d(\text{low})) / \text{low}^2$. Now, let me walk you through this. It's a very silly thing to say. It's like, "Oh, low, d-high minus high d-low all over low squared." It's very catchy and quick to remember, ultimately. But what do I mean by this?
- 2.67** Low is saying that's $g(x)$. That's my $g(x)$. $D(\text{high})$ is saying what? What does the "d" represent? It's saying take the derivative of high. So, $d(\text{high})$ is saying $f'(x)$ —what is high? High is $f(x)$. Again, $d(\text{low})$ —what is $d(\text{low})$? D means take the derivative. So $d(\text{low})$ going to be $g'(x)$ all over low^2 . That's a very quick way to remember it.
- 2.68** If I have this formula, then I can say, "What is the $d/dx [\tan(x)]$? Okay, so you're like, "I took geometry at some point and the teacher told me that $\tan(x)$ was just $\sin(x) / \cos(x)$, so maybe I can start there. Maybe I can start there. I know that $\tan(x) = \sin(x) / \cos(x)$. You're like, "You just made your life worse. You gave me originally just $\tan(x)$, and you made it a quotient. Now it's a fraction." You're like, "I just made my life worse." It's like, yes and no.
- 2.69** I've expanded $\tan(x)$, I've made it bigger, and that's intimidating. But now I've given myself an avenue to solve the problem because now I can stand back and I can say, "Oh, this is a quotient to function. It's $f(x) / g(x)$. Do I have a rule that tells me how to

differentiate quotients? Oh, sure. The quotient rule will help me. I go to the quotient rule and I say, "How do I actually work this out?" I need to find my designated f and g associated to these things. So f is the numerator, that's the top function, in this case, f would be my $\sin(x)$, and $g(x)$ is my denominator. And so $g(x)$, in this case, will be designated as my $\cos(x)$. I can say, "If I know that $f(x) = \sin(x)$ and $g(x) = \cos(x)$, now I can run through and compute this."

- 2.70** Let's see how that works. Well, it's going to be low—that's my $g(x)$. I'm going to get $\cos(x) \cdot d(\text{high})$. That's $f'(x)$. So I need to say $f'(x)$, and I have to go back and say, "What is the derivative of $\sin(x)$? That's $\cos(x)$." So $f' = \cos(x)$. So I get $\cos(x) \cdot \cos(x)$ — the top $f(x)$, which is $\sin(x)$. So I get $\sin(x) \cdot \text{what?}$ — I have to look and I say — $d(\text{low})$, d low is the derivative of $g(x)$. So I go to $g(x)$ and I say, "What is $g(x)$?" That's $\cos(x)$. I have to go back and say, "I remember the derivative of $\cos(x)$, it was equal to $-\sin(x)$. I get that. I get $-\sin(x)$ here. All divided by what? All divided by $g(x)^2$. Well, $g(x)$ is the denominator. The denominator in our case for $\tan(x)$ is $\cos(x)$, so I say divided by $\cos^2(x)$.
- 2.71** Now I can do some simplification. $\cos(x) \cdot \cos(x)$ —whenever I have an object times itself, it's the square of that object. If I have $x \cdot x$, it's x^2 . If I have $3 \cdot 3$, it's 3^2 . This is $\cos^2(x)$. Now, what do I see here? I see a minus sign there and I see a minus sign there. What's negative times negative? It becomes positive. I'm going to get a positive. I have object times itself, so $\sin(x) \cdot \sin(x) = \sin^2(x)$, all divided by $\cos^2(x)$. Then I have to remember there's this identity that says if I take $\cos^2(x) + \sin^2(x) = 1$. That's just a standard identity. This is $1 / \cos^2(x)$. Then I have to say, "What is $1 / \cos^2(x)$?" Well, $1 / \cos(x) = \sec(x)$. $1 / \cos(x)$ is equal to $\sec(x)$. That's one of our other identities. What is this? This is $\sec(x) \cdot \sec(x)$, which is $\sec^2(x)$. That's exactly what we had for our definition of $\tan(x)$.
- 2.72** What I'm trying to illustrate is you can memorize all of these things all day long, and that's completely fine, and that'll serve you well. Ultimately, at the end of the day, that will serve you well. But if you're like me, and I don't like memorizing things necessarily, I like to understand the process of getting to places, then I'm just going to say, "As long as I understand the derivative of sine is cosine, the derivative of cosine is negative sine, all I have to do is understand the quotient rule, how to apply it, and I can get everything else." Does that make sense, ultimately? You can use those principles to get that.
- 2.73** Those are the derivatives of our trigonometric function. Now we have the derivatives of power functions, x^n . We have derivatives of trigonometric functions. That's fine. Let's just write down a couple of derivatives of other functions that show up quite regularly, and then we'll start figuring out how we can actually use this to try and get to the original problem at hand, which was understanding what the derivative is telling us.
- 2.74** The other functions that show up, and they're mirrors of each other because they're inverses of each other, are exponential functions and logarithmic functions. Logarithms are, again, one of those things that make people bristle a little bit when they see them.

They're these things that you were subjected to with all sorts of things in school. That's okay.

- 2.75** How do I take the derivative with respect to x of something that looks like e^x ? Now, remember, e is a special number. I mean, yes, it's a letter, but it's representing a specific number. That number is roughly 2.718 and change. It's some special number that Euler we came up with at some point. Ultimately, this is saying it's an exponential function. It's the reverse of a power function. The power function—and this is a very common mistake, actually, and it's natural to make this mistake—the power function was what? It was a variable raised to a number. This was variable raised to a number. That's what its structure was. The power rule said, "Oh, I know how to take the derivative. Bring the number down, rewrite the variable, and then decrease the number by 1." That was our rule.
- 2.76** Think about what this is saying. This isn't saying a variable raised to a number. This is flipping the script. This is saying, "I have a number raised to a variable" in this case. It's plausible, at least, to say that the power rule probably isn't going to apply here. A very natural mistake is to say, "Oh, I know what this derivative is. I bring the x down and then I just decrease the x by 1 because that's what you told me how to do with things that look like that." I'm like, "No, that's not what I said. What I said was when you have something that looks like variable raised to a number, that's when the number comes down and you decrease it by 1. I didn't say that when I have number raised to a variable, that that rule is the same." Does that make sense? The functions look very similar, but they're very different in their overall makeup. So this is not the correct thing.
- 2.77** What is it? In this case, this one's beautiful. It's literally itself. That's all it is. It's one of the easiest things. Aside from constants, it's the easiest thing in the world to differentiate. It's like, what's the derivative of e^x ? It's itself, e^x . I don't even really have to think about it all that much. It is what it is. If I want to generalize this and say, "That's all fine and good, but I want something that looks like this." I have a different base that's not e in this case. So b could be 2, b could be 10, b could be 50 million. It doesn't matter what it is. I want to say, "What is this derivative?" The derivative is b^x , but then I have to multiply by $\ln(b)$. I'm going to have $\ln(b)$.
- 2.78** Now, this $\ln(b)$, it's like, "Why isn't that occurring here? Why isn't there a $\ln(e)$ here?" There is. This is technically $e^x \cdot \ln(e)$, but $\ln(e) = 1$, so it just doesn't show. Technically, that natural log is still there, but we don't write it because it's equal to 1. For example, using this one: If I said, "What's the derivative with respect to x of the exponential 2^x ?" This is saying for every x you give me—if I go $x = 1$, $x = 2$, $x = 3$, I'm just going to double each time. What is this going to be? Well, I get $2^x \cdot \ln(2)$. The $\ln(2)$ is not equal to 1, so $\ln(2)$ stays. At the end of the day, it stays.
- 2.79** But please keep this in mind: This particular function is a number raised to a variable, not a variable raised to a number. That's a very key difference. Those two functions are

vastly different from each other. If I look at the functions x^2 versus 2^x , even though they look very similar to each other, they're vastly different. If I plug in $x = 10 \rightarrow 10^2 = 100$, and if I plug in $x = 10$, I'm going to get 10,024 or something? No, 1,024, something along those lines. I'm getting a massively different number. So if their functions themselves differ in their inputs, then it stands to reason that their rates of change are going to vary significantly. If the functions themselves are different, then their rates of change, their derivatives, are probably going to be different at the end of the day. That's what we're getting at. That's how you differentiate exponential functions that look like b^x .

- 2.80** The last one is: How do you differentiate their counterparts, meaning their logarithms? Exponentials and logarithms are inverse functions to each other. For example, if I take $\ln(e) = 1$. If I take e to the $\ln(1)$, I'm going to get 1, things like that. They're inverse operations to each other. The derivative of $\ln$, that's how you read this, "natural log", is 1 divided by whatever its input was. In this case, its input is just x , so it's going to be $1/x$. That's fine.
- 2.81** What if I had a general one? What if I had the derivative with respect to x of $\log_b(x)$? Just like b^x is the generalized exponential function, what would $\log_b(x)$ be in this case? Again, I get my $1/x$ still, but I also have a $1/\ln(b)$. That goes along for the ride, which is why this one— $\ln(x)$ —is actually $1/x$, if you think about it, because the definition of $\ln(x)$ is $\log_e(x)$. So if I apply this rule to $\log_e(x)$, what am I going to get? $1/\ln(e)$. Well, what was the $\ln(e)$? That was just 1. So the $1/\ln$ doesn't play. This is just equal to 1, and I'm left with $1/x$. That's the generalized version of logarithms.
- 2.82** Those are our building blocks. Those are our elementary functions to some extent. Now we can say, well, let's take the rules that we had for derivatives and apply it to these elementary functions and see what happens. Let's see what happens. Can we actually compute them now? Now that we have all these rules. Let's see how this works. I'm going to start throwing the chain rule at you and all things along those lines because the chain rule is used every single time you compute a derivative. Here we go.
- 2.83** Here are our examples. Let $f(x)$ equal, we'll start off relatively nice, $\sin(x^2)$. We want to try and understand how to find $f'(x)$. So find $f'(x)$. Now, again, the motivation for this is, "You're just having me push symbols around on a board." To some extent, yes, I am just having you push symbols on a board right now. But what if this $\sin(x^2)$ is representing a model that I have for a particular patient's heartbeat in a very small range? In a very small range of time, a patient's heartbeat is being represented by $\sin(x^2)$. After all, I told you that waves—in the beginning, I told you that waves can actually be modeled using sine and cosine associated to these things. The wave that's coming out of my mouth regarding the sound that's being produced, that can be modeled using sine and cosine.
- 2.84** It stands to reason that if you have something that has a wave pattern, like a heartbeat, you could probably model that using sines and cosines. Why would it be relevant to know the rate of change if this is representing a small part of a heartbeat? You want to

understand how is a patient's the start rate changing in time. You want to understand, is it speeding up? Is it slowing down? What's actually happening? So find $f'(x)$. How do we work through this?

- 2.85** So solution—that's my abbreviation for solution, function—I have to look at this and say, "I see a sine function, but I also see this x^2 function, and they're nested. It's like the x^2 is inside of the sine function. So the first observation I have to make is I have to say there's a composition going on here. There's two functions that are being composed. You give me an x , what's the first thing that I'm going to pass it through? You give me x , I'm going to say I'm going to square it. I'm going to turn it into x^2 . Then I hand off the x^2 to you and you say, "Okay, now that I have the x^2 , I'm going to go and stick it inside of the sine function, and that's going to give me the output of f ." You see that it's a two-step procedure. You give me x , I then square it, I take that square, and then I put that inside of the sine function. There's a composition going on.
- 2.86** But what is the outermost function? What is the last function that I do anything with? It's that one—that's my sine function. Let's go back to the chain rule. The chain rule said what? If I take the derivative with respect to x of $f(g(x))$, that was equal to $f'(g(x)) \cdot g'(x)$. Yes? It was the onion analogy. I start with the outermost function f . I differentiate that, and then I go to the next layer, g , I differentiate that, and then I keep working my way down. I say, "What's the outermost function in this case?" The outermost function is sine. I say, "What's the derivative of sine? Oh, it's cosine. I know that.
- 2.87** Then f' , I say that's going to be cosine. But cosine of what? $\cos(x)$, $\cos(x^2)$, cosine of what? What am I actually going to evaluate it at? Let's go back to the definition. The definition says whatever the inside was $g(x)$, you take the derivative of f and you evaluate it at $g(x)$. In our case, $g(x) = x^2$. The derivative of f is cosine. I evaluate it at x^2 . That's this part of the chain rule, that's this part, that's the outermost layer.
- 2.88** Then I cover this up and I say, "I'm done with sine. What's the derivative of the inner part?" That's x^2 —the derivative of x^2 . I use the power rule, and the power rule tells me the derivative is going to be $2x$. So I'm going to multiply this by $2x$. Now, that's perfectly fine. That's the correct answer. But depending on how you write things—if you put the parentheses here or if you don't write it with parentheses, it's safer to simply move this $2x$ in front and write this as $2x \cdot \cos(x^2)$. That's our final answer at the end of the day.
- 2.89** The reason why I say moving the $2x$ in the front is actually safer is because if you don't have parentheses here, it's very easy to make the mistake of saying, "Oh, $2x \cdot x^2 = 2x^3$, so this should be $\cos(2x^3)$. No, the $2x$ is multiplied on the outside of the function. You're not multiplying the inputs. You're not multiplying the inputs. There's an example.
- 2.90** Let's do another one. Let's do another example. Let $g(x)$ —we're going to make this one pretty horrible. We're going to say this one is equal to $\sec^2(x^2 + 1)$. Actually, let's make the inside—let's do $((x^2 + 1) \cdot e^x)$, which is pretty terrible. This is going to blow up very quickly. This is going to be pretty horrible. But it illustrates the point: As long as you

follow basic principles step by step, you can conquer hard problems as long as you're sinking your way through the problem.

- 2.91** Let $g(x)$ equal this. Find $g'(x)$. Find $g'(x)$. Okay. What's my solution? How do I think about a problem like this? I have to say, "Wait a second. This is a composition. You give me an x , what am I going to do? I'm going to stick it into this. I'm going to square it, add one, and multiply by e^x . That's the first part. Then I'm going to take that and I'm going to feed it into the secant function, which is a disaster in and of itself. Then whatever that output is, I'm going to square the result. I'm going to square the result. There's a three-fold—there's three functions being composed in this particular thing. It's a three-layered onion to some extent.
- 2.92** The first thing I'm going to do is I'm going to rewrite this and write this as $\sec((x^2 + 1) * e^x)^2$. That's what we get. That's what this notation is. When you have secant raised to the power of two, it's really saying $\sec(x) * \sec(x)$. It's $\sec^2(x)$ squared associated to that thing. I stare at this for a long enough time and I say, "What's the outermost function?" It's some mysterious object squared. Do we agree? The outermost thing is something squared. I have to go back and I have to say, "Wait a second. Do I have a rule that tells me how to differentiate something that is blank squared? Oh, power rule. Any day of the week."
- 2.93** Variable squared—power rule says, "Bring the power down, decrease it by 1." So I can say, All right, at the very least, $g'(x)$ will be equal to what? The two comes out—I get a 2. I rewrite whatever this mysterious thing is. This is $\sec((x^2 + 1) * e^x)$. Then I decrease that by 1. Now, I'm not going to write " $2 - 1$ " because that's is equal to 1. I'm not going to put a 1 here. Why? Because if I'm a little fast and loose with my notation, I might think that that's a prime symbol and it's telling me to take the derivative when in actuality just a 1. I'm not going to rewrite that. My outermost function was something squared. Fantastic.
- 2.94** My next layer is what? My next layer is this secant layer, and I have to say, "What is the derivative of secant in that case?" The derivative of secant is what? Secant went to secant times tangent. I'm going to have to multiply by $\sec((x^2 + 1) e^x) * \tan(x^2 + 1) * e^x$. Now my secant layer is done.
- 2.95** Now I'm down to the final layer, which is this product, and I have to say, "What's the product?" That's product rule. I told you this one was going to blow up very quickly. Now I'm going to have to multiply by the product rule of this. I have to think back, "What was the product rule?" It was the first term times the derivative of the second. The first term is $(x^2 + 1) * d/dx$ of the second term. The second term is e^x , and that was the best derivative ever because its derivative was itself. Okay, so that's just e^x . Plus what? The second term, e^x —I'm going to have to go down a line. So $e^x * d/dx$ of the first. That's a power rule, so that's going to give me a $2x$, and we're done.

- 2.96** Just a recap. We have these basic rules which tell us how to compute derivatives, and we can combine those rules in any particular way with compositions, with products, with quotients, with all these different things to compute derivatives. Now, why do we care about derivatives? Well, we'll get into that. We'll get into that. Why are derivatives relevant? We'll get into that. It all stems from optimization, and that will be the topic for what we do going forward.
- 2.97** Can you briefly dissect Euler's number and its importance?
- 2.98** Yes. So Euler's number. That was his number e . I told you this number e was special and Euler's number was 2.718 and change and all of that. Where does it come from and what was the motivation for it? What Euler was doing was he was looking at the following situation. He was saying, "If I graph the function that's $y = b^x$, so an exponential function, where $b > 1$, where $b > 1$. So $b > 1$. If I graph that, any graphing utility, whatever you want to use, this is my graph of $y = b^x$. Here's my x-axis, here's my y-axis. Here's the point. If I plug in $x = 0$, what do I get?" What's b^0 ?—this point here is 0, 1.
- 2.99** What Euler was asking, he was asking the following question. He was saying, "Look, if I draw straight lines, if I draw straight lines there, that line has a slope associated to it, and it's going to dictate how steep it is, how flat it is, maybe it's negative, things like that. That's what the slope is going to tell me." He started asking the question. He asked a very simple question, "What b -value makes that line have slope equal to 1?" That was question he asked. Why did he ask it? No clue. No idea. But at the end of the day, he was like, "Is there a b -value that would make this have slope equal to 1?"
- 2.100** He started saying, "If $b = 2$, the slope is something. If $b = 3$, the slope is something. He realized that the slope at $b = 2$ was not quite right. The slope at $b = 3$ wasn't quite right, but it was somewhere in the middle. It was somewhere in the middle, and that's where he found the particular number. He went through and he said, "The answer was that $b = e$, which is roughly this 2.718 and so on, is the right number. That's the right number."
- 2.101** Now, we can see that to some extent. We'll get into this, but the slope of this line, the slope of this line is the derivative evaluated at $x = 0$, because that's the point where I'm considering this. I can evaluate that now and I can be like, "All right, wait a second. So this function is $f(x) = b^x$. Then we had a rule for its derivative: $f'(x) = b^x \ln(b)$ —that shows up again—it was equal to b^x , but I had a $\ln(b)$ there as well. I had that. If I know that, then I can say, "Oh, cool. If I evaluate this thing at $x = 0$, What am I going to get? I'm going to get the following thing.
- 2.102** What I want is $f'(0)$. I want that to equal 1 because $f'(0)$ is the slope of whatever this line is. The line has a name. The name is called the tangent line at $x = 0$. But that's beside the point. If I evaluate $x = 0$, it gives me the slope of the line. I have a formula for my function: my function is $f'(0)$. So plug in $x = 0$. What do you get? You get $f'(0) = \ln(b) \cdot b^0$. Okay, $b > 1$. It's a number. So think about b is like five or whatever— 5^0 is what?

It's equal to 1. It's equal to 1. So this thing is equal to $\ln(b)$. So I'm saying, "What is the b-value necessary for that to be equal to $b = 1$?" And the b-value is e. The b-value is e. So $b = e$. That was the motivating question that he had.

2.103 What his motivation for was for asking that question? No clue. This is pretty common with mathematicians. We ask questions all the time, and whether the question has an application or not is irrelevant. But that's where it came from.

Lecture 3: Optimization in Action

At a glance — Optimization in action, building on derivatives.

- 3.1 All right, so now that we have some of the basics of derivatives and computing them under our belt, we want to understand why are derivatives relevant, why are they important, and then try and understand what methods there are to be able to find where they're equal to zero, ultimately. Finding where the derivative is equal to zero is relevant for the following reason. This is "why" derivatives, if you will.
- 3.2 If you remember, I said, effectively, everything that we do is an optimization problem. When you start viewing the world through that lens, it's frustrating, ultimately. You go to the supermarket now and you realize that all of the soup cans have the same shape, and you're like, "Wait a second, why?" Then you realize why—because there's a mathematical problem underlying it that says that's the optimization, that's the optimal thing that they should be doing. You ruin shopping for yourself to some extent, ultimately.
- 3.3 But everything that we do, to some extent, is an optimization problem. Where I want to get lunch, that's a constrained optimization problem. I have a certain amount of money to go and do that. Where I want a vacation, where I want to—how do I get from A to B? I'm trying to get from my job back to my home, and the freeway shut down because of an ice storm or whatever the case may be. So now I have to take backroads and all of this. I want to optimize—I want to get home as quickly as I can, but I have constraints that show up to these things. Everything we do is an optimization problem.
- 3.4 Even, and this is ironic, even taking exams or anything along those lines. Let's say you're taking a certification exam of some type and they say passing is 75%—75% above is a pass for the certification. You study really hard and you work really hard and all that, and then you get to the exam and you don't realize that it's an actual optimization problem. You scan through the exam and you say, "Oh, I can do these 60 problems easy out of the 100 problems that there are." Let's say there's 100 problems on the exam. You look through and you're like, "Oh, these 60? I can knock out of the park. Easy. I get all those correct." That's fine.
- 3.5 Then in the back of your head, you're like, "I got to get that 75% cut off, though. What am I going to do?" Well, the ones that I may not necessarily know, then I start trying to optimize like, "I think I know this one. I think I know that one. Maybe I can eliminate some choices and all that." What are you trying to do? You're trying to optimize your score. That's ultimately what you're doing. You want to get the highest score possible. That's an optimization problem at the end of the day. Everything we do can be couched in an optimization problem.
- 3.6 So optimization boils down to—if I give you a function, for example, let's say I give you this function, Whatever it is, I give you the graph of it, let's say. Let's say the function

does something along these lines. It does something like, something like that. I don't know. That's a function. That's fine. Frankly, I don't know what its equation is, but at the end of the day, it doesn't matter what the equation is. But that's what it is. Maybe this thing has a physical interpretation in the real world. Maybe this is something where it's my revenue in my business or something like that at some point. That's what this is modeling.

- 3.7** Then it's relevant to say the following things. It's relevant to be like, "Where does this occur?" That's some peak behavior. That's like a maximum to some extent. That's where locally my function is maxing out. If I narrow my input to say, "I'm only allowed to look at input values in that range," maybe that's the first quarter of the year, that's my range. I want to say that max is out there. How do I characterize that maximum? Then maybe I move over to the next one and I say, "That bottoming out, that's relevant, too. I want to know what that's about because that means I didn't have that great of a quarter, potentially. I bottomed out. I didn't get as much revenue or profit as I wanted to." The same thing goes for this. Same thing goes here. Same thing goes here. Same thing goes here. Hopefully, this isn't your business, like down here. You got negative revenues. That's not great.
- 3.8** But I want to classify. If you gave me the formula, if you told me $y = f(x)$, and you said, "Look, Robbie, here's $f(x)$. Find out where those peaks and those valleys are. I want to know where they are because that's relevant to my business. I want to know what are the input values that maximize this because I want to keep doing those. I want to keep doing those." What are the ones that give me the lowest value? I want to avoid those at all costs. So this is relevant, yes? Okay.
- 3.9** So what we have to understand is that the derivative has an interpretation. The derivative represents a rate of change. It also represents the slope of— so new vocab word. These lines, if I looked at this top thing and I just drew this entire horizontal line right there, do we agree that that line isn't fluctuating up or down? It's completely flat. It's completely flat line. This line has slope equal to what? The slope tells me how pitched the line is in this case. If the slope—if the line is completely horizontal, then the slope is nothing. The slope is zero. That's the slope = 0.
- 3.10** Now, what's special about this line is that it just kisses off the graph in this one point. Whatever this x -value is down here, Whatever that x -value is, if I evaluate something at that x -value, I should get zero, ultimately. What is that something that I'm evaluating that? That's $f'(x) = 0$. This also represents the slope of the tangent line—tangent line, that's the new vocab word. That's what these horizontal lines are. They're tangent lines to the graph. It means that it just bumps off the graph at this one point.
- 3.11** Now, they don't all have to be horizontal. They don't have to be horizontal. You can look at, say, this point right here, and you can say, "I can draw a line that's going to just bounce off of that at one point. That's a valid tangent line. That's completely fine. But it doesn't have anything to do with the derivative being zero. In this case, if the slope of

this line is given by the derivative evaluated at whatever that point is, we would say it's negative at this point. Maybe it's -2 or something like that. That's not necessarily a relevant point for us necessarily right now because it's not characterizing this valley or one of the peaks associated to it. It also represents the slope of the tangent line at the point x .

- 3.12** There's multiple representations of what is the derivative, just like in ordinary language. In ordinary language, words can mean—you can take a specific idea or a specific object, and you can have multiple words for describing that particular object. Do we agree? The derivative is no different. The derivative, we can say it's a rate of change. That's fantastic. We can say it's the slope of the tangent line at the point. That's completely fine. We can say it's the instantaneous rate of change of whatever the object is. It has a lot of different interpretations. But the one that we're going to really care about is that it's the slope of the tangent line at that point.
- 3.13** So what we notice is that—here's an observation. I have to say, mathematics, it has a stigma about it that it's so difficult or that it's so hard. It's all these things. To be brutally honest, it is. It is pretty difficult at the end of the day. But one of the most important things you can get into your head is that you don't have to be an expert to make observations about something.
- 3.14** For example, you don't have to be an expert in art history or something along those lines to walk into a gallery and look at a painting and say, "That color is yellow." You can make that observation. It seems silly and it seems trivial and it seems completely inconsequential to go in and say, "Oh, that color is yellow." We get in our heads and we're like, "No, if I can't go and say that this is from the impressionist era, or that's a Monet, or that's a Van Gogh, or something like that, then somehow those observations don't matter to some extent."
- 3.15** In mathematics, it's completely different. In mathematics, making some of the most "trivial observations" can sometimes be the most important observations. If I give you a graph like this and somebody stands back and they're like, "Oh, the slope zero seems to be relevant." Somebody might look at that and be like, "That's a trivial observation." No, it's not. It's a very important observation because when the derivative = 0, I get this local max type behavior, like in this little window, that's a max, that's a min, max, min, max, min, and that turns out to be extremely relevant and extremely important.
- 3.16** Even some of the most trivial observations can actually be the most profound. The observation that we're making here is that $f'(x) = 0$ is relevant for maximum/minimum behavior. Do we agree? The derivative = 0 has something to do with some maximum and minimum behavior. What I want to get at today is saying the following: How do I figure out when functions equal zero? How do I actually do that? Because we now have this observation and this motivating problem of saying, "If I can find the x -values for which the derivative = 0, then I can find x -values where I'm going to have maximums or minimums. I can find values like that."

- 3.17** What's the relevance in the real world about finding maximums and minimums? We'll go back to what we talked about earlier. We talked about saying in machine learning and data science and things like that, you're trying to model based on the data that comes in. You're not going to have a perfect model at the end of the day. If I'm buying something online and my user history is telling me, "Here's a recommendation of what you might buy," I may not even buy those things. At the end of the day, I may not buy those things. Just because you recommend it to me doesn't mean that I'm going to buy it.
- 3.18** The model that you're giving me isn't perfect. It's not saying "By the way, because you bought these three textbooks, you should buy this one." "No, I'm not going to buy that." The model isn't perfect. It's not predicting exactly what's going to happen. It's just giving me a recommendation. So there's some error in the model. The idea is that you want to minimize that error. The smaller the error, the more accurate the model. If my business is selling products to somebody, then I want that error as small as possible so I can make the most recommendations to my consumers so that I can generate more revenue at the end of the day. If I want to figure out minimum type behaviors, I'm trying to figure out where the derivative = 0. This is a relevant question, when I'm looking for a particular function and setting that equal to zero.
- 3.19** So what I want to go through today is one of the most important algorithms. What is an algorithm? It's effectively a list of instructions. It's a recipe of sorts. That's how I like to think about algorithms. It tells me, "You do this set of instructions and I will give you this output." At the end of the day, that's what an algorithm is. You give me a recipe, you put that recipe together, and I will give you an output, whatever that is. All right?
- 3.20** So one of the most important algorithms in mathematics or in the world in general is what's called Newton's method. Newton's method. This is by far one of the most important of all time. And what's interesting is that it's actually a very simple idea. I think that's what makes it so beautiful is that it's one of the most profound applicable ideas in the world today. This isn't just math that you learn that you're like, "This was created back in the 300 BC in the Greek era, and you're learning mathematics from thousands of years ago. No, this is relevant today.
- 3.21** Newton's method, is the following thing. If someone were to come up to you on the street and say, "Hey, what's Newton's method?"— this never happens, by the way. Like ever. This never happens. No one ever comes up to you on the street and says, "Explain a math concept to me." But if somebody were to come up to you and say, "Hey, what is Newton's method?" The idea is it's trying to figure out where a function = 0. That's the idea. It's a very simple idea. The idea is where does some function—I'll call it $g(x)$ —equals zero? That's all it's asking.
- 3.22** Now, for us, we're going to apply it and say, "The function $g(x)$ that we're interested in is f' , the derivative, because I want to know when is the derivative equal to zero? But this isn't simply just the derivative is equal to zero. This is more broad than that. It's

saying any function you can think of whatsoever, where is it equal to zero? Is there a method to actually do that?

- 3.23** Here's the beautiful thing. The idea behind it is you guess, which is nice. You guess, and then you check to see if it actually works in a systematic way. That's fantastic. Now, if you're anything like me, I wasn't the greatest of students to some extent. In elementary school, when I would be given the math problem of, "Guess and check." The teacher's like, "Guess and check." I'd just throw out a number. I'd be like, "Seven." The teacher's like, "That doesn't even make any sense. What are you talking about? Seven?" I was like, "You said to guess, and now I'm going to check. Seven doesn't make any sense." Then they would say, "That doesn't make any sense. Think about it." I'm like, "All right, 12." I didn't like the method of guessing and checking, but that's effectively what Newton's method is.
- 3.24** Here's how we do it. I give you the graph of the function $g(x)$, whatever it is. Let's say it does something like this: $y = g(x)$. Now, it crosses the axis right here. What does that mean? Remember, a function takes an input x or whatever you want to call the variable, t , θ , y —it doesn't matter what it is. But it takes the input and then it gives me an output. This is saying, "If I want to figure out where $g(x) = 0$, I'm saying, 'Find the input, find the input x such that $g(x) = 0$ in that case.'" The input is x , the output is the y value— $g(x) = 0$ in that case. That's what I'm trying to find.
- 3.25** What Newton's method does is it says: choose a random point. Choose a random point and see what happens. You choose a random point. Let's say I choose this point right here. You label it with a subscript. It's x_0 to indicate that that's your first guess, x_0 . Newton had the amazing insight where he said, "The first thing I'm going to check is I'm going to take x_0 and I'm going to plug it into g . Maybe I'm incredibly lucky. Maybe I'm incredibly lucky and I plug it into g and $g(x_0) = 0$. It's a miracle. It's a miracle. I actually guessed what the answer was off the top of my head." That never happens at the end of the day. It very rarely happens.
- 3.26** He says, "Here's the thing. If I go to x_0 and it's not zero, then that means that it's either up here or it's down here." Do we agree? It's one of the two. It's either positive or it's negative. If I'm not zero, then I'm either positive or I'm negative. Those are my only possibilities. In this case, I go up to the curve and I say, "There's the point. There's the point." Now, what are the coordinates of that point? It has an x -coordinate and it has a y coordinate. The y coordinate is the output. And so this coordinate, these points here look like $(x_0, g(x_0))$. That's what I have, where $g(x_0) \neq 0$.
- 3.27** I look at that and I say, "What can I do?" I can look at the tangent line to the graph at that particular point. Now, it's not clear why that would be the right thing to look at, but I can at least look at it. Let's say I draw the tangent line associated to this. I'll do my best. Here's the tangent line. That's not too bad. There's the tangent line. Can I come up with the equation of that tangent line? Sure.

- 3.28** Each line—what are the ingredients you need for a line? For a line, you need two things. You need to know the slope of it, how tilted is it at the end of the day. You need to know how tilted is it, and you need to know at least a point that it goes through. If you tell me the line looks like this and it goes through this point over here, then I can simply be like, "There's my line." I need a point and I need a slope. For a line, you need a point. That's good. I have one. I have the point $(x_0, g(x_0))$. So my point is $(x_0, g(x_0))$. But I also need the slope. How would I get at the slope? That's what the derivative does. The slope is going to be $g'(x_0)$. Sometimes "x zero" is called "x naught". So now I have my slope and I have my point.
- 3.29** Then I have to go back and be like, "If I have my slope and I have my point, what's the equation of the line?" There's a nice formula for this. It's called the point-slope. Point-slope formula. This is one of those things that if you're like me, I have a hard time remembering things. I have to remind myself, "How do I actually get to this thing? How do I actually get to this particular formula?" I'm going to give it to you. The point-slope formula says the line will look like this. Let me break this down for you. Let me break this down.
- 3.30** It says the following: m is going to be my slope, so that's how tilted the line is at that point; m is my slope. The point (x_0, y_0) , that's this point here. It's the point that the line goes through. It's the point that the line goes through. For us, this looks like what? This looks like y — what's my y coordinate in this case? My y coordinate is that guy. That's my y_0 . That's my $y - g(x_0)$. That's $y - g(x_0) = \text{my slope}$.
- 3.31** I have to say, "While I'm looking at a line, what line am I looking at?" I'm looking at the tangent line. The key word here is tangent. Tangent is queuing you in and saying that the slope is equal to the derivative. That's the word association that you should see. If I'm thinking tangent line, I'm thinking derivative. Tangent—it's a tangent-derivative, tangent-derivative. m is going to be $g'(x_0) * (x - x_0)$. This equation right here is the equation of this line right there. That's the equation of the tangent line.
- 3.32** Now, what Newton said was he said, "If that line isn't horizontal, isn't horizontal, then it has to cross the x -axis somewhere." Think about that. You draw a line on the ground. You draw a line on the ground. You say, "As long as I'm not parallel to that line, I have to cross it somewhere." Do we agree? The only way the two lines don't cross is if they're parallel to each other. So if this line—if this tangent line right here is not horizontal, then it's not parallel to the x -axis, which means it has to cross the x -axis somewhere. That was the insight that Newton had.
- 3.33** He says, "If it crosses someplace, then what is the y coordinate at where it crosses? How high is the y ? It's equal to zero. It's equal to zero. So I cross the x -axis, crosses the x -axis when $y = 0$. $y = 0$. My question to you is, "If $y = 0$, what is the x -value for which $y = 0$? I'm saying set $y = 0$, solve for x . That's what I'm saying."

- 3.34** Let's see what happens. Let's see. Let's do it right here. If I set $y = 0$, I get: $-g(x_0) = g'(x_0) * (x - x_0)$. Solve for x . How am I going to do that? Remember, g' is a number. This is really like $2 * (x - 5)$. Solve for x . What I could do is I could divide both sides by $g'(x_0)$. Divide both sides by $g'(x_0)$. Those will cancel. Then I can add x_0 to the other side. So I add x_0 to the other side. What do I end up getting? I get $x = x_0 - g(x_0) / g'(x_0)$. Do we agree? I divide by the g' , that cancels here, and then the x_0 , I just add to the other side. That's what I get.
- 3.35** Here's the beautiful thing about it. What is that x -value? That x -value is right here. It's that x -value. What's the property that that x -value has in terms of finding where $g(x) = 0$? It's closer. It's closer than x_0 was. Do we agree? g crosses here. I'm trying to figure out what is that x -value? I start with a random x_0 . I ask myself, is $g(x_0) = 0$? The answer is, no, it's not. I say, "Okay, go find the tangent line to it." So I go up, I find the tangent line, and I say, "Where does that cross the x -axis?" If it crosses the x -axis—that's a big if. If it crosses the x -axis, then I can solve for what it is. Here's the point. I can call this x_1 . This will be my x_1 . I can say x_1 is closer to where $g(x) = 0$. Then I can just reiterate the same process. I can do this over and over and over again.
- 3.36** Now that I have x_1 , I ask the question: Does g vanish at x_1 ? When I plug in x_1 , is $g(x_1) = 0$? If the answer is no, then what do I do? I go up to the curve, I look at this point now. I find the equation of the tangent line associated to that. I find out where it crosses, I call that x_2 . I then go back up to the curve here. I draw a line. I see where it crosses. I call that x_3 . What are the x s doing?
- 3.37** x_0, x_1, x_2, x_3 . What are they getting close to? They're getting close to precisely where $g = 0$. They're getting close to precisely the point that I'm interested in. That's the beauty of Newton's method. Newton's method, it's an iterative method. Iterative meaning that I start with x_0 , I run Newton's method once, I get an x_1 . I start with x_1 , I run Newton's method, I get an x_2 . My next iteration of Newton's method gives me a new x -value each time, at least in theory. There are pitfalls where this fails, and we'll talk about where this does fail. But at least in theory, I get a sequence x_1, x_2, x_3, x_4 , where those things are getting closer and closer and closer to where $g = 0$. Do we agree that at least picture-wise, proof by picture, which is never a good thing, but proof by picture, this concept seems relatively solid. It might generate a method for figuring out where a function vanishes, is equal to zero.
- 3.38** There's a couple of things I want to talk about now. What do I mean by— what is my formula iteratively? If I've run this 12 times, what's the 13th step? What is x_{13} ? If I've run it 70 times, what's x_{71} ? How do I figure out what those are? From this formula, you can figure out what the corresponding thing is. Again, this is one of those observations. If I look at it, I say my next iterate x_1 is equal to some formula. Looks relatively complicated, but it's some formula on the right-hand side, but notice x_0 occurs there, it occurs there, and it occurs there.

- 3.39** I can ask myself the question, "What would x_2 be? What would x_2 be?" Think about it. What are you doing for x_2 ? You're literally doing the exact same thing you did for x_0 . If this is my formula for getting x_1 , given that I had x_0 , what do you think your formula is going to be for x_2 ? I'm probably—we're going to have x_2 is equal to what? Well, now x_0 plays the role of x_0 , or x_1 plays the role of x_0 . This becomes $(x_1 - g(x_1)) / g'(x_1)$. Do we It's going to be that that's at least plausible? That that's at least plausible. x_0 , again, was playing the initial guess. When I did that, I got x_1 , which was given by this particular formula.
- 3.40** Now, if I start with x_1 —that's my initial guess for the second round of Newton's method — x_1 is my initial guess. Then it stands to reason that x_2 is going to be $(x_1 - g(x_1)) / g'(x_1)$. If I say, "What's x_3 ?" Now x_2 is playing the role of my initial guess, and so my formula is going to look like that. I would guess that x_3 is going to be $(x_2 - g(x_2)) / g'(x_2)$ and so forth and so on. Yes? That seems plausible, at least.
- 3.41** It turns out that Newton's method says the following: Start with some x_0 . You get to choose that. This is the beautiful thing about it, is that you get to choose what (x_0) is at the end of the day. If you have some intuition where you're looking at your function and you're like, "Oh, my goodness, I think $x_0 = \frac{1}{2}$ is probably close to where my function is zero," then guess $x_0 = \frac{1}{2}$. That's fine. And run Newton's method. If you have no idea what the value is going to be close to, then just choose 1 and run the algorithm and see what happens.
- 3.42** You start with some x_0 , and then subsequent iterates are given by what's called given recursively. I'll explain what that means by the following, the $n + 1$ iterate. The notation is a little confusing, but all it's saying is, "The next iterate that you get will be equal to the previous iterate minus the value of the function at your previous iterate, divided by g' at that iterate." Exactly like we had down here for x_2 , x_3 , and so on.
- 3.43** The n is a little confusing because when you see this notation, you're like, "Wait a second. What is n ?" You get to choose what n is. For example, if $n = 0$, If $n = 0$, what do I get? $0 + 1 = 1$. So this is saying x_1 on the left-hand side is equal to $(x_0 - g(x_0)) / g'(x_0)$. That was what x_1 was equal down here. That's fine. If I say $n = 1$, then this is saying $x_2 = (x_1 - g(x_1)) / g'(x_1)$. That's what this one's telling me. If I say $n = 2$, so you see how this works, right? You get to choose what the n is, and then this formula is telling you what the next one is going to be.
- 3.44** This is going to generate a sequence. This generates a sequence of x -values, and the hope—I'm going to put that in quotes—the "hope" is that eventually, you hope that $(x_n) = 0$, or let's say is approximately zero at some point. What do I mean by that? There exists some n value large enough—let's say $n = 1,000$ —such that when I take $g(x_{1000})$, it's like 0.0000000001 . It's close enough to zero. I may not be able to get it on the nose. I may not be able to say $x = \sqrt{7} / \pi$ gives me $g(x) = 0$ on the nose. But at the end of the day, do we really care? Do we really care if I get the exact value, or do I care that I'm close enough?

- 3.45** Let me illustrate that. We all have GPS in some form on our phones, something like that. We're going to a new destination, whatever, we punch in the address and our GPS says, "Go to this location." Is GPS 100% accurate? No, it's not. But is it enough? Absolutely. If my GPS says, "Hey, I'm looking for a particular business," I punch in the address and it says, "Hey, it's right here," and I'm standing with my phone and I'm standing right here and I'm like, "It doesn't look like it's right here," but it's 10 feet away—that's good enough. That's good enough. Do we agree? Knowing exactly where $g = 0$ may not necessarily be necessary. But knowing that I'm close enough to where it's equal to zero is relevant, and that's very important.
- 3.46** Newton's method gives us a way of actually being able to approximate that. That's all fine and fun and games. That's the theory of it all. Let's actually do this and figure it out. Let's actually do this and figure it out. Let's do an example. Let's do an example. Then I'll show you how we can apply this to derivatives.
- 3.47** Here's my example. This is the beautiful thing about mathematics is I think math suffers from the following thing is that we are taught that there has to be a correct answer, and there's only one way to get to that correct answer for a math problem. That's really frustrating to me because that's a fallacy. It's completely false. There are so many ways to get to the right answer, whatever that may be.
- 3.48** Think about it from a GPS standpoint. If I start from here and I want to go to a city that's 50 miles away, how many different ways are there of getting to that city? Tons. I could take backroads, I could take the direct route, I could take all sorts of different ways to get there. At the end of the day, I get to that city, my destination. Now, if you restrict that question a little bit and say, "I want to get to that destination as fast as possible." Now that's different. I want to get the optimal route to that place. That's different. But at the end of the day, there's multiple ways of getting to that particular destination. Okay, great.
- 3.49** There's multiple ways to solve math problems. There's not a "the way." What I'm really interested in is getting you to think about the problem, getting you to make observations about, "This observation helped me to get to that," and "This step helped me to get to there." Not necessarily thinking what is the next step, but what is a next step to move forward in the problem. Does that make sense? All right.
- 3.50** So here's the example. I want you to approximate the $\sqrt{2}$. That's a lame question, to be brutally honest. At the end of the day, I'm just going to go find some online calculator, whatever the case may be. I'm going to plug in the $\sqrt{2}$, it's going to give me a decimal, and I'm going to say that's good enough. That's really what I'm going to do. But what I'm trying to do is illustrate a principle, not necessarily that the answer is the most important thing, it's that the principle is the important thing.
- 3.51** One way I can do this is I can say, "To approximate the $\sqrt{2}$, I can turn that into a problem of looking for where a function is equal to zero. How would I do that? I could

say, "Here's an approach. Here's an approach to it. I'm trying to figure out what that thing is. I'm going to give it a name. I'm going to call it x . It's a variable." At the end of the day, I don't know what the $\sqrt{2}$ is, which is a lie. We know what it is, but I'm trying to illustrate the principle. I'm trying to figure out what x is.

- 3.52** What I could do is I could square both sides. I could raise both sides to the square. So square, square—what does that do? That tells me $x^2 = 2$. So $x^2 = 2$. Still doesn't help. It doesn't really help that much. I'm still racking my brain like, "Wait a second, the $\sqrt{2}$? Okay, 1, 2—" That's not helpful. I start plugging in values. But then I can say, "I can move the 2 over and I can say that that's $x^2 - 2 = 0$. Now, just by some simple algebra manipulating symbols, I've now put this in the context where Newton's method can apply. I can say, $g(x) = x^2 - 2$. I'm trying to figure out where $g = 0$. That's what I'm trying to find. Does that make sense? Just making a very simple observation, some very basic algebraic manipulations.
- 3.53** Now it fits into Newton's framework. It's like those little children toys, where you have a square peg and you stick in the round hole thing and you're like, "This isn't working." Now I've taken the problem and I've stuck it in a position where I actually have an algorithm, a method for attacking this. Now I have a method for attacking this. Okay, great. Let's try and run Newton's method on that.
- 3.54** What does it tell me? Here's my formula. I'll leave the formula up. There's my formula. It says: start with some x_0 . You get to tell me what (x_0) is. You get to tell me what (x_0) is. This is the nuance. This is the beauty of Newton's method. There's these little salient features to Newton's method that it's like, "If I start with x_0 equal to this, it works. If I start with x_0 equal to that, it doesn't. If I start with x_0 equal to this, it works, but not quite." Those are some of the pitfalls that we want to figure out with Newton's method.
- 3.55** Here's the point: if I know what the graph of that looks like, that can inform my guess for x_0 . That can inform my guess. If I go and I graph this thing—now, whether you know what the graph of this looks like by memory or not is irrelevant. I don't care. There's so many utilities out there. You can just use a free utility, graphing utility whatever it is. It'll graph it for you. You get a picture. Fantastic. I look at this thing.
- 3.56** Now, what its graph looks like is this: when $x = 0$, I'm down here at -2 . This is $(0, -2)$. Then it's going to come up like this and come up like this. It's going to cross here and it's going to cross here. That's what it's going to look like. It's a parabola. It's a degree-two thing— x^2 . It's a degree 2, it's a parabola. It looks like this or it looks like this. It's either up or down. The fact that the coefficient is a 1—positive—means it's going to look like this. If it were negative, it would look like this. Now, what are the coordinates of this point? Well, the x -coordinate is the $\sqrt{2}$, the y -coordinate is 0. The coordinate over here is $-\sqrt{2}$, the y -coordinate is 0.

- 3.57** Here's the thing. If you think about Newton's method, if you think about Newton's method, if I guess certain points, Newton's either going to work, fail, or converge to something else. If I choose a point, let's say right here. If I choose a point right there—if that's my x_0 , what's going to happen? I go down to the curve right here. I look at the tangent line to it, which is here. My next point will be x_1 , which is right there. Do we agree? There's x_1 . If I then go back up to my curve, what does it do? It comes up here. I draw the tangent line. I'm right here. It looks like that sequence x_1, x_2, x_3, x_4 is going to get close to $\sqrt{2}$. Do we agree? It looks like it'll work.
- 3.58** Can you find a point where it fails? The observation that we're making is that if x_0 is positive, it looks like Newton works. It looks like Newton will work. Okay, great. What happens if $x_0 = 0$? What happens there? Okay, so if $x_0 = 0$, what happens? I start with this point right there at the origin. There would be my x_0 . I go down to my curve, which is right here, and I say, "Look at the tangent line associated to it. What's special about that tangent line? It's horizontal. Is that ever going to cross the x-axis again? No. It's going to fail. It's going to fail. Why? Because what is $g'(x_0)$ in this case? Well, it's a horizontal line, which means its slope is zero.
- 3.59** So if I go over to my formula and say, "What is x_1 ?" $x_1 = (x_0 - g(x_0)) / g'(x_0)$, you're dividing by zero, which you're not allowed to do. You can't divide by zero. That's one of the pitfalls of Newton. Maybe my initial guess is where $g' = 0$, in which case, toast. Newton doesn't work. Newton doesn't work. Conversely, what happens over here? What if I choose x_0 to be here? What's going to happen? I go down to the curve, I get a tangent line that looks like that. x_1 will be right here. What am I going to converge to? I'm going to go to $-\sqrt{2}$. You have to be very careful. The initial guess of Newton is very, very important.
- 3.60** We're going to guess $x_0 > \text{zero}$. I'm going to choose $x_0 = 1$. $x_0 = 1$. Let's say $x_0 = 1$. Then what does my formula tell me? My formula says then $x_1 = (x_0 - g(x_0)) / g'(x_0)$. What do I need? I need a formula for g , and I need a formula for g' . Well, I have a formula for g — $g(x) = x^2 - 2$, but no one has told me the derivative yet. Well, this is why we got those computational tools last time is so that we can actually compute g' .
- 3.61** What is g' in this case? g' is what? Let's go back to it. I have the difference of two functions. I have $f(x) - g(x)$. The derivative—sorry, not $g(x)$. Let's say $f(x) - h(x)$, whatever, if you want to give them labels. $f - h$, the derivative should be $f' - h'$. h is a constant. What's the derivative of a constant? It's zero. What's the derivative of x^2 ? Power rule told me that was $2x$. So $g'(x) = 2x$.
- 3.62** Now I can go back from my formula and I can plug it in here. I can say $x_1 = x_0 - \frac{g(x_0)}{g'(x_0)}$ —we called that $1 - \frac{g(1)}{g'(1)}$. What is this? $1 - \frac{g(1)}{g'(1)}$. If I plug in 1 for g , I get $1^2 - 2$. So that's $1 - 2$. That's -1 . My numerator becomes $-1 / g'(1)$. g' was $2x$. So if I evaluate at 1, I get $2 * 1 = 2$ in that case. So that's equal to $1 - (-\frac{1}{2}) = 1 + \frac{1}{2}$, which is $\frac{3}{2}$, which is 1.5, which is freakishly close to the right answer. If you go into a calculator and you punch in

$\sqrt{2}$, the answer is like 1.4141 and additional stuff. So I've already gotten close to 1.4 by simply doing one round of Newton.

3.63 What happens if I did it again? What happens if I did it again? The unfortunate thing is you never do this by hand. You never do this by hand. This is why you need a computer to run these things. So you can run many iterations very quickly. Let's see what's going to happen. So x_1 is 1.5. My formula tells me what? Let me leave my formula up. I'll put it in a box so it doesn't really get in the way, even though it is in the way. $x_1 = 1.5$. That means that x_2 — trace the n. If I'm looking at x_2 , what is n in this case? n has to be 1. I get x_2 . If $n = 1$, this is going to be $(x_1 - g(x_1)) / g'(x_1)$.

3.64 Then I have to go back and say, "Do I know what x_1 equals?" The answer is yes. You computed it last time, so you know what its value is. It's equal to 1.5. So when I plug that in, this is equal to what? 1.5 —what is $g(1.5)$? I go back and I find $g = x^2 - 2$. I plug in 1.5. Very ugly. So what is this? $1.5^2 - 2$. No idea what that value is. Frankly, I don't care. Again, we're illustrating principles here. You would do this with a machine. You would not do this by hand at all. That's $g(x_1)$. You divide by $g'(x_1)$. So $g'(x_1)$ — x_1 is 1.5, so that's $2 * 1.5$. You work that out, whatever it is, this thing will turn out to be, roughly speaking, 1.4 and change, which is pretty amazing given the fact that if you were to go to a calculator and plug in the $\sqrt{2}$, it's 1.414 and then stuff. Newton's method is actually giving you that x_2 is equal to 1.4 and change simply in two iterations. Simply in two iterations. It's an incredibly simple idea, but an incredibly powerful one all the same.

3.65 Now, why is this relevant for optimization and derivatives for us? Why is this relevant from that stage? Think about it for a second. Look, this is my formula when I was simply trying to find what? When $g(x) = 0$. That was my formula. If I back up a little bit and I say, "What about $f'(x) = 0$? What about when $f'(x) = 0$? I'm trying to figure out, what about $f'(x) = 0$. Then what am I going to do? This becomes g . That becomes my $g(x)$. So think about that for a second. If I plug that into my formula, Newton is telling me— Newton says: x_{n+1} should be equal to x_n — what is $g(x_n)$? If $g(x) = f'(x)$, then $g(x_n)$ should be $f'(x_n)$. That's what it should be. Do we agree? If I simply plug in, if $g(x) = f'(x)$, then when I evaluate $g(x_n)$, I'm evaluating $f'(x_n)$ because they're the same thing. What does my numerator become over here? It becomes $f'(x_n) /$ —

3.66 Now, here's something that's important about derivatives is, look, when we talked about derivatives, we simply said, "I'm going to give you a function, and I'm going to teach you some rules to compute the derivative of that particular function." Fair? If I gave you something like this—this is just an aside. If I gave you $h(x) = x^3$, then we said, "The power rule applies to get me the derivative $h'(x) = 3x^2$." Do we agree? The 3 comes down, you decrease it by 1, you get a 2, so $3x^2$. Fine.

3.67 Why stop there? Why stop at one derivative? After all, here's a function. I could just give you the function $3x^2$ and say, "What's its derivative?" That's a valid question. I don't care that $3x^2$ is equal to the derivative already. I don't care that it's equal to h' necessarily. $3x^2$ in and of itself is a function, so I can differentiate it. How do I denote

that? If I took one derivative and I put one slash, then the next derivative, I'll probably put two slashes. At least it stands to reason. I put two slashes there. You call that the second derivative. If I differentiate that again, I'll get the third derivative. If I differentiate that, I'll get the fourth derivative and so forth. You can keep going. You can keep going.

- 3.68** So $h''(x)$ is going to be what? How do I do this? The constant stays there. Remember? Constants come out. I'm going to forget about the 3 for a second, and I'm going to say, "What's the derivative of x^2 ?" Power rule again: the 2 comes down, and I decrease by 1. So what do I get? The constant $3 \cdot 2x = 6x$. That's my second derivative.
- 3.69** If I come over here and I say, "What is g' ?" What would g' be? Well, if I put one tick mark to indicate the derivative there, that means I put another tick mark here, which means this is the second derivative of f . g' is the second derivative of f . So $g'(x_n)$ goes in the denominator, so this means this has to be $f''(x_n)$. That would be the formula via Newton's method. We're finding where g , which is equal to f' , is equal to zero. So this method right here—this finds x -values where peaks and valleys occur.
- 3.70** When this is valid, when this is valid—when would it fail, for example? Well, it fail if $f''(x_7) = 0$. If f'' at anything is zero, this method fails because why? I have a horizontal line. It's never going to cross the x -axis. So the method fails. That's why I'm saying this will occur—this will actually converge to the points at which $f' = 0$. So that'll give me where I have a peak or where I have a valley associated to that. Does that make sense, how this works? How we're applying it to the function $g = f'$ and seeing how that is capturing the peak or valley associated to this?.
- 3.71** Let's figure out what are the pitfalls. What are the pitfalls of Newton? What are some of the downsides to it We already saw one: horizontal tangent line. If I have a horizontal tangent line, then that thing is never going to cross the x -axis, and so it's never going to give me a next iteration. It's never going to give me x_2 or x_3 or whatever may happen. How do I get around that?
- 3.72** Let's say I start with x_0 and I get a horizontal tangent line. So what do I do? Do I just put my hands in the air and be like, "I guess I'm done. Failed." That'd be a natural thing to do. It'd be a natural thing to do. But the fact is this: You get to guess what (x_0) is. You get to choose what (x_0) is. So if you happen to choose that (x_0) gave you a horizontal tangent line, guess a different x_0 . Guess a different one and run and see what happens.
- 3.73** The second one is that it can give you an infinite loop. It can give you an infinite loop, and it never converges down on something. What do I mean by that? What do I mean by that? I could have some behavior like this. I could have some behavior like this. Let's say my graph is something—I'm going to do my best. That's okay. That's not bad. That's not bad.
- 3.74** But I could have something like this. If I were trying to converge down to, say, this point here— I want to try and find that particular point. What if I chose this point as my

starting point x_0 ? What could happen? Well, I come down here, I look at my tangent line. My tangent line does something like that. I would say, "Here's x_1 ." I then go down to my curve and I look at the tangent line, and maybe I hit x_0 again. Then I come back down and I go back over to x_1 , and I come back down, I go back over to x_0 , and so on. What's it doing? It's just bouncing back and forth between x_0 and x_1 . It's never going to zoom in on the origin. Do we agree?

- 3.75** How do you get around that? This is something that I'm going to address in terms of coding this up, because at the end of the day, you don't do this by hand. You do this by coding it up. When you see this, what happens is you start with x_0 , and then you just run the program for a little while, and you see what happens. If you notice it's like $x_0 = -2$, and $x_1 = 2$, and $x_2 = -2$, and $x_3 = 2$, and it's just bouncing back and forth, you realize you're stuck in a loop. It's never going to do anything. What do you do? You say, "Okay, don't guess $x_0 = -2$. Guess $x_0 = -1.9$. And then run the algorithm and see what happens. Suddenly you see like, $x_{11} = 0.000000005$. You're like, "I got it. I got it."
- 3.76** Newton is very temperamental to where you start with x_0 . It's very temperamental to where you start. Why is this ultimately relevant? It's ultimately relevant for the following reason. We want to know when $f' = 0$ because that gives me peaks and that gives me valleys. It tells me what my maximums are and what my minimums are. In the back of our heads, we're saying, "I want to apply this to models where I'm bringing in data and I'm trying to get predictive power from that particular data." I'm trying to say, "I bring in all of this data" and I want to say, "What is the next move I need to make in my business?" Does that make sense?
- 3.77** Then I have to take that data, create a model based on that, and then minimize the error of that model. That's the idea. I want to minimize the error. I want to find a valley of whatever the error function is. Newton can help me do that. Newton can help me do that. You can do this in a single variable math class. You don't need some advanced degree. You don't need anything like that. What do you need? You need derivatives and you need a picture. That's it. You need a machine that you can program it on.
- 3.78** I'm going to show you how to do that. I'm going to show you how you can program this very simply, and you can run the model, and then I'll show you pitfalls and where you get something that's just bouncing back and forth between -2 and 2, or you get something where the derivative is equal to zero, and you get a horizontal tangent line or things like that, and how to get out of those. But ultimately, Newton's method is one of the most important algorithms today, not just hundreds of years ago, today, in both machine learning, data science, and AI.
- 3.79** We've talked about approximating a value. How about approximating a function and its applications to potential optimization of a data set, whatever?
- 3.80** Yeah, great question. A great question. We've talked about approximating a value where those values are relevant for finding maxes and mins, little peaks and valleys.

But what if I want to say something along the lines of—something like this? If I want to do something like this, and this has a really cool application associated to it.

- 3.81** Let's say I give you something that looks like a heartbeat, for example. You hook a patient up to a heart rate monitor, and they're in the hospital, and you see the little machine. It's like little blips, and it just keeps going, and it monitors the pumping of the heart. Let's say you have something that it's up, and then it does this, and then it does this, and it does this, and so on, and it keeps doing that. Well, hopefully it doesn't do this. It's like, dot, dot, dot. Ultimately, you don't want it to go flat. You don't want it to go flat. It just keeps doing that same thing over and over and over again.
- 3.82** You can ask the question, you can be like, "Can I actually write that down as a function? Can I write that down as a function?" The answer is yes, surprisingly. You can write this down. As a function— we first have to ask the question, "Is it a function? Is it a function?" Is it the graph of a function? Can I write this as $y = f(x)$ for some function f ? The answer is, "As long as this doesn't have any vertical points where it hits itself multiple times, the answer is yes."
- 3.83** We can see that. If I were to just draw vertical lines over and over again, I'm never going to intersect this graph in more than one point. That corresponds to, if I'm looking at the heart, the heart is beating in time. It's going to go up, down, and it's moving constantly as you're going to the right—up, down, up, down, up, down, and so on. This will actually be the graph of a function.
- 3.84** Now, there's this branch of mathematics, which is incredibly versatile and very useful, called harmonic analysis. It's called harmonic analysis, where there's this very important technique called Fourier series, which says that if you have some function and it satisfies various properties—it's continuous, for example. There's no gaps. There's no gaps. It doesn't just jump from this value at -5 to this value at 7, just like that. There's no gaps to it. Then you can approximate this thing with sines and cosines. You can approximate it with sines and cosines. That's incredibly powerful. This gives you sines and cosines.
- 3.85** Sines and cosines approximate this curve, whatever that is. What I mean by that is I can take some coefficient like $2 * \sin(x) + 3 * \cos(x) + 5 * \sin(2x) + 10 * \cos(2x)$ and so on. I just keep changing the frequency of the sines and cosines. I make them speed up really quickly when I'm in something like this. Then I make them slow down when I'm in these little horizontal parts where the heart is resting. I can use that in a lot of different ways to approximate this particular function.
- 3.86** So one application, which is super useful to this, is noise cancelation. You put your ear buds in, whatever it is, and suddenly it's like the world goes dark. It's like all the sounds disappear, and you're like, "What the heck is going on?" At the end of the day, what you're doing is you're saying—all of the ambient noise on the outside, you're trying to cancel out that wave to some extent. It's like when you're jumping on a trampoline as a

kid and you try and launch somebody, they're jumping on the trampoline and you try to give them a boost. At the end of the day, what you're trying to do is you're trying to impact the wave that's associated to their jumping on the trampoline. That's what you're trying to do.

- 3.87** So noise canceling headphones, noise cancelation in any form, takes advantage of being able to approximate using these particular functions, sines and cosines, in a very precise way, and then say, "At a certain frequency, I want to turn everything off. I want to turn everything off." So the human ear can only hear up to a certain frequency. Dogs can hear certain higher frequencies, things like that. So why am I keeping the frequencies that I can't hear? Why don't I turn those off? That'll help cancel out some of the ambient noise. Then when I have a wave that comes in like this, I can say, "Oh, all I have to do is analyze the wave, look at what it looks like, and then create a wave, create a sound, that's the opposite of that so that it cancels out the waveform."
- 3.88** It would basically be like if you were to take a completely flat pond and drop a pebble into it, You see the ripples as they go out, the waves. It would be that if you could somehow generate a wave that when the water is going up like this, the other part of it is coming down. So what happens? The water can't go up anymore. It has to fill the trough. And so what is it going to do? It's going to go flat, completely. So yes, you can approximate functions, but it lives in this domain. It lives in the harmonic analysis domain. Yeah, great question.
- 3.89** As you give a machine learning program more and more examples, it can become more accurate, right?
- 3.90** Potentially, yes.
- 3.91** Potentially. Is there ever a point when it has enough examples that it can be perfect, or is it going to be like it's getting infinitely closer to perfection but never reaches it?
- 3.92** Great question. Short answer is no. Short answer is no. This gets into something that's called overfitting your data to some extent. I can illustrate this from a very simple idea of if you give me a whole bunch of dots on a board. You give me a whole bunch of dots. Let's say this. Let's say you give me dots that look like—here's one, here's one, here's one, something like that. You give me those dots. I don't care what their relevance is. Maybe these are geographic locations, maybe these are purchasing transactions, it doesn't matter what it is. But the question is: Can you give me a straight line that goes through those points, which is as close to all of the points as possible?
- 3.93** The game would be like—if we just eyeballed this, this would be something like- ish, something like that. That would be the idea. Now, does that line go through all of the dots? The answer is—the way I drew it, it doesn't go through any of them at the end of the day. But it's still giving me a decent approximation. It's saying that if you were to give me a new dot, you were to give me a new dot, I can say, "Whatever the value is on

the line that corresponds to whatever the input was, that's going to be my guess that gets closest to that dot."

- 3.94** Now, what I could do alternatively—and this is getting to your question—is saying, "What if I don't care that it's a straight line? What if I just want a function that goes through every single black dot?" I could say, "I'm going to do something that it comes up here, it hooks down, it comes up, hooks up, does this, does something very violently here, and then it does something maybe—" That's my best approximation to it.
- 3.95** But what do I notice? If I give you a dot that's like, I don't know, let's say I give you a dot that's like down here, or maybe I give you a dot that's down here, it's like the black curve is doing a terrible job of getting close to this thing. So yes, if you give me the entire data set, I can, more often than not, give you a machine learning algorithm that will just tell you, if you gave me a point from that data set, the machine learning algorithm will be perfect in guessing that that was the right data point. That's called overfitting. But if you give me a brand new data point, that overfit model turns out to be a terrible approximation to whatever that new data point is.
- 3.96** Its predictive power is awful. That's the game you have to strike by saying, "Look, I want a model that gives me predictive power that's accurate, but I also want it to get all the data that I can." You're straddling this line of you want predictive power, but you want accuracy as well, and you have to play those two off of each other. Very rarely are you ever going to get a machine learning algorithm that you run, and then you throw a random data point at me, and it says, "This is what the output should be."
- 3.97** For example, this is completely relevant—medical diagnostics. If you're looking at tumors, for example, you're looking at, say, a data set of 10,000 biopsies of a particular tumor. You look at those 10,000 biopsies and you say, "7,500 of them were malignant, so they were cancerous. The other 2,500 were benign." You could create a model that says, "These ones—maybe you color them red—those are the malignant ones. These ones are blue. You color them blue and you say, "Those are the benign ones." You would naturally see in your data set that the red live over here, the blue live over here.
- 3.98** But then the question becomes: What happens if you get a biopsy that's square in the middle? What is my model going to do? Some models might take that and be like, "It's benign. We'll color it blue." Other models might be like, "No, it's malignant. We need to color it red." You're not going to get a perfect model that's going to give you perfect predictive power, more often than not. Yeah. Great question.

Lecture 4: Introducing Limits

At a glance — Introducing limits and finding where functions vanish.

- 4.1 All right, so we've been discussing how to find where certain functions vanish, where they equal zero. That was one of the big topics we've been talking about. If I give you some particular function, whatever that may be, whether it's $f(x)$ or whether it's the $f'(x)$ or anything along those lines, how do I actually determine where that is equal to zero? We talked about Newton's method and how we could potentially approach that.
- 4.2 One of the biggest concepts in mathematics is trying to understand when is a function close to something—its output values—when is it close to something, when its input values are close to something as well. This gets at the notion of a limit, the limit of a function. What is a function getting close to when its inputs are getting close to something else? We're going to discuss that today, and we're going to see how limits naturally fit into this framework of trying to understand certain characteristics of your function.
- 4.3 If I have a function, for example, that blows up as I get close to a certain point, it gets arbitrarily large, or maybe it gets really close to zero or something along those lines, we're going to make that rigorous today. In the course of doing that, we're going to figure out another method for being able to find out when a function is equal to zero, given this certain f has a certain quality associated to it, which is going to be that the function is continuous. We'll make all of this rigorous today.
- 4.4 But ultimately, I want you to keep in mind, today is all about creating another algorithm, effectively, for finding when a function is equal to zero. That's really the motivation for what today is. Remember, the motivation for figuring out when a function is equal to zero is that if this function is equal to the derivative of a function, then that was helping us find our peaks and our valleys. So we're setting the stage for the introduction to optimization, ultimately. That's the goal for today.
- 4.5 I need to introduce the notion of what is a limit. So limits of functions. There's two ways to think about limits: there's the intuitive way to think about it, and then there's the very rigorous mathematical way. I think for me, personally, the intuitive way is really the way I like to think about these things. So I'm going to give it to you in two ways. We're going to have the intuitive notion, the intuitive way, and then we'll have this rigorous way. And then I'll explain both.
- 4.6 The intuitive way. The first thing that I need is I need a notation. I mean, mathematics is a language after all. So I need to be able to say, "How do I actually denote limits of functions and things like that?" The notation is—I'll give it a name. I'll call it L for limit. That seems like a reasonable thing to do. Then the notation on the right-hand side is really what we're getting at. This will be $L = \lim (x \rightarrow a) f(x)$. The way that you read this is you say that this is saying that the limit of $f(x)$ as x approaches a . Remember, x is our

input variable. I plug that into the function f . That gives me $f(x)$, that's my output, whatever that is. I claim that I'm going to give that thing a name. I'm going to call it L . The intuitive way of thinking about this is saying when x gets close, and I'm going to say really close to a , $f(x)$ gets really close to L . That's the idea.

4.7 The picture that you want to keep in mind is something along these lines: If this is a , and my function is doing something—I don't know what it is—let's say this is $y = f(x)$, something along those lines. Then it's saying that as x gets really close to a —the really close isn't rigorous at all. What do you mean by "really close"? Somebody could say, "I live really close to my friend because he lives three blocks away from me," whereas somebody else might say, "I live really close to my aunt because they're four miles away." What does "really close" actually mean? You have to quantify it. This is what I mean is that it's an intuitive notion. It's not rigorous at all.

4.8 But the idea is that you want to think about getting really close to a as being effectively at a , but not necessarily equal to a . If a were the value 2, or whatever the case may be, then thinking really close to 2 could be 2.1 or 2.01 or 2.001, or 1.9, or 1.99, or something. You think like, "I'm in a very tight window around a ." That's the idea. So if I look in some very tight window around a , I want to say, "What is my function doing in that window?" So you imagine you put these blinders on and you're allowed to consider input values that are that close. If I look at that and I say, "What's my function doing?" It's like, "That looks like whatever that value is, that's going to be L ." That's going to be the limit.

4.9 Now, there's some nuances to this because you want to say, "If x gets really close to a —" I can get close to a in a couple of different ways. I can say, "I'm getting close to a if you imagine walking along this line and I'm walking towards a from that side." Or I could say, "I'm walking towards a along this side." There's some directionality that we have to handle this. Am I approaching a from the right or am I approaching a from the left? So there's some nuances there.

4.10 But you can imagine these functions can have all sorts of behavior at a . What if my function were blowing up to infinity in that case, or it was coming down to minus infinity at that case, or what if my function comes up to a over here and then suddenly at a , it jumps all the way up here and then it comes down, something along those lines. All this behavior can play when we're computing limits. But right now, we're just trying to develop the intuition for this. That's the intuitive notion for what this is.

4.11 The rigorous way is the following. You say that the limit exists, so the limit of $f(x)$ as x approaches a is equal to L . It exists if you give me any tolerance whatsoever. You give me any tolerance whatsoever. You say, "I want a window range, if you will, on your output values of 0.01," something like that. You give me any window imaginable, and you get to choose that. You get to choose the tolerance. So you give me some window here in your output range, then I can guarantee you that there's a window around my

domain values for which the values of this function in that domain window don't get larger than the tolerance you gave me with an L .

- 4.12** The idea is the following. You give me a window. I don't care what the window is. You're going to give me this little window here, just to say, for example. If I look in this particular window out here, you get to control that. Then I can guarantee you come to me and you say, "I want my window to be within $1 / 1000$." I say, "Fine." Then I can say, "I can produce to you a window such that when I'm in that window of domain values, my function values do not get out of this box. They do not get out of this little range." Does that make sense, the intuitive notion of it? You get to control how close you want to be to the output, and then I can guarantee the input values for which you are within that range.
- 4.13** The rigorous way to say it is: for any tolerance—and mathematicians use Greek letters to denote these things, so it's a little intimidating to some extent—but for any tolerance, which is epsilon (ϵ)—I like to think about epsilon as an error to some extent. It's giving me some error window. So for $\epsilon > 0$, there exists another tolerance, $\delta > 0$, such that the following is guaranteed to happen. If x is within δ of a —the way that I capture that is I say, "If I look at $|x - a| < \delta$, the distance between x and a is less than delta." That's my window in my domain coordinates. If that's true, then my output values $|f(x) - L| < \epsilon$.
- 4.14** Let me illustrate that with a picture. What am I really saying? Again, here's my setup. Here's a . Here's my function, whatever that is. Let's say this is my value L here. It says: you get to choose your tolerance epsilon. What does this thing actually mean? If I go down epsilon here— so let's say the epsilon comes down to here, so that value would be $L - \epsilon$. I come up epsilon—so this value here would be $L + \epsilon$. There's my window. That's what I have. You get to choose that.
- 4.15** Then the burden for me is that I have to produce a tolerance within my domain values. For those values in that window, my function stays in that range. So I'm going to choose a delta such that this thing here is $a + \delta$, this thing here is $a - \delta$, such that in that range, I'm guaranteed that the value of f and $|f(x) - L| < \epsilon$. That's the burden of trying to understand limits. If you want to do this rigorously and prove that a limit is what it is, then you technically have to do this.
- 4.16** But again, we're looking for intuition. How do I actually compute these things? How do I get my hands dirty with these things very quickly? How can I actually compute limits without having to prove these things necessarily? We're going to go through some techniques in order to do that today overall.
- 4.17** But that's the notion of a limit. This is effectively what motivated calculus to some extent: taking certain objects that we all know and love, that we've had for mathematics for hundreds of years, and turning that into saying, "What happens when I go to the infinitesimally small type thing? What happens when I take a limit as things

get arbitrarily small or things get arbitrarily large or things like that? Can I actually say something concrete quantitatively and mathematically regarding things like that?" That's the notion of a limit. The derivative is tied in with the limit, and we'll get into that in a little bit.

- 4.18** Let's do some examples. Let's figure out some examples. Again, I want you to think about this intuitive notion of limits, that when x gets really close to a , what is $f(x)$ getting really close to, ultimately? So limits of functions. Let me give you some limits. We'll start off relatively small and we'll build from there. Then ultimately, I'm going to give you a technique that is very versatile that will allow you to compute limits very easily now that we have the notion derivative.
- 4.19** Here we go. Number one, the first thing, I'm going to say, "What is $\lim (x \rightarrow 2) x^2$?" Second one, I'm going to say, "What is $\lim (x \rightarrow 0) 1 / x^2$?" The third one, I'm going to say, "What is $\lim (x \rightarrow 0) 1 / x$?" Then let's do the fourth one, which is $\lim (x \rightarrow 0) \sin(x) / x$. I know a lot of these are $x \rightarrow 0$, but ultimately, it doesn't really matter what x is approaching. The idea is the same.
- 4.20** Now, what's nice about this is that if you know what the graphs of these things look like, if I can actually graph each and every one of these functions, then I can just look at the graph and say, "What am I getting close to when x gets close to zero?" That's a great way to approach computing limits, but it's unrealistic, to some extent. Realistically, you give me some function, you give me something crazy: $y = (x^2 + 1) * e^x / \tan(x)$, I have no idea what that looks like. I literally don't. However, if I were to say, "What's $\lim (x \rightarrow 4)$?" That's a valid question, but if I don't have the graph and I don't have any other techniques to evaluate limits, then I'm stuck. I don't have any way out of computing limits in that case.
- 4.21** But for now, we're just going to look at the graphs of these things. If I look at this as a function, so x^2 . What does that thing look like? If I look at $y = x^2$ —it's the last time we talked about that, that's my parabola. It looks like this. It's opening upwards. If I try and draw that as best I can. Just do your best. Something along those lines. That's what it is. What I need to figure out for my limit is I need to say, "What is my input value?" My input value is $x \rightarrow 2$. So I'm going to come over here and I'm going to say $x = 1$, $x = 2$. There's 1, there's 2.
- 4.22** I need to ask myself the question, "What is my function getting close to when $x = 2$?" I imagine myself as if I'm living on the x -axis and I'm walking towards 2, and I can walk towards 2 in various ways. I can walk towards 2 from the right-hand side, or I can walk towards 2 from the left-hand side. I consider both of those cases.
- 4.23** So if I'm walking towards 2 in this direction, what are the values of my function approaching? I imagine for each dot that I put on the x -axis, I imagine my dots on the y -axis. What are my function values getting close to? They're getting close to this value 4 right here. You imagine if I plug in 1.9, 1.99, 1.999, and if I had a calculator with me,

then I could just plug those values in and I would see that the values start getting close to 1.9^2 , 1.99^2 , 1.999^2 , and that's getting closer and closer and closer to 4.

- 4.24** Now, if I approach from the right-hand side, I have these little dots. I start moving this way. Then what am I doing? My function—the dots are approaching down here. But again, ultimately, what am I approaching? This point here on the graph is (2, 4). So what are the output values of my function approaching? They're approaching 4.
- 4.25** If I go back to my limit definition, the limit definition would say, "As I get arbitrarily close to 2, what is my function getting arbitrarily close to?" It would be 4 in this case. This limit actually equals 4 on the nose. That's my answer: $\lim (x \rightarrow 2) x^2 = 4$. That's great.
- 4.26** Let's look at this one. If I know what the graph of this thing looks like, thankfully, I do, but that's fine. If I look at the graph of that, this is $y = 1 / x^2$. It looks like this volcano. It looks like this, symmetrically. Now, I spot my input values. There's $x = 0$ there. I have to start asking myself the question, "As x gets close to zero, what are the output values getting close to?" This one is very different. Do we agree? The behavior of the function is very different.
- 4.27** The behavior of the function over here was, yes, my function is still going up, that's fine, but that was a finite value. That was just the number 4. That's fine. But if I come over here and I start looking at, "What's going to happen around zero? What does it look like my function is doing?" It looks like my function is blowing up to be arbitrarily large at that point.
- 4.28** Now, notice there was a slight difference up here. Really, when I computed the limit, there's really no difference that for me, just taking this value and plugging it into x , I get 2^2 , $2 * 2 = 4$, and that happened to be my answer. Maybe that was a fluke. Maybe that was actually a legitimate mathematical principle. I don't know. We're trying to investigate this. But notice that that doesn't apply here. I'm not allowed to do that in this case. I'm not allowed to say, "Oh, I'm going to take $x = 0$. I'm going to plug it in for x . That's going to give me $1 / 0^2$. My answer is clear. No, you're not allowed to divide by zero.
- 4.29** The world explodes when you divide by zero mathematically. That's one of the fundamental principles is that you're not allowed to divide by zero ever, but that still doesn't mean—remember back to the definition of the limit. The definition of the limit was when you're close to $x \rightarrow 0$, your output values are close to something else. It does not say when $x = 0$, the value of the function is something. It doesn't say that at all. It just says when you're close to it.
- 4.30** So even though I can't plug $x = 0$ in directly, I could plug in something that's close to it, like 0.1 or 0.01 or 0.[50 zeros]1, something like that. I'm allowed to plug those in because those aren't going to make the denominator zero. It'll make me arbitrarily close to zero for sure. If I plug in $x = 1 / 1000$, that's pretty small. When I square it,

that's going to give me a ton of zeros and then a 1. When I plug that in, what's actually going to happen? What's happening in this case? Let's look at the graph.

- 4.31 If I start walking towards zero from the left-hand side, what is my function doing? My function values look like they're coming up here to plus infinity. If I start marching from the right-hand side, the function values look like they're going to the same place. That's one of the important properties of a limit is that you have to take the directionality into play; you have to take that into account. If I approach from the left-hand side and I'm going to a certain value and I'm approaching to the right-hand side and I'm approaching a different value, that's concerning. But if they're both converging to the same point, that's a good thing. That's what you want.
- 4.32 In this case, graphically, we can say that the limit will be plus infinity. My function is tending to go to the same point. It's going to plus infinity. Whereas for this one, we run into the directionality problem from the right and from the left, because what does the graph of this thing look like? If I look at the graph on this one, this one looks like it comes down like this, comes down like this. That's what the graph of $1/x$ looks like. You can see we're going to run into a problem at zero.
- 4.33 What's the problem? I have $y = 1/x$. I spot my value for my input that I'm trying to approach, which is $x = 0$. Then I try and figure out what is this function tending to as I get close to zero. Again, that intuitive notion. What are the values getting close to when $x \rightarrow 0$? Again, let's look at how can you approach zero.
- 4.34 There's two ways. You can approach from the left or you can approach from the right. Let's see what happens. If I start marching in from on the left, what are the values of my function doing? They're going down here. That's going to be my minus infinity. They're going all the way down here. Whereas what happens when I approach from the right? As I approach from the right-hand side, the values of my functions are going where? They're going up to plus infinity. Are those two things equal to each other? No, they're literally as far away as humanly possible from each other. You've got one at minus infinity, you've got one at positive infinity.
- 4.35 So think about the intuitive notion. The intuitive notion of a limit says: if I'm standing close to zero, then whatever my output values should be in my function, they should be close to something. So think about it for a second. If this is zero, so here's zero, this function right here is saying, "If I stand barely to the left of zero, I'm basically as far negative as I could possibly be. I'm at minus infinity. But if I go back to zero and I move just ever so slightly to the right, suddenly I'm infinitely high."
- 4.36 So think about the intuitive notion. If I'm standing near zero, am I close to something? The answer is no, I'm not, because on this side, I'm infinitely far away. On this side, I'm infinitely far away. Those two things are not close to each other. So what does this limit equal? It doesn't exist. This does not exist based on that. Does that make sense? If I'm standing at a point, $x = 0$ in this case, and I wiggle to the left and to the right, and I say

that my output values don't change all that much, then my limit should exist. But if my output values are vastly different, then my limit can't possibly exist.

4.37 Think of it over here. If I'm standing at $x = 0$ and I move to the left, I'm infinitely high. If I move to the right, I'm infinitely high, then I'm close to being infinitely high. That's exactly what the intuitive notion of a limit would say. That's why this limit being equal to plus infinity actually makes sense, as opposed to this one, the limit does not exist.

4.38 Now, this one's a little bit trickier. The graph of this one, it's a little subtle, but it looks like this. If I were to just look at that as a function and say, let's just plug in the value. After all, that's I did up here. I took $x = 2$. I plugged it in $2^2 = 4$, and magically, that was my limit. That's fantastic. If I were to try the same exact technique down here and I just plug it in, I would get $\sin(0) / 0$. This is one of those annoying nuances—trigonometric functions always sneak up on you at the last moment.

4.39 It's like, "Okay, what is $\sin(0)$? I got to go back and figure out what is that value." You check and you look, you plug it into a calculator, you Wikipedia, I don't care, whatever you do to look up the value, what is $\sin(0)$? $\sin(0) = 0$. If I were to plug $x = 0$ into my function, if I were to plug it into that, I'm getting $0 / 0$. What the heck is that? I'm not allowed to divide by zero, number one, so that's a problem. But does the zero in the numerator somehow cancel out the zero in the denominator? $5 / 5 = 1$ after all. Is $0 / 0 = 1$ in some way? The answer is no, it's not.

4.40 In this example, though, if I look at the graph of what's happening at $x = 0$, the function is not defined. I'm going to denote that with an open circle. The open circle means that my function value is not defined there. But what's interesting is that if you look at this graph, it comes down and then it oscillates, but it tapers off. That's ultimately what's happening with this graph. You plug it into any graphing utility or whatever you want to use. This is what the graph of $y = \sin(x) / x$ is equal to.

4.41 The question is, spot the input value. That's $x = 0$. That's right here. That's $x = 0$. Start asking yourself the question, "If I'm close to zero on the left and the right, what are the values of my function getting close to?" You start marching I'm in from the left. I come in from the left. What's happening? The dots are coming up here. It looks like I'm approaching whatever that value is. But as I approach from the right, what's happening? The values are coming up and they look like they're approaching the same thing.

4.42 So even though the function is not defined at $x = 0$, the limit doesn't care because the limit just says: if I'm close to zero—I'm not equal to zero, but I'm close to zero—what is my function getting close to? I can be close to zero in the sense of $x = 0.0001$, I'm close to zero, or $x = -0.0001$, I'm close to zero, but I'm not equal to zero. If you figure out what this value is, this value is actually equal to one in this case.

4.43 Again, this is a graphical thing. I'm going to give you a technique in five minutes to figure out how to actually do this computationally so you have this method overall. But

what I notice is that if I approach from the left, the values of the function are getting close to 1. As I approach from the right, the values are getting close to 1. That means that this is equal to 1. Even though the function is not defined at $x = 0$. That is a key difference. Even though $\sin(0) / 0$ makes zero sense. It doesn't make any sense. It's what's called an indeterminate form, meaning you cannot determine what its value is. You have to investigate it further. But even with that, I get this is equal to 1.

- 4.44 Now, you might be thinking, "Robbie, you told me that earlier $5 / 5 = 1$, and $2 / 2 = 1$. And so now you're telling me that $0 / 0 = 1$," fair point. That's fair. But if I simply modified this slightly and wrote this as $\sin(3x) / x$, it's still presenting to me as a $0 / 0$, but now the answer will be 3. No, $0 / 0$ is not equal to 1 always. In this case, it is, but I can give you a very simple case in a couple of minutes where I can show you that the limit is not equal to 1.
- 4.45 Let me give you a method for doing this. This is one of the most important methods in computing limits, which is why I didn't want to motivate the derivative as a limit, which we will talk about in a little bit. I didn't want to motivate the derivative as a limit to introduce it. I wanted to introduce the derivative as a computational tool because if you have that first, then computing limits is much simpler. It's much simpler. You don't have to memorize all these obscure limit techniques for all of this. You get this massively important tool for computing limits right out of the gate.
- 4.46 There's this notion that's called L'Hôpital's rule. Some people call it L'Hospital's rule because this little circumflex, actually, in the French, this becomes an "os" so people call it L'Hospital's rule. That's nonsense. The way you pronounce this is this is L'Hôpital's rule. What it says is the following. I want to give you the conclusion of the theorem. I want to give you the conclusion of the result. It says that if I have a limit of a quotient, $f(x) / g(x)$ as $x \rightarrow a$.
- 4.47 There's going to be some conditions on this. There's going to be some conditions. I need $f(x)$ to be differentiable, meaning that its derivative exists. I need to be able to compute $f'(x)$. I need to be able to compute g' . So both of these need to be differentiable. I need this to present to me in an indeterminate form, but a very specific one. I mentioned that $0 / 0$ is an indeterminate form. There are other indeterminate forms as well. But I want to get the conclusion of what this result is saying. This is equal to $\lim_{x \rightarrow a} f'(x) / g'(x)$. The conditions that I need in order to make this statement rigorous is the following.
- 4.48 I need to say both $f(x)$ and $g(x)$ are differentiable. They're differentiable near $x = a$, meaning that their derivatives exist. It makes sense to compute f' . It makes sense to compute g' . I need that to be true. I also need, when this presents—this needs to present as either a $0 / 0$ or $\pm\infty / \pm\infty$. If it doesn't present in one of those forms—what do I mean by presents in one of those forms? Think back to this example. Think back to this example: $\lim_{x \rightarrow 0} \sin(x) / x$. Think back to that.

- 4.49 The first thing you should do with any limit whatsoever is simply take the x value and plug it in and see what happens. If I take $x = 0$ and I plug it in directly and I get the number 7, my limit is equal to 7. That's fine. That's exactly what we saw over here when I have $\lim_{x \rightarrow 2} x^2$. If I just took $x = 2$, I plug it in for x , I notice that that's 2^2 . 2^2 makes sense. That's 4. And so that means that that's my limit.
- 4.50 That's the first thing you should do with any limit whatsoever—plug it in and see what happens. In this case, if I plug in $x = 0$, what do I get? I get $\sin(0) / 0$, and we said that's $0 / 0$. And we say, "We don't know what that is." That's fair. I don't know what that is. Nevertheless, it is one of the forms that's covered by L'Hôpital's rule. It is one of the forms that's covered by L'Hôpital's rule. That's what I mean by this limit presents to you in this form, a $0 / 0$ or a $\pm\infty / \pm\infty$.
- 4.51 Now, these don't have to match, by the way. If the numerator goes to $+\infty$ and the denominator goes to $-\infty$, that's fine. They don't have to be $+\infty / +\infty$ or $-\infty / -\infty$. They can differ. But that's what I mean by the limit presents to you as this indeterminate form: $0 / 0$ or $\pm\infty / \pm\infty$.
- 4.52 L'Hôpital's rule says: if that happens, then as $x \rightarrow a$, I can write this limit as (differentiate the numerator) / (the derivative of the denominator). Now, notice I'm not using the quotient rule when computing my derivative. I'm simply differentiating the top, differentiating the bottom. How would that apply here? How would that apply here with L'Hôpital's rule? I first look at it as $x \rightarrow 0$, I get $0 / 0$. Fantastic—L'Hôpital's applies. L'Hôpital's rule tells me: take the derivative of the top. So I have to go back and say, "Last time we talked about derivatives of trigonometric functions. What's the derivative of $\sin(x)$?" I go and I look that up. That's $\cos(x)$. I get the limit back, $x \rightarrow 0$. The top is a $\cos(x)$.
- 4.53 I say, "What's the derivative of x ? x is a power rule. It's x^1 . How do I differentiate x^1 ? If I have x^1 , what do we do? That exponent comes down. I'm going to get a 1, and then I decrease the power by 1. This is really $1 * x^0$, which is just 1. The derivative of the denominator is just 1. That's fine. This is $\cos(x) / 1$. I reevaluate. I say, "It's a limit, so I'm just going to plug in the value and see what happens." Let's see what happens. As $x \rightarrow 0$, this is $\cos(0)$. I got to look it up again. I got to figure out what are the values. This is equal to $\cos(0)$. That's equal to 1, which is exactly why I gave you that limit via the graph of $\sin(x) / x = 1$. But now we actually have a concrete, algebraic, analytic way of evaluating a limit using L'Hôpital's rule.
- 4.54 Let me convince you that I can actually give you an example where $0 / 0 \neq 1$. I told you that I could do that. I should stand by my word on that. Here's the other example. If I give you $\lim_{x \rightarrow 0} \sin(6x) / x$. A very small modification of the $\sin(x) / x$. I'm just speeding up the sine a little bit by multiplying it by 6. Again, if I plug in $x = 0$, what do I get? I get $\sin(6 * 0) = \sin(0)$, so $\sin(0) / 0$, and that's $0 / 0$. Even though that's indeterminate, that's a good thing because it presents to me in the right form for L'Hôpital's rule. It allows me to use L'Hôpital's rule.

- 4.55 L'hôpital allows me to do the following. It says this is the limit as $x \rightarrow 0$. I have to take the derivative of the numerator and divide it by the derivative of the denominator. The derivative of the numerator is a little confusing because it's not just a $\sin(x)$; it's a composition of things. You give me an x value. I'm first going to multiply it by 6. That's the first function I'm going to do. That's going to give me a $6x$, and then I'm going to take that $6x$ and I'm going to feed it into $\sin$. That's actually a composition. You have to think back—if it's a composition, then how do I actually differentiate compositions of functions? That's the chain rule. The chain rule is how I do it. It's the onion analogy that I had. You work your way from the outermost layers to the innermost layers.
- 4.56 So I look at this function and I say, " $\sin(6x)$ —what's the outermost function here?" The outermost function is the $\sin$. I say, "What's the derivative of $\sin$?" We just figured that out. The derivative of $\sin$ was $\cos$. This top will be $\cos(6x)$. But now once I've broken through that $\sin$ layer, I need to say, "What's the derivative of the innermost layer?"—the next layer of my onion— which is my $6x$, and the derivative of $6x$ is just 6. I'm going to get $\cos(6x) \cdot 6$ —I'll put the 6 in front— / the derivative of x , which was equal to 1.
- 4.57 Now I reassess. What happens when $x \rightarrow 0$? When $x \rightarrow 0$, what do I get? I'm going to get a $6 \cdot \cos(6 \cdot 0) / 1$. We said $\cos(0) = 1$, so now the answer is 6. Even though the original limit presented to me as a $0 / 0$, and we saw in the first case with $\sin(x) / x$, that was equal to 1, the reason why it's indeterminate is because, look, here's another one that presents to me as $0 / 0$. I'm not saying the answer is 1 because it's not. The answer is actually 6. That's why $0 / 0$ is indeterminate. You don't know what it is.
- 4.58 You can use this example—simply replace 6 with any real number imaginable. You replace it with π , you replace it with $\sqrt{7}$, you replace it with 50 million—it doesn't matter what it is. You can now create a family of limits that present to you as $0 / 0$, but their value is whatever the real number was that you gave me. That's why this is indeterminate. You can't determine its value without further investigation, overall. But L'Hôpital's rule is by far one of the most versatile tools for computing limits going forward. All right?
- 4.59 Now that we have L'Hôpital's rule, we can compute all sorts of limits. We can go through, we can figure out how to do all this, but only if the limit originally presents to me in one of those two forms. I'm not allowed to use L'Hôpital's rule if I plug something in— so for example, one of the previous ones that we had was this. One of the previous ones we had was $1 / x^2$. This is the volcano one that we just did: $\lim (x \rightarrow 0) 1 / x^2$.
- 4.60 If I plug in 0 directly, what do I get? How does this present to me? It presents to me as a $1 / 0$. That's not a L'Hôpital form. That's not an indeterminate form. It doesn't make sense for me to simply say, "This is equal to the limit of the derivative of a constant—that's 0—over the derivative of x^2 , which is $2x$ as $x \rightarrow 0$." That's not legitimate because L'Hôpital's rule doesn't apply in those cases. That's one very easy pitfall to fall into is that suddenly every limit that crosses your path is a L'Hôpital limit. That's a fallacy. That's not true.

- 4.61 You have to have that the limit presents to you as a $0/0$ or $\pm\infty/\pm\infty$. If it doesn't present to you this way, then you have to resort to other techniques for actually evaluating the limit, such as looking at the graph and saying, "Oh, this goes off to $+\infty$ in that case." You might have to look into graphical analysis and things like that. Does that make sense? L'Hôpital's rule is not this cure-all technique for evaluating every possible limit you ever encounter. But it does handle a lot of them, especially the ones of $0/0$ or $\pm\infty/\pm\infty$. If I can get my limit to look like one of those, then I can actually use the technique.
- 4.62 Last time we were talking about this number e when we were talking about derivatives and how that comes up. Interestingly enough, L'Hôpital's rule can allow me to derive what e is. I want to use this economics example really quickly to motivate this. If I decide I'm going to take a chunk of money and invest it or let's say, just put it in a bank, something like that. I'm going to take a chunk of money, plug it in a bank. Well, you're effectively giving the bank a loan. That's effectively what you're doing. As a result, the bank is going to provide you some incentive for giving them your money at the end of the day. That's what your interest rate is.
- 4.63 Let's say I take, for easy numbers, let's take \$1,000, whatever it is. I go to a bank and the bank says, "I'm going to give you—" if you choose a bank that has a decent interest rate, you might get something like, I don't know, 3 or 4% these days, something around those lines. Other banks, they'll give you 0.1% or things like that. It depends on the bank. But if I give them that money, then they're like, "We're going to give you a certain amount of interest for giving us effectively a loan." That's great.
- 4.64 But how often do they give me that money? That's how many times it's compounded. Certain banks, they'll compound at the end of every single month. They'll say, "Every single month, we'll give you whatever interest is, we'll apply that to your account." If they give me 4% on this, then if they do it 12 times throughout the year, $4/12$ is how much they're going to multiply my principle—the amount that I gave them—to give me my interest at the end of the day. There's a formula that governs this.
- 4.65 The formula is you have your initial principle that you give them—that would be my \$1,000 in this case. That's going to be multiplied by what? That's going to be multiplied by $(1 + R/n)$, n is going to be the number of compounds. If they compound monthly, n would be 12 in this case because there's 12 months in the year. The rate is going to be whatever the interest rate was. If it's like 4%, you convert that to a decimal, it'd be 0.04 — that's what R would be. Then you raise this to the number of compounds times the number of years that this is going to be invested. But that's only if they're giving me compound interest every single month. Every single month.
- 4.66 What I want to know is: what if there existed a bank out there that actually gives me interest—it compounds for any possible subdivision of time I can think of. You think of time, I mean, time is a continuous thing. Even from me saying a word now to saying a word now, there is some continuum of time that took place in between those words

being said. Do we agree? It's not like I say a word now, and now I say another word, and suddenly it's, "No, here was one time period, and then there was this gap, and then here's another time period." That time is continuous.

- 4.67 The question I can ask is, "What if I compound infinitely often? What if I compound infinitely often? What actually happens?" The question I want to understand is: What happens when I take the limit as $n \rightarrow \infty$? I have infinitely many compoundings that are happening of this quantity. So $A_0 \cdot (1 + r / n)^{nt}$. One nice property with limits is that if you have a constant here, that can come outside of the limit.
- 4.68 That makes sense if you think about it. If I just multiply a function by 2, that's not going to affect the values of my function when I evaluate at x . It's just going to take whatever value I plug into my function f and multiply that by 2. I can take constants outside of limits. I get an A_0 here. I get a limit as $n \rightarrow \infty$. Now, this thing I'm going to write as $((1 + r / n)^n)^t$. This is just an algebraic exponential rule that if you take something raised to a power that's a product like this, then that's the same as how would I do this? The t comes in and it multiplies this exponent and gives me an $n \cdot t$. It's a property of exponents. It's an algebraic rule.
- 4.69 Now, the beautiful thing about this is what is this limit here as $n \rightarrow \infty$? It's tempting to be like, "As $n \rightarrow \infty$, r is fixed; r is our rate. That's 0.04 if we're saying 4% interest. That's a fixed number. The denominator is getting infinitely large. What's happening as the denominator gets infinitely large? Where is that number going? Let's think about this for a second.
- 4.70 If I had $1 / 1$, $1 / 2$, $1 / 3$, $1 / 4$, and I keep increasing the denominator, those numbers are getting smaller. Do we agree? One-half is 0.5, one-third is 0.33333, one-fourth is 0.25, one-fifth, and so on—they're getting smaller and smaller. If you think about it as powers of 10, eventually, I'm going to have a $1 / 10$, then I'm going to have a $1 / 100$, then I'm going to have a $1 / 1000$, and things like that, those are just 0.1, 0.01, 0.001, 0.0001, and so on. So what does it look like? What are these values going to go to as the denominator goes to infinity? It's going to go to 0.
- 4.71 It's tempting to look at this and be like, "A finite number— $0.04 / \infty = 0$." So this number is really 1, and I'm taking 1^∞ . It's like $1 \cdot 1 \cdot 1 \cdot 1 \cdot 1$ —the number is 1. It's a very easy trap to fall into. That number is not 1. This number here is actually e^r . It's actually very far away from 1. Remember e was our 2.718. That was this number that Euler created when he was looking at this. I raised that to the r power in this case. For example, if I just wanted to know what is the number e , I would take $r = 1$. Then I take this entire thing here, I raise it to the t power, and I multiply by A_0 , and that's what my limit is.
- 4.72 What this is saying is if I were to invest \$1,000 into some fictitious bank that compounds infinitely often for any subdivision of time imaginable, any nanosecond of time, it's compounding all the time, it's saying that the return on my investment is

going to be governed by this particular equation: $A_0 e^{rt}$. You might look at that and be like, "Whoa, that's absolutely fantastic. This is an exponential function. Exponential functions blow up to infinity incredibly fast." Whoa, what's the rate at which it's doing this? The rate is a really small number: $r = 0.04$.

- 4.73** This function is not growing as we like to think about exponential functions. We say, "Oh, something is growing exponentially. It's out of control. It's growing so fast," and all this stuff. It's like, "No, I can temper its growth by having a very small exponent so that this thing doesn't grow too quickly." But this L'Hôpital's rule will allow you to prove that this limit— $\lim (n \rightarrow \infty) A_0 \cdot (1 + r / n)^{nt}$ —is actually equal to your $e^{rt} \cdot A_0$. L'hospital's rule will allow you to do that.
- 4.74** Even though it doesn't present to you as $0 / 0$ or $\pm\infty / \pm\infty$, you can algebraically manipulate this thing to turn it into something that becomes a $0 / 0$, and then L'Hôpital's rule applies. But there's a good four or five steps to do that. That's beside the point. I wanted to bring this in from what we did earlier when we were talking about this number e and how it can arise as a limit in that case. All right.
- 4.75** So let's get to limits and try and understand how do limits characterize a property of functions called continuity. Continuity. What does it mean for a function to be continuous? What does it mean for a function to be continuous? We say that a function f is continuous at a point $x = a$ if the following condition holds. If the following condition holds. If $\lim (x \rightarrow a) f(x) = f(a)$. We say that it's continuous at $x = a$ if that is true.
- 4.76** Now, if you were to look this result up, typically this is broken down into saying that a function is continuous at $x = a$ if three conditions hold. Those three conditions are really encapsulated in this statement alone. I want to break that down. What this is saying is f is continuous at $x = a$. In order for me to even write down $f(a)$, implicitly that's saying that $f(a)$ exists. It actually makes sense to plug $x = a$ into the function f . So $f(a)$ exists. The value at $x = a$ of my function exists. That's something that's important to handle.
- 4.77** The second thing that's happening is that if the thing on the right-hand side exists, then this limit should exist. It's not like this $1 / x$ function where one side was going to infinity, the other side was going to minus infinity where the limit didn't exist. This is implicitly saying that this limit actually exists. When I compute the limit, the limit is equal to 5, it's equal to 4, it's equal to π , it's equal to whatever. The limit actually exists.
- 4.78** Then furthermore, the last thing is that those two things have to coincide. Those two things have to coincide. The limit has to equal the value of the function. That's getting at the notion of continuity. If you look at what does it mean to be continuous—it's funny because if you ask somebody, "What does it mean for a function to be continuous?" Nine times out of 10, they're going to tell you, "I can draw it without lifting the pencil." That's a great definition, actually. It's not mathematically rigorous, but it makes a lot of sense. I don't have any gaps. If I have to pull my pencil off the page

to actually move to another part to start drawing the graph, then there's some gap that's happening there. There's no continuity of my function.

- 4.79** But we can have behavior that does something like this. We can have behavior like this where what? Whatever this value is, let's call this value L , the limit exists because as I move to $x = a$ from the left, the values of the function are getting closer to L . As I move to $x = a$ from the right, the values get closer to L . So the limit exists. But what do I notice? The function isn't even defined here. So my function is discontinuous. It's not continuous there. And that makes sense if you think about it.
- 4.80** If this is a trail that I'm walking on, and suddenly the trail is fine, the trail is fine—the second I get to a , suddenly the trail disappears. Well, I have to walk over whatever is the disappearing part. There's a gap. The continuity has failed. The continuity is not there. I could have behaviors like this, or I could even say, maybe I define the value of my function at $x = a$ to be this dot up here. Still discontinuous. It's not a continuous function. If I wanted to draw this, I'd have to draw it here, draw a circle, lift the pen, draw the point, then come back and draw the rest of it. That's not continuous. But the limit exists. But the value of my function does not. That's the key difference.
- 4.81** Continuity is saying: no, no, no, no; that's not allowed. Continuity is saying: the limit has to exist and the function value also has to exist. So limits will allow me to characterize the behavior of being continuous at a point simply by using limits in the value of function. The most important thing that comes out of this is what's called the intermediate value theorem. This is where we're going to get our next algorithm. We're trying to find where functions are equal to zero.
- 4.82** The intermediate value theorem is a very simple theorem to understand intuitively. Mathematically, it seems a little strange. I'm going to write down the mathematical version of this, and then I'll give you the intuition for how you think about this. The intermediate value theorem says the following thing. It says: if f is continuous on the closed interval $[a, b]$ —this is just a notation for an interval from a to b , and I have brackets meaning that it's closed. If f is continuous on $[a, b]$, and if M is any value between $f(a)$ and $f(b)$, then there is some $c \in (a, b)$ such that $f(c) = M$.
- 4.83** If you're anything like me, you've lost all attention span at this point, and you're looking at it and you're like, "I have no idea what you're telling me right now. You're just writing symbols on a board." And that's fair. You have this massive statement. "What is this actually trying to tell me?" It's telling you what young children understand very simply by going out to the playground with a jump rope. Again, I told you guys earlier that I interpret things very simplistically. It's the way I learn. I have to have a very simplistic model in my brain to actually understand the principle.
- 4.84** If you give me something like this, I'm not going to understand it. But if you tell me that if you draw a line on the ground, if you draw a line on the ground and you go out to recess. My son goes out to recess with a jump rope, and he sees a line on the ground.

He takes that jump rope and he just tosses it on the ground. The jump rope does something like this. Let's say here's the end of the jump rope, here's the other end of the jump rope. He tosses it on the ground. The one end of the jump rope is there, the other end of the jump rope is here, and the line is somewhere right there. Then he can look at it and he can say, "It crosses the line." It has to cross the line. Do we agree? It has to cross the line.

- 4.85** If this is one end of the jump rope, this is the other end of the jump rope—the jump rope is a continuous thing. There's no gaps in it. You've never jumped a rope before with something—I mean, maybe you have. You cut the jump rope in half and you're literally just swinging things now. You're not jumping over anything. It's a broken thing. It's discontinuous. That means that it's got to cross the line somewhere. Do you agree?
- 4.86** Now let's put this in the framework of what the theorem is actually saying. The theorem is now saying this. It's saying: let's say here's my x-axis. Here's my y-axis. Over here is a. Over here is b. At $x = a$, this value here is $f(a)$. When $x = b$, this value is $f(b)$. So the theorem is saying the following. It's saying: if you give me any number in between $f(a)$ and $f(b)$ —so here's $f(a)$, here's $f(b)$ —you give me any number in between, here's M . There's my value of M , and I look at $y = M$, a horizontal line. It says: if you have a function which is continuous on $[a, b]$, and here's its graph—here's $y = f(x)$, that graph has to cross M at some point. It has to cross M at some point. It's the jump rope. The jump rope has to cross the painted line on the ground at some point.
- 4.87** Where does it cross? It crosses right here. And what's the input value? That's the c value that they're telling me. If I take $x = c$ and I plug it in, I should get $f(c) = M$. That's what the intermediate value theorem is saying. But it's a schoolyard theorem at the end of the day. That's how you can remember it. If I draw a line on the ground, and I take a piece of rope, and I toss it down on the ground, I can say that rope has to cross that line.
- 4.88** We're going to develop another algorithm, which is called the bisection method, which gives us another way for determining when a function is equal to zero. Because if $M = 0$, then it's saying: whatever the function is, it has to cross zero. I'm going to leverage the intermediate value theorem with continuity of a function to be able to compute when does $f(x) = 0$? When does that $f(x) = 0$?

Lecture 5: Limits in Action

At a glance — Limits in action and the intermediate value theorem.

- 5.1 Now that we have the intermediate value theorem, we're going to create another algorithm, which allows us to find where a function will vanish. Remember last time we talked about—earlier, we talked about Newton's method, and it's this very versatile method for finding where a function crosses the x-axis, where it's equal to zero. We said that we could leverage that to figure out where the derivative of a function vanishes as well. That was where our peaks and valleys are coming from.
- 5.2 But we said that there were certain aspects of Newton's method where we had pitfalls associated to it. On one of them, if my derivative was ever $= 0$, then the tangent line was now horizontal, so it wouldn't make sense to actually say it crosses the x-axis. It destroyed the iterative nature of Newton's method. We also had that if I somehow chose a point and my function started oscillating between the two, it could be $x_0 = 2$, and then I get $x_1 = -2$, and then $x_2 = 2$, and then $x_3 = -2$, and now you're just going 2, -2, 2, -2, 2, -2—you're just bouncing back and forth. It's not actually narrowing in on what the corresponding root is at any point. So there were some downfalls to Newton's method.
- 5.3 Now that we have the intermediate value theorem, we can come up with the bisection method. The bisection method, which is another method for trying to determine when a function is equal zero. Remember, that's our motivation for this entire thing, trying to figure out when does $f(x) = 0$.
- 5.4 Remember that the ultimate goal is we have a function that we're trying to set equal to zero where that function is going to be the derivative of something because that's going to tell me where I have horizontal tangent lines, and that's going to be where I have my peaks and my valleys at the end of the day. That's still the motivation for all of this. We're aiming towards getting optimization, and we're going to get there very soon. But the bisection method is very beautiful. It takes advantage of the fact of what the intermediate value theorem says.
- 5.5 Let me remind you what the intermediate value theorem says. This was that schoolyard thing that I mentioned last time. The intermediate value theorem. This said what? If f is continuous on a closed interval $[a, b]$, and M is any value between $f(a)$ and $f(b)$, then there exists some c in the open interval (a, b) such that $f(c) = M$. Again, if you're anything like me, once I start reading a theorem like this, my brain wanders and I don't understand anything as to what it's actually saying. But the picture is really all you need to know.
- 5.6 Let's say I have something like this. Let's say here's a , here's b . $f(a)$ is maybe—let's say I'll put it down here. Here's $f(a)$. $f(b)$ —maybe I'll put it up here somewhere. $f(b)$ is up here. It says that if you give me any value of M in between $f(a)$ and $f(b)$, that means

that, for example, if I chose M to be here— so here's $y = M$. So M is this value on the y -axis right there—it says that if f is a continuous function and it starts here and it ends here, then it has to cross the horizontal line. That's what the intermediate value theorem says at the end of the day.

- 5.7** So you can say, "If I do something like that, I have to cross $y = M$. That's really where the continuity is I'm running through. Last time we talked about continuity as saying $\lim_{x \rightarrow a} f(x) = f(a)$. It's saying that the limit exists and it equals the value of my function. I can't have any gaps, because you can see quite easily how you could make this fail if you're allowed to say that your function is discontinuous.
- 5.8** Think about the schoolyard analogy that I had. If I drew a line on the ground and I took a jump rope and put it on the ground, then it's going to have to cross that line as long as one part of the jump rope is on one and the other part of the jump rope is on the other side. But you can imagine if I draw a line on the ground, I can start walking up to the line and then step over the line. I never have to cross the line necessarily. If I'm drawing something with my foot, I don't necessarily have to cross that line at some point. You can imagine if I relax continuity, I could draw the line, and then the second I get over here, I could jump over it and then finish getting to $f(b)$ at that point.
- 5.9** Continuity is really crucial for the intermediate value theorem. The way that we're going to leverage this—again, we're trying to figure out where $f(x) = 0$ — is the following thing. If I modify the intermediate value theorem slightly and I say, "Iff (a) and $f(b)$, one of which is positive and one of which is negative—" what do I notice? Here I said, " M can be any value in between this." I take $M = 0$. I take the x -axis. $f(a) < 0$, $f(b) > 0$, that means it has to cross the x -axis somewhere. Notice—it does. It has to cross at some particular point.
- 5.10** If we modify this slightly—we apply the intermediate value theorem when what? When—one way of capturing that $f(a)$ and $f(b)$, one of which is negative, one of which is positive, is to simply say, "If I take the product of their values, it's negative." Does that make sense? For example, $f(a)$ can be negative, then $f(b)$ would have to be positive, or $f(a)$ could be positive, and $f(b)$ could be negative. I don't want to have to say, "If this, then that," or, "If that, then this." I'd like to be able to just have one statement.
- 5.11** The easiest way to say it is simply to say, "If I take $f(a) * f(b)$, it's negative." If I take two numbers and I multiply them and it's negative, one has to be positive, one has to be negative, but it doesn't tell me which one has to be which. This statement here, $f(a) * f(b) < 0$, captures both. Either $f(a) < 0$ and $f(b) > 0$, or $f(b) < 0$ and $f(a) > 0$. It captures both statements here.
- 5.12** We apply the intermediate value theorem when $f(a) * f(b)$ is negative. So what's the conclusion? It says: there exists some c in (a, b) such that $f(c) = M$. What is M that I'm going to choose? We're going to choose $M = 0$. The conclusion is there is some c in (a, b) with $f(c) = M$, which is 0. That's exactly what we're trying to do. That's what we're

trying to accomplish. We're trying to figure out where does the function equal zero. The bisection method is basically an algorithm for finding when the function actually crosses the x-axis. When does the function cross the x-axis?

- 5.13** I want to illustrate this in the following way. How do I do it? I'm going to give you a picture first to motivate it, and then we're going to do some concrete examples. We're going to use an example that we did last time with Newton's method for trying to approximate $\sqrt{2}$ that we did during the Newton's method section. The picture is going to be the following. I have my interval here. Let's say here's a and here's b . To illustrate the property, I'm going to say that $f(a) < 0$ and $f(b) > 0$, but you could reverse the roles. You could say $f(a) > 0$ and $f(b) < 0$. It doesn't matter. The same technique is going to apply.
- 5.14** Let's say $f(a)$ is down here, $f(a)$, and $f(b)$ is up here—there's $f(b)$. Again, I'm thinking of a , b as living on my x-axis. Those are my input values. That's my closed interval from $[a, b]$ that lives on my x-axis. Now, if my function is continuous, then the intermediate value theorem tells me that it's going to have to cross the x-axis at some point. This value here is the c that I'm looking for, such that $f(c) = 0$. That's guaranteed to me by the intermediate value theorem. One thing that you need to be aware of is that the intermediate value theorem simply asserts the existence of a c value on $[a, b]$. It doesn't tell you that there's only one.
- 5.15** Think about it. If I gave you the function $\sin(x)$, for example— $\sin(x)$ oscillates, does something like this. I'm going to do my best. There's $\sin(x)$. And I gave you the closed interval, say, here's a and here's b . How many times does the graph cross the x-axis? It crosses once, twice, three times in that range. The intermediate value theorem would simply say: there exists a point at which you cross the x-axis, but it doesn't say there exists three points at which it does it. It's simply an existential statement. It's just saying there exists at least one, but there could be more. There could be more, but that's fine, because at the end of the day, we only care about finding when a function equals zero. We don't care how many times it equals zero necessarily. We just care, "Can we find when it equals zero?"
- 5.16** Here's what the bisection method does. Let me illustrate this for you. Here's the bisection method. All right. It says the following: look at the interval from a to b . You know that $f(a)$ in this example is negative and $f(b)$ is positive. Then you say: take the interval and look at its midpoint. Look at the middle of the interval. The middle of the interval is right here. We have a formula for it. When you have an interval from a to b , the midpoint is—you add the endpoints and divide by 2: $(a + b) / 2$. That's the middle of the interval. You can convince yourself of that to some extent.
- 5.17** If I gave you the interval from 0 to 4, what's the middle? 2. Okay, if I take $(4 + 0) / 2 = 4 / 2 = 2$. That's giving me the midpoint of that interval. You can convince yourself that $(a + b) / 2$ is actually the midpoint. Then what do I do? Compute $f((a + b) / 2)$. Compute the value of your function at the midpoint. Now, I don't necessarily care what the

actual number is. I just care: is it positive or is it negative? If I plug in the midpoint and I get the value of 7.3, I don't care that the value is 7.3. I just care that $7.3 > 0$. That's all I care about.

- 5.18** What do I do? I evaluate at that point. In this picture, this value is my $f((a + b) / 2)$, which happens to be positive. It happens to be positive. So what do I do? How do I actually get the bisection method to work in my favor? Think about it for a second. The intermediate value theorem told me initially, because $f(a)$ was negative and $f(b)$ was positive, there had to exist a c in between $[a, b]$ such that $f(c) = 0$. But what did we just figure out? We said, $f(a) < 0$ and $f((a + b) / 2) > 0$. My function is still continuous on that interval, so the intermediate value theorem still applies.
- 5.19** The intermediate value theorem says: I don't need to search for c over here anymore because I know that $f(a) < 0$ and $f((a + b) / 2) > 0$, so c has to live somewhere in between those things. So it takes my interval $[a, b]$, it cuts it in half, and says: you don't need to search in the entirety of $[a, b]$. You can now search in one half of it. Then what am I going to do? You repeat this.
- 5.20** In this example, I repeat on what interval? $[a, (a + b) / 2]$. I can throw away the other half of my interval because the intermediate value theorem doesn't guarantee me anything on the other interval. The other interval says f of the midpoint is positive; f of the right endpoint is positive. It doesn't tell me anything about maybe somewhere in there I'm negative. Maybe I am; maybe I'm not. It's possible my function does something like this: it's possible my function comes up here, dips down, and then comes back up, in which case it would actually cross the x -axis. But this doesn't tell me that. This doesn't tell me that at all. If I'm positive here and positive here, I am not guaranteed to be zero anywhere in between those two things. Does that make sense?
- 5.21** If I'm at 2 at the midpoint—if I give you that this is 2 and this is 5—the intermediate value theorem just tells me: you're guaranteed to take on any value in between 2 and 5. It doesn't tell me I'm going to dip down and go to 0 or hit 0 somehow. It doesn't tell me anything like that. So I can't leverage it to actually create an algorithm. But now that I do know that f of the midpoint was positive here and $f(a)$ was negative here, the intermediate value theorem says: there has to be a c value in between that's 0 somewhere. Now you repeat the process.
- 5.22** You look at this thing and you say, "Let's find the midpoint." c actually looks like it's in the midpoint, but I'm going to pretend that it's not. I'm going to say, "Let's say I look at this point—that's my next midpoint there. What is that value?" That value—how do you do it? You take this guy plus this guy and divide the result by 2. That's the midpoint there. This is my midpoint, let's say. That's my midpoint. What do I do? I evaluate my function at that midpoint. This picture looks like it's down here somewhere. It looks like that's f of midpoint.

- 5.23 Let's give this thing a name. Let's call this x_1 , $f(x_1)$. It looks like that's negative based on the picture. What can I say? I can say, "If that's negative"—I know that $f(x_1) < 0$, $f((a + b) / 2) > 0$ —I can say, "I don't need to search in this interval anymore either." Then you keep repeating this. I now search in this interval. I take its midpoint. The value of my function is positive. I throw this out. I take the midpoint here. The value of my function is positive. I throw this out. Take the value of my midpoint. Throw this one out.
- 5.24 What can you see? You're literally trapping the value $x = c$ by simply taking these smaller and smaller and smaller intervals. You do this long enough, and what's going to happen? You choose a point in each of these intervals, eventually, it'll converge down to c , which is precisely what you're trying to figure out. That's what the bisection method is.
- 5.25 But notice—the bisection method, nowhere in here did I say that f' can't be 0 or that f has to be differentiable or anything like that. Things that Newton's method required, I'm not even using here. So if there are certain pitfalls for Newton's—I get this alternation back between -2 and 2, -2, 2, things like that—this doesn't suffer from that. It doesn't suffer from the derivative being zero or something along those lines. It doesn't suffer from that. So in applications where Newton's method doesn't necessarily apply, this method would. It gives us another algorithm for doing it.
- 5.26 Now, this method is slower than Newton, in fact, much slower than Newton, but it's still relatively fast. It's a logarithmic convergence associated to that, whatever that means. But it means that it effectively converges relatively quickly. Not as fast as Newton. Newton is like the prime standard for the most part. However, this one does converge when applicable.
- 5.27 Let's do an example. We talked about trying to approximate $\sqrt{2}$ using Newton's method, and we saw that it converged really quickly. So let's do the same thing using bisection. Here's the example. Let's approximate $\sqrt{2}$ using bisection method. Using bisection. Now, remember how we did this for Newton. We have to formulate this problem as taking a function and setting it equal to zero. That's effectively what we're getting for every single one of these questions. We're trying to formulate it as a function equals zero.
- 5.28 Remember for Newton, what did we do? My solution was I set $x = \sqrt{2}$, and we said that then that means that $x^2 = 2$. To turn it into a function equals zero, we said to move the 2 over. Now I'm looking at $x^2 - 2 = 0$. This is going to be $f(x)$. $f(x) = x^2 - 2$. That's my function. That's the little trick to turn it into the framework of getting a function, setting it equal to zero at the end of the day.
- 5.29 What do I do now for bisection? Well, bisection, you have to produce an interval from a to b , and you have to guarantee that your function is continuous because you have to invoke the intermediate value theorem. The intermediate value theorem says: you

need two ingredients. You need the function to be continuous, and you need an interval, closed interval $[a, b]$. You need that to be true.

- 5.30** When finding the interval from a to b , this is the hardest part, and this is where people typically ask, "How did you know a was this and how did you know b was this?" The answer is, "I didn't. I didn't at the end of the day." What I really need to step back and look at is—remember, I need to produce an a such that $f(a)$ is either positive or negative. Then I need to produce a b such that $f(b)$ is the opposite sign of whatever $f(a)$ was. If $f(a) > 0$, I'm looking for b such that $f(b) < 0$ or vice versa.
- 5.31** One of the nicest things to check is to plug in $x = 0$ and see what happens. Plug in $x = 0$. That's usually a safe value to plug in. If I take in $f(0)$, what do I get? I get $0^2 - 2 = -2$. That's negative. That's negative. Now the goal is choose another value such that f of that value is positive. I don't have to look too far. I could look at this and say, "Okay, $x = 0$; that was a nice point. What if I plug in $x = 1$?" $f(1) = 1 - 2$. That's still negative. Not good. Okay, $x = 2$. That would be a natural next choice. So you choose $f(2)$: $f(2) = 2^2 - 2$. So that's $4 - 2$, which is 2 , which is positive. Bingo. I have my interval. My interval is going to be $[0, 2]$. There's my interval.
- 5.32** But there's no secret sauce to this whole thing or whatever to actually figure out what is the interval. It's a guess and check to some extent. You start plugging in values and see, "I'm positive here. I'm negative here. Therefore, intermediate value theorem tells me I have to be zero somewhere." Then you run bisection in that interval. Notice that somebody could have come in and said, "I'm going to plug in $x = -5$. $f(-5) = 25 - 2$, that's positive. Then somebody comes in and says, "Then plug in zero." $f(0) < 0$. That means that my interval could be $[-5, 0]$, and I could run bisection method there, but I wouldn't actually be getting $\sqrt{2}$ in that case. I'd be getting $-\sqrt{2}$ because I'm in a negative part of my real axis. You have to look back and say, "What am I actually trying to compute?" I'm trying to compute $\sqrt{2}$. That's a positive number. So trying to choose 0 and something else is a safe bet. That's a safe bet. You have to keep the context of the problem in mind when you're doing that. There's my interval.
- 5.33** Now, I have to assert that f is continuous. I need to be able to say something about the function being continuous. Well, the function is $f(x) = x^2 - 2$. These things have a name; they're called a polynomial. So f is continuous because f is a polynomial. That feels like cheating to some extent. You're like, "Oh, it's continuous. I'm not going to check the property. I'm just going to invoke the fact that it's a polynomial, and then it's a well-known fact that polynomial are continuous." That's fine.
- 5.34** But if you wanted to prove that f is actually continuous, you'd have to prove that $\lim_{x \rightarrow a} f(x)$ for any point a is equal to the value of the function at that point. You'd have to prove that result. So we're just going to use the fact that f is a polynomial, and polynomials are known to be continuous. Therefore, by the intermediate value theorem, since $f(0)$ was negative and $f(2)$ was positive, there has to be some point in between 0 and 2 for which $f(c) = 0$. If I plug in c for x , that's saying that $c^2 - 2 = 0$, i.e. $c =$

$\sqrt{2}$. Now we're going to use bisection to trap the value c and figure out how close we can get to it.

- 5.35** Here we go. Let's run the algorithm. Here's how I like envisioning all of this. Here's 0. Here's 2. I know that at $f(0)$, I'm negative down here somewhere. Here's my $f(0)$. That's negative. I know that at $x = 2$, I'm positive. I'm up here somewhere. Here's $f(2)$. Now, I know that it crosses somewhere. It crosses somewhere. Let's say—I don't know—something like this. You don't actually have to draw the graph of the function, after all. You just have to draw that it's positive here, it's negative here, and it has to connect the two somehow. You could literally draw a straight line between the two. It doesn't matter. It doesn't matter.
- 5.36** What does the bisection method say? It says: find the midpoint. What's the midpoint here? I'm going to take the left endpoint plus the right endpoint: $(0 + 2) / 2$. What am I going to get? That's a 1. The middle of that interval $[0, 2]$ is 1. It says: evaluate the function at 1. So compute $f((0 + 2) / 2) = f(1)$. All right, what is $f(1)$?—let's compute that up here. Let's give us some room up here. Let's compute $f(1)$ and see what we get.
- 5.37** All right, so $f(1) = 1^2 - 2 = 1 - 2 = -1$. That's negative. Okay, so that's negative. I can draw the picture here and say, "Here's $f(1)$. $f(1) < 0$." What can I conclude? I can say, " $f(0)$ was negative, $f(1)$ was negative." The intermediate value theorem doesn't tell me anything on this interval. It doesn't assert that there has to be a 0 here. I can say, "Throw this out." I no longer have to search in that range. I no longer have to search in that range.
- 5.38** Now I'm searching in this range—in between 1 and 2. I apply bisection again. It says: let's look at the midpoint. The midpoint is right here. What is that? That's $(1 + 2) / 2$, so that's $\frac{3}{2}$ or 1.5. It says: evaluate the function at 1.5 and check—is it positive or is it negative? So let's see how that is. If I take $f(1.5)$, that's what? $1.5^2 - 2$. I don't like dealing with decimals, personally. It's easier for me to handle fractions: $1.5 = \frac{3}{2}$. So this is really $(\frac{3}{2})^2 - 2 = \frac{9}{4} - 2$. What is that? 4 goes into 9 twice with a remainder of 1. So this is like 2 and a quarter. This is 2.25. This is $2.25 - 2 = 0.25$. Again, I don't care what the value is, that it's actually 0.25. I just care if it's positive or negative at the end of the day.
- 5.39** I come over here to my 1.5. I computed the value. I got $f(1.5)$. That was positive. What can I say? Which interval do I throw away? I'm positive here. I'm positive here. Intermediate value theorem tells me nothing here. So I throw this interval out. This one gets thrown out. You keep doing this.
- 5.40** We'll do it one more time. I'll do it one more time. This is 1.5, this is 1. I look at the midpoint here. What is this? This is $f((1 + 1.5) / 2)$. What's that? $2.5 / 2$, which is what? One and a quarter: 1.25, I think. You have to check me. Keep me honest on this. I think that's right. $2.5 / 2 = 1.25$. Okay, that's fine. I evaluate my function at that midpoint now. What is $f(1.25)$? I take $f(1.25)$. Okay, 1.25 as a fraction is what? That's five

quarters. Again, simplistically, I have to think about it that way. If you hand me five actual quarters, like coins, what do I get? Four of them give me one whole, and then I give another quarter. I have five quarters; that gives me 1.25. Okay, so that's fine.

5.41 That's $\frac{5}{4}$. I get $(\frac{5}{4})^2 - 2$. So what is this? This is $\frac{25}{16} - 2$. It's like, "Is that positive or is that negative?" 16 goes into 25 once and change. It doesn't go in twice because if it went in twice, this would have to be 32 or higher. It goes in once and change, and I'm subtracting 2. So this is like 1 point something minus 2—it's negative. It's negative. Okay, fine. This is negative. Again, at the end of the day, I don't care what the actual value is. I just care if it's positive or negative. That's it. I evaluated at 1.25, I got that I was negative. This was $f(1.25)$. Then I throw out this interval because I'm negative here, I'm negative here. Intermediate value theorem tells me nothing here. Then you just repeat this.

5.42 But notice what's happening. We first started in the interval $[0, 2]$. That was where we started. Then what did we do? Here, let me draw an arrow saying we run bisection method once and it carves down the interval to something else. What interval were we now playing in? We went from $[0, 2]$. We said we didn't need $[0, 1]$. That took me to $[1, 2]$. Once I looked in $[1, 2]$, I said I didn't need $[1.5, 2]$. That means I'm looking in the interval from $[1, 1.5]$.

5.43 Then I said, "If I'm in the interval $[1, 1.5]$, I don't need the interval $[1, 1.25]$." Now I'm in the interval $[1.25, 1.5]$, and then I'm going to keep doing this over and over again. But remember, last time we said $\sqrt{2}$ was equal to what? This is roughly 1.414 and change. Look at how amazing this is already doing. It's saying: start in this large interval— $[0, 2]$ —carve that down to half of it— $[1, 2]$ —carve that down to $[1, 1.5]$ —great—carve that down to $[1.25, 1.5]$. It's literally narrowing in on this number. As I keep carving this interval down and further and further and further, I'm getting closer and closer to 1.4 and change. Does that make sense?

5.44 It's magical at the end of the day, that this is actually working to some extent. But it's literally just exploiting the fact that the intermediate value theorem says: if you're negative here, you're positive here, you have to cross. If I lay a jump rope on the ground and there's a line in between the two and the jump rope is continuous, it has to cross the line somewhere. That's it. It's not telling us anything about the derivative or anything like that.

5.45 Okay, that's all fine and great, but at the end of the day, remember—we're trying to figure out when does the derivative equal zero? When does the derivative equal zero? The motivation for this entire thing is what? We want to be able to get to a place where we understand optimization because applied mathematics, machine learning, data science, AI, all of that, at some point, there is some governing function for your model that you want to optimize. You want to minimize the error associated to it. We are trying to get to optimization, ultimately. We want to figure out when does the derivative equal zero. That's what we're trying to get to.

- 5.46 Now, analytically, sometimes finding the derivative is relatively simple. In practice, in the real world, it's never simple. I can give you—let me compare and contrast this. Let me compare and contrast this. Then I'll explain what I mean by the real world. Here's a simple example. When you learn this material, it's very simple to fall into this false sense of security of being like, "All of the functions you've ever given me, I've been able to just compute the derivative using basic algebraic principles, plug in values, and actually find out where it's equal to zero." It's very simple to be like, "That's what it's like in the real world." It's like, "That's not what it's like in the real world. In the wild, the functions that you are dealing with are not functions that are very easily manageable. They're not functions like this."
- 5.47 If I gave you a function $f(x) = x^3 - 6x^2$ or something like this, and I say, "Where is $f'(x) = 0$?" Where is $f'(x) = 0$? That's the game. Then I would say, "Let's take its derivative and see if I can set it equal to zero by hand and see what happens along those lines. A natural thing to do would be to say, "What's $f'(x)$?" I have to step back and be like, "What are they asking me to do? They're asking me to find $f'(x)$. Do I know rules that tell me how to find $f'(x)$ in this case?"
- 5.48 The answer is, "The power rule on x^3 tells me: bring the 3 down and decrease the exponent by 1." So that's going to become a $3x^2$. This one says: it's a difference of functions. I have $x^3 - 6x^2$, and we said the derivative of a difference is the difference of the derivative. So I just need to go term by term to differentiate. I don't have any products of things. I don't have an $x^2 * \sin(x)$, things like that. I do have a product with the 6, but remember, constants can come out when they're multiplied in. That was one of our derivative rules associated to it. Now I say, "This is going to be $-6 * d/dx(x^2)$ —power rule again—that's going to be $2x$. This whole thing is $3x^2 - 12x$, and I'm setting that equal to zero.
- 5.49 I say, "What rules do I have for setting things equal to zero?" I could say this is a quadratic, so I could hit it with a quadratic formula or something along those lines. I could look up the formula and figure out how to set it equal to zero, and that's fine. Or I can say, "Does this thing factor? Do they have anything that are common?" I say, "I have a 3 and 12"—3 comes out from both of those. I have an x^2 and an x , and x comes out from both of those. Then what do I need in here to fill in the actual factorization? When I bring this in here, the result has to be a $3x^2$ —so this has to be an x . When I bring this in here, it has to be $-12x$. I don't need any more x s, I just need more 3s. I get that in the form of a $4 - 3 * 4 = 12$, so that's going to give me my factorization.
- 5.50 Then because these are real numbers, I say, "When I have the product of two real numbers is equal to zero, that means either one of them is zero: $3x = 0$, or the other one is zero: $x - 4 = 0$. That means either $x = 0$ or $x = 4$. But that's incredibly specific. That's incredibly specific. This function, one, its derivative was relatively simple to compute, and two, it factored nicely, and I was able to say $x = 0$ or $x = 4$. That's very contrived. It's very contrived. What if I simply tweaked this? Here's example one.

- 5.51** What if I gave you another example and I said something of the form, "What about $g(x) = \sin(x) - x^2$?" I'll give you this function. It doesn't seem that much different than the previous one. After all, this one I had two terms, this one I have two terms, so that's not too bad. Yeah, I've changed $\sin(x)$ quite a bit. It's not x^3 anymore. But I'm still trying to ask the question, "What about $g'(x) = 0$? Does that have a solution? How would I find it?" All of that.
- 5.52** I can start the same way. I can say, "For my solution, I can say $g'(x) = 0$ —" I need to figure out what the derivative is, so I have to go back and say, " $\sin(x)$, is it a composition, number one?" The answer is, "No, it's straight up $\sin(x)$." That's fine. It's not $\sin(6x)$ or $(\sin(x))^2$ or anything like that. So what's its derivative? Derivative of sine is equal to cosine. I look at that, I get $\cos(x)$. Then I have the difference of functions. Then the derivative of this is the difference of the derivatives. So I'm going to get minus the derivative of x^2 , which is $2x$, and I say that's equal to zero. Good luck with that. Good luck with that.
- 5.53** At the end of the day, up here, when I said this was equal to zero, I said, "Let's find the values of x for which that's true." I got all the x s on one side, I factored them out, and I said, "The product of numbers being zero means that one of them has to be zero," and I simplified from there. All right, fine. Factor x . I can't. Solid. That technique's out. What do I do? I'm completely stuck. There's nothing I can do at this point.
- 5.54** I can try and say, "Add $2x$ to the other side." I'm saying, "Does $\cos(x) = 2x$?" That's maybe helpful. I don't know, but it doesn't really seem very helpful at this stage. Why? Because I don't know whether $\cos(x) = 2x$ or not. I've got no clue. I can stare at this for a little bit and say, "Does it have a solution, though?" I may not be able to actually find what x is analytically, being able to factor things out and all of that stuff, but does it actually have a solution?
- 5.55** Let's look at the geometry of it. Let's see, what is the graph of $\cos(x)$? $\cos(x)$ is an oscillating wave that looks something like this. Don't make fun of me. I'm trying to do this as best I can. Here we go. That's not bad. $y = \cos(x)$. So $y = \cos(x)$. That's this one. If I graph $y = 2x$, what does $y = 2x$ look like? That's just a line. That's just a line. It has to go through the origin and it has a slope of 2. It's going to go through here and it has a slope of 2. Here's $y = 2x$. Again, I ask you the following question, "Does that have a solution?"
- 5.56** It has a solution if those two graphs intersect each other. So does it intersect? Yes, it does. It intersects right there. So then the question is, "If I drop that down, what is that c value for which $2 * c = \cos(c)$, or equivalently evaluating this function and setting it equal to zero." So geometrically, I can say, "Yes, there's a solution." So that's a start. I know a solution has to exist, but how do I find it? How do I actually find it? And that's where the bisection method comes into play. That's where the bisection method comes in.

- 5.57** So let's run bisection method on it. Remember, bisection method tells you: you have to invoke the intermediate value theorem. The intermediate value theorem says you need two ingredients. You need the function is continuous. In this case, what is my function? The function is this guy, $g'(x)$. That's the function. It happens to be a derivative. That's fine. But at the end of the day, I just care about whatever the function is. I need to assert that $g'(x) = \cos(x) - 2x$ is continuous on $[a, b]$ in order for the intermediate value theorem to apply. I also need that $g'(a)$ is either negative or positive, and $g'(b)$ is the other one. If $g'(a) > 0$, then $g'(b) < 0$ or vice versa. I need that in order to invoke intermediate value theorem.
- 5.58** I need to now construct an interval $[a, b]$. Now, $\cos(x) - 2x$ —cosine is a trigonometric function which is continuous everywhere. Here's its graph. There's no gaps to this thing. It waves the entire time, so continuous. $2x$ is a line—that's continuous everywhere. Continuous minus continuous. This is continuous for everything. It's continuous on the entire real line, on the entire x-axis. I just really need to find a and b now such that $g'(a)$ is either positive or negative, and $g'(b)$ is the reverse. Again, plugging in zero is usually a safe bet. Let's try that. If I take $g'(0)$, what do I get? I get $\cos(0) - 2 \cdot 0 = \cos(0) - 0$. I have to look up what is $\cos(0)$ —that's 1. So this is 1. That's positive. That's good. I know that one point gives me a positive value. I need another point which is going to give me a negative value.
- 5.59** Now, remember, these are trigonometric functions. You're going to have to evaluate these at points. Choose points that you know some values of the trigonometric function. For example, I know what $\cos(\pi)$ is. $\cos(\pi) = -1$. Maybe trying π might work. Maybe it doesn't. I don't know. Let's see. If I take $g'(\pi)$, what do I get? I get $\cos(\pi) - 2\pi = -1 - 2\pi$. I start at -1 and I subtract more. That's going to be negative for sure. Now I have my interval: $a = 0$, $b = \pi$. This interval is going to be $[0, \pi]$. That makes sense from the picture if you think about it right here. π is right here. π is there, and 0 is here, and c lives somewhere in between 0 and π .
- 5.60** Now we're going to try and run the bisection method to try and trap what c is equal to. Let's do that. I got to erase some of this, but let's figure this out. Here we go. Here is 0 , here is π . I know that $g'(0) = 1$, so that's positive, so that's something up here. So $g'(0) > 0$, and I know $g'(\pi) < 0$ at someplace down here. Now, I'm not going to draw the line, but I know that g' is continuous, so it connects this point to this point, so it has to cross somewhere here.
- 5.61** So what do I do? I look at my midpoint. My midpoint is what? $(0 + \pi) / 2$. So this point is $\pi/2$. I plug that into my function and decide—is it positive or is it negative? We're going to do that computation over here. We have $g'(\pi/2)$. That's equal to $\cos(\pi/2) - 2 \cdot \pi/2$. That's nice— 2 s cancel. That's nice. $\cos(\pi/2) = 0$, and I'm getting $-\pi$. That's negative. That's negative. I know at $\pi/2$, the value of my function is negative. $g'(\pi/2) < 0$. Which of these two intervals can I get rid of from the bisection method? This one goes from

positive to negative, so I know it has to cross. This one goes from negative to negative—I don't know anything. So this one has to be thrown out.

- 5.62** Does that make sense how this is working? You find the midpoint, check—is it positive, negative—compare it with one of the others, and then invoke intermediate value theorem on that. Next one, I find the midpoint here, $(0 + \pi/2) / 2 = \pi/4$. So this is $\pi/4$. I check the value at $\pi/4$. What is $g'(\pi/4)$? What do I get? That's going to be $\cos(\pi/4)$ —this is getting more difficult because these values are not very clean the further you go— $2 * \pi/4$.
- 5.63** Thankfully, we actually know what $\cos(\pi/4)$ is. I can look it up on the unit circle or look up trig tables or whatever the case may be for trigonometric functions. $\cos(\pi/4) = 1/\sqrt{2}$ —and yes, you are allowed to divide by $\sqrt{2}$. You're allowed to have square roots in the denominator. That's allowed. That's a mathematical thing— $2 * \pi/4 = \pi/2$. Now, without a calculator, it's hard to be like, "Is this positive or is it negative?" But think about this for a second. π is like 3.14 and it keeps going. If I took the number just 3— and $\pi > 3$ —but if I took the number 3 and I divided it by 2, do I get something bigger than 1? Sure. If I take $3 / 2$, I get 1.5. If I took $\pi/2$, I'm going to get something bigger than 1 for sure. Do we agree? 3.14 and change / 2, definitely going to be bigger than 1. This quantity on the right is bigger than 1.
- 5.64** But look at this thing here: $\sqrt{2} > 1$. We said $\sqrt{2} = 1.4$ and change. If I take 1 divided by something bigger than 1, this number is less than 1. This number is less than 1, and I'm subtracting something that's bigger than 1. Is this positive or negative? It's going to be negative in this case. That's a quick hack if you don't have a calculator on hand. Just compare the numbers. Is it bigger than 1? Is it less than 1? I'm taking a number that's less than 1, and I'm subtracting a number that's bigger than 1. It has to be negative. That's at $\pi/4$ —the value is negative.
- 5.65** Again, $\pi/4$ —what do I get? I get the point down here, let's say $g'(\pi/4)$ is again negative. I'm down here. Which of the two intervals can I get rid of? I can get rid of this one because this one is negative to negative. Intermediate value theorem tells me nothing. This one is positive to negative. Intermediate value theorem says I have to cross here. I would check at $\pi/8$. $\pi/8$ —I would check the value. If it's positive, then I would not use this interval. If it's negative, then I would use this interval. You keep going from there and you eventually trap the root for where $g'(x) = 0$.
- 5.66** This gives you a method for actually determining when do functions equal zero. In this case, it happens to be the derivative I'm looking at, but it actually gives you a concrete method. Now, you don't do this by hand. You code this up every single time. You code this up, and we'll go into this on how to code this from basic principles. But you code this up to determine what is the value that actually makes the derivative vanish. You don't do this by hand.

- 5.67 Now, in the wild, when you're doing these particular things, the functions that you're coming up with, each of these only had two terms. The two examples that we had, they just had two terms. That was it. The first one had an x^3 , and what was it? It was $-6x^2$ or whatever it was. I had two terms, and I found the derivative very simple to solve. This one had two terms: $\sin(x)$ and x^2 . I took its derivative, and suddenly it was impossible to solve.
- 5.68 The functions that you deal with in the wild, when you're dealing with neural networks or you're dealing with things like that, these things have thousands of variables associated to them. They have thousands or ten thousands of worth of terms, each and every single one of them. So good luck looking at this by hand and taking its derivative by hand. Number one, even be able to write that down would be an accomplishment, let alone being able to set it equal to zero and solve for it. That's the key principle with all of these is that if I can code this up, then you give me any function whatsoever. As long as it's continuous, I can stick it into this, code it, and go from there. That's the value here. We're illustrating it with simple little models, but in the real world, you can apply this to functions. As long as these properties are satisfied, you can apply them all over the place. That's their value.
- 5.69 I want to give you an example of where this actually applies that's not even in a function theoretic framework. This is simply a computer science type statement. But it illustrates the versatility of the intermediate value theorem and how you can apply it to different scenarios and things like that. I want to ask you the following question. I'm going to give you a list of numbers. It doesn't matter what the list contains. I could give you a list of letters, a list of words, a list of something. The same principle can be adapted accordingly to actually solve this problem. But I'm going to give you a list of numbers, and I am going to tell you that that list of numbers is ordered. It's ordered from least to greatest.
- 5.70 As an example, let me give you a list. Let's say L is equal to the following list. Let's list off numbers: $L = [2, 4, 5, 9, 11, 13, 14, 16, 20, 32, 39, 41, 80]$ or something like that. There's my list of numbers associated to it. They're listed from least to greatest, so it's an ordered list. It's ordered simply by least to greatest. I want to ask you the following question. I want you to find a number in there quickly. I'm going to give you a number. I'm going to say—let's say the number is 39 for the sake of argument. I want you to come up with a method which finds 39 quickly. I want you to find 39 quickly.
- 5.71 This is like one of those funny memes that you look at and you say the question says, "Find x ," and the person just circles it and says, "There it is." That's not what I'm asking. I'm saying, "Give me a method for finding 39 relatively quickly." A first naive approach, which works, but it may not be efficient, is to simply say, "I know that it's ordered, so start at the first element and ask yourself the question, 'Is that 39? No. Is that one 39? No. Is that one 39? No.'" You just keep doing this. Eventually, I'll get to 39. But how many steps does it take to get me there? It takes me a lot. It takes me step 1, 2, 3, 4, 5,

6, 7, 8, 9, 10, 11. It takes me 11 steps to simply start from the beginning and go to the end.

5.72 It's like in the olden days when you had a phone book and you're trying to find a business's phone number or something like that. A terrible way to do this would simply be like, "Start on the first page, look at the first number and look at that business. Is that the business I'm looking for? Nope. Second business—is that the business I'm looking for? Nope." That's a terrible method for doing that. It's going to take forever. What do you do? Well, the book is alphabetized. So you first find what is the name of the business, and you start looking there. You narrow in locally into something that actually allows you to search it much faster. Nowadays, we just punch it into our phone and it gives us immediate answer.

5.73 So if I do this linearly, meaning just check the first, check the second, check the third, and go forth—how many steps do I have? Well, I take steps = 11 in this case. Is that what we said? 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11—it takes me 11 steps to get to where I want to go. But if I step back and I say, "Maybe this intermediate value thing, the principle of it—of taking the interval, cutting it in half and searching either half—maybe I can somehow use that to my advantage." This is what's called binary search or bisection method, bisection search in computer science for searching ordered lists and see what happens.

5.74 What does that say? It says: take your list and find the midpoint. Find the midpoint. So let's see—how many elements do we have? We have 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13. That means if I go six here and six here, then the one in the middle should give me what I want. Do we agree? So what's the midpoint? I say: 1, 2, 3, 4, 5, 6. 1, 2, 3, 4, 5, 6. So what's my midpoint? 14 is the middle.

5.75 Now, you can argue, you could say, "Your list had an odd number of elements. What if it had an even number of elements? There's not going to be a midpoint. I would have six here and six here." I agree with you. There isn't a midpoint. If, let's say, I didn't have 80 here, then the midpoint would be somewhere in between 13 and—let's see: 1, 2, 3, 4, 5, 6—it'd be somewhere between 13 and 14. You have to make a choice at this point. You say, "If there's nothing here, then choose the left or choose the right." That's the point that you're going to investigate. For us, we don't have to worry about that right now because there's actually a midpoint.

5.76 Then you ask yourself the question, "Is 14 bigger or smaller than 39?" You say, "It's smaller than 39." That means since this is ordered, all of the elements on this side have to be less than 14. If $14 < 39$, I don't need to search here anymore. I can throw all that out, just like we did with the intermediate value theorem. We checked what is the value of my function at the midpoint—it's positive or negative—and then I can either throw out the left or the right interval. This is a very similar adaptation to the intermediate value theorem.

- 5.77** Now it says: search in this interval now. Search in the interval. So how many steps have I done here? This is my binary search. Binary search, or you could call it bisection if you want, because it's very similar to what we did with the intermediate value theorem. Now I'm looking in this list from 14 up to 80. If you're clever about it, you can say, " $14 < 39$. Why am I even including 14 anyway? Why don't I actually just look in the interval—at this list from 16 up to 80?" Because $14 \neq 39$, so I throw it away.
- 5.78** I look in this list and I say, "Let's do the same thing. I've got 1, 2, 3, 4, 5, 6 elements. This is going to illustrate the principle that we had that I mentioned earlier, that if your list has an even number of elements, what do I do? I find the midpoint. That means I got 1, 2, 3, 1, 2, 3—the midpoint is right there. I have to make a choice. Let's suppose I choose the left. If I choose the right, I win. If I choose the right, then I'm at 39, then I'm done. Let's say I don't choose the right because I want the worst case scenario to some extent. Let's say I choose the left. I look at 32 and I say, " $32 = 39$?" We say, "No. It's less than 39." Because it's less than 39 and everything to the left of it is ordered—this has to be less than—I can throw out this entire interval.
- 5.79** I've done two steps in doing this, and I now have three elements left to compare. I look at this interval—I look at this list, I find it's midpoint. It's 41. Is $41 > 39$? It's bigger. So I throw out 41 and everything to the left of it, and now I have 39 on the nose. How many steps did that take me? It took me three. I had the first step gave me this one, the second step gave me this one, the third step gave me this. I guess technically four, because the fourth step, I finally got my answer. Fourth step, I got my answer. In binary search, it was steps = 4.
- 5.80** This seems fairly contrived. It seems fairly contrived, but it's simply just an adaptation of what we were doing with the intermediate value theorem. I check something at the midpoint. I compare it—is it positive or negative? Is it bigger or less than? Is it alphabetically larger, lexicographically? If I'm thinking about this as looking at alphabetical letters—I give you a list of letters and I tell you that they're in alphabetical order. You can adapt this in so many ways. It's so versatile. But it gives me a way of searching for an element in here, just like the intermediate value theorem told us how a function can be equal to 0 simply by subdividing the interval. We're right there for optimization, and we're going to get into that really quickly.
- 5.81** You already mentioned using—what's it called—limits to figure out interest rates. I was wondering if there are any other applications, if you have a Calculus I education that you could do in real life.
- 5.82** With limits?
- 5.83** Limits, approximating functions, integrals, derivatives, whatever have you.
- 5.84** Yeah, of course. That's a good question. Limits effectively form the foundation, ultimately, in a lot of mathematical principles. For example, we talked about approximating functions using sines and cosines to some extent. I gave this example of

using heartbeat where I have this—if I gave you a graph that looked something along these lines— let's not put axes to it. Let's say I've got my heartbeat that's doing something like that, and then dot, dot, dot. You have this. We said that we could approximate that with sines and cosines.

- 5.85** What did I really mean by that? I meant that if I take "enough" sines and cosines and I combine them in a way—I'm adding them—then I can actually approximate this function. What I'm really saying is that if I take the limit of these functions, I take an infinite number of those functions, then I can actually get this to be approximated on the nose if I'm careful. That's where limits show up.
- 5.86** Another thing that you can do is you could do something along the following lines—and this is getting into the integrals that you mentioned as well. If I want to ask the question something along the lines of—let's say I have from a to b up here, and I want to say, "What is the area of that particular thing?" What's the area of that particular thing? What I could do is I could say, "I don't have a formula for this because I don't know what that curved shape looks like." I can't just look this up and be like, "That's a trapezoid," or, "That's a triangle," or, "That's a square or a rhombus," or whatever, and say, "There's the formula for what its area is."
- 5.87** But what I can do is I can say, "I know what rectangles look like, and I know what areas of rectangles are." So I can do the following. I could say, "Let's stick some rectangles in here." Let's say I take a rectangle that looks like this and then another rectangle that maybe looks like this. I look at these two rectangles here, and I can say, "I can compute the areas of those," but it's not going to give me the whole area because I'm missing some parts to it. So I can say, "Well, let's not abandon that idea right away. Let's see if I can modify it a little bit."
- 5.88** Can I take that and I can say, "Two rectangles was nonsense. It didn't really give me a very good picture. What if I took 20 rectangles? What if I took more?" If I took more, then I'm saying, "Cut this up into a bunch of pieces," and now I'm starting to look at things that look like this. I've got a bunch of rectangles and so on that look roughly like this. I'll do my best. That's all you can do. That area, if I add all those areas up, that's going to give me a better approximation. It's like, wait a second. "What if I took infinitely many rectangles?" What does that tell me? That gets into the definition of integrals, which actually tell you what that area is on the nose. If I take the limit as the number of rectangles grows to infinity, in a very precise way using limits, I can actually compute the area of that particular shape. So limits show up everywhere in mathematics in this particular way.
- 5.89** Other ways along the lines of what you mentioned earlier from an economics standpoint is you can say things like—if I want to look at, say, $\lim (x \rightarrow \infty) f(x)$, what am I trying to understand with this? Well, if this is equal to a value— whatever this is, let's call this M —let's say that that's equal to a particular number, then that's saying that my function is somehow tending— if I looked at it graphically, if this is M , $y = M$, I don't

know how my function gets to M , but it tells me that as I go further and further and further out in x , maybe my function does something like this. Maybe it's oscillating, but it's narrowing in on M . It's crossing it occasionally, but it's eventually narrowing in on M . That's telling me the long term behavior of this function as $x \rightarrow \infty$.

5.90 This is incredibly valuable in so many different ways. Things like what are called SIR models, which talk about the spread, the infection, the recovery rate in terms of a virus or something like that and how it spreads and whatnot. It's vital for us to be able to understand what is the long term behavior of that function. What's the long term behavior of how many people are going to get infected, how many people are susceptible to it, how many people are going to recover from the virus and things like that. These limits are incredibly important, especially from trying to project out what is the long term behavior of whatever the function is modeling in that case. So yes, limits show up all over the place.

5.91 If that limit doesn't converge in a hypothetical disease context, that's a pretty bad disease. That's what that's indicating.

5.92 Yeah, that's a great question. If this limit doesn't converge down to something, that is very concerning. That would mean that either one of two things is going to happen. If this limit doesn't exist, then what's really happening? That means either the thing is blowing up to infinity, it's going off to infinity, it's going down to minus infinity, or it's just oscillating. It never narrows in on something.

5.93 Remember, the limits say the following thing: it's as x gets close to whatever this is. What does that mean? x gets close to infinity—it means x is arbitrarily large; x is larger than we can even fathom. It's saying that the values of this function are getting arbitrarily close to this thing. For this limit not to exist, you can either go to infinity, go to minus infinity, or it can oscillate back and forth. If I said, "What's the limit of $\sin(x)$?" As x goes to infinity, it's nothing. It's just going back and forth. It never narrows in on anything.

5.94 In the case that the limit goes up to positive infinity—if this thing from the infection standpoint that we were talking about, if that thing blows up to infinity—it would say that the number of people being infected will eventually be the entire population. Saying that the limit of this is infinity, if this is modeling infection, it doesn't make sense to say infinity because there's only a finite number of people in the world. At the end of the day, it's saying: the limit of this being infinity is saying everybody's going to be infected at some point.

5.95 If it says something about negative infinity, then the context doesn't make any sense because it's saying iff (x) is modeling infected and I get negative five people are infected, that doesn't make any sense. The context is out. That doesn't make any sense.

5.96 If the limit simply just oscillates back and forth and it says the limit doesn't exist, that's really more what we see in the real world to some extent, if you think about it. There's

going to be spikes where there's high infection, and then there's going to be lows where people are not as infected, and it's going to oscillate back and forth. Different times of the year, flu season peaks, we get more people are infected, where then in the summer months, flu rates go down, things like that. That's going to oscillate back and forth.

- 5.97** The limit can tell us a little bit. It can try and give us a sense as to how many people are infected or not infected or things like that. You could also throw this in where this function is particularly modeling like— let's say you take ibuprofen or something like that for inflammation. You got injured and your doctor says, "Take some ibuprofen" or something along those lines. You want to understand—how is that going to metabolize in your system? Why is it that x number of hours later, I can take another dose of the ibuprofen? It's because it's metabolized, it's half life, It's gone through your system. There's not enough of it in there. It's saying: the limiting factor is after a certain period of time; you're at a safe level to where you can take another dose. That's where these limits are also coming into play.
- 5.98** The more inputs that you add, you get different results. For example, the virus—if you add an input, like an antivirus, eventually, you will stabilize the result.
- 5.99** Exactly. The more inputs you have, the more data that you have, to some extent, the better your predictive power should be. For example, with the virus side of things, if I'm only taking into account— let's say there's a virus such that a certain age range of individuals is more susceptible to it for whatever reason. Maybe it's an older population, and they're more susceptible to being infected by this virus, things like that. If my model is only considering, say, college-age students—so let's say ages 18 to 25, somewhere in there—if my model is only including that, then that's probably not a great model if I'm trying to model how it's going to spread in populations that are 60 years plus.
- 5.100** It would be better if I included more input and I said, "No, I want to sample people that are 60 years plus because I want to incorporate that into my model to get a better, more predictive power and say, 'Whoa, this is actually going to infect their population way more than it is for somebody who's a younger generation,'" things like that. The more input you have, typically, the better your predictive power is going to be.
- 5.101** For example, theoretically, we would all agree to some extent that if I gave you a fair coin and I flipped it, then it's 50/50; it's going to be heads or tails at the end of the day. But technically speaking, if I knew all of the data that went into that—if I knew how the quarter sat on my thumb, the velocity at which I flipped it, the angle at which I flipped it, the turbulence of the air through which that coin is going through, the surface of the ground when it hits the ground, what angle it hits the ground, how it bounces, how hard the ground is versus how bouncy the ground is, things like that, and I knew every single bit of data, the second I flip the coin, I can say it's heads before I even observe it. I know all the data associated to it. Is that realistic? Of course not. Of course not.

5.102 There's a balance to it. How much data do I incorporate into the model? How much do I not incorporate into the model? There's a fine-tuning that goes into that to actually get the optimal conditions. With viruses, it's very difficult because there's so much dynamic with it. But typically, I would say on average, the more data you incorporate, the better things are going to be. That was a great question.

Lecture 6: First and Second Derivatives

At a glance — First and second derivatives for analyzing functions.

- 6.1 All right, so we've been exploring different methods for trying to find when a function is equal to zero, ultimately getting to where the derivative equals zero because we're aiming for optimization overall. We're finally there.
- 6.2 We have one more method we're going to get to, and we'll get to that later. But the two methods that we've come up with so far have been Newton's method and the bisection method. They were all about trying to find where the derivative is equal to zero or where a function is equal to zero for that matter.
- 6.3 We need to understand the basics of optimization. What is the language of optimization? What are we trying to do at the end of the day? Now, optimization is all around us. In everything we do, we effectively try and optimize our situations, whether or not I'm trying to optimize the amount of money that I make, or I try and optimize the food choices that I have, or optimize how quickly I get from point A to B or something along those lines. I would argue that almost everything that we do can be couched in terms of an optimization framework.
- 6.4 It sounds funny, but I think there's a lot of truth to this. Even the relationships that we have are optimization problems to some extent. When you start looking at the world as these optimization problems and everything, it opens your eyes into realizing that these things are incredibly important. But we need to understand the language that we're talking about in terms of these things.
- 6.5 Let's look at optimization. What are we getting at with this? We're trying to understand these peaks and these valleys associated to that, and we're trying to understand how can I characterize those things in terms of a given function. Some of the language that we're going to use with this is the following. This is really terminology so that we're all speaking the same language. But the terminology is going to be the following.
- 6.6 Number one, if $f'(x) > 0$, what can I say? What does that mean? What does it mean that the derivative of my function is greater than zero in some interval? If $f'(x) > 0$ in some interval, then the original function f is increasing. What does that actually mean? We've talked about f' as representing the slope of the tangent line to the graph at that point so let's think about why this actually makes sense.
- 6.7 If I have a graph, or whatever that is, and I look at a certain point x , I look up here, and I notice that the derivative is positive. So my slope of that line is up. It's from bottom left to top right. It's up. If that's a tangent line to my function, then it stands to reason that my function is going up at that point. My function is increasing.
- 6.8 The rate at which it's going up is different. You can have a function that's either going up like this or up like this, and those two things are vastly different. One seems to be

growing faster and the other one seems to be growing, but at a slower rate. We'll get into what that actually means. But at the very least, I can say that if the derivative is positive, then f is increasing in that interval. And vice versa—if I look at my function where it's negative—let's say I look at this point and I look where the derivative is negative, that means that my slope is pointing downward, and so it stands to reason that my function is going to be decreasing when my derivative is negative. That's the second part.

- 6.9** You say: if $f'(x) < 0$ in some interval, then f is decreasing. Why would that be useful? At the end of the day, I want to be able to think through the problems and why these things are useful. I don't necessarily just want to memorize facts. I want to actually teach you how to think through these things. Why is increasing and decreasing important?
- 6.10** Think about if you have a peak—so you're trying to figure out some maximum to your function—what's actually happening here? If I have some peak, then I notice that if I look at slopes, it's positive, positive, positive, positive, zero, negative, negative, negative. So my function is doing what? It's increasing, and then it plateaus, and then it's decreasing. Somehow knowing where the derivative is positive or the derivative is negative, that's cuing me in as to saying something about: maybe I have some peak, maybe I have some valley. Just like over here, the valley is saying: negative slope, negative slope, negative slope, zero slope, positive slope, positive slope, positive slope. Understanding f' is telling me something about these valleys and these peaks. That's what we want to try to understand.
- 6.11** What else do we need to understand the language of optimization? Well, we haven't dealt with the second derivative much, but what does the second derivative actually tell us? I'm going to have a very similar statement here. If $f''(x)$ —so the second derivative, again, we talked about this—if I give you a function f , let's say it's $\sin(x)$. I say, "Can I take the derivative of $\sin(x)$?" Sure— $d/dx[\sin(x)] = \cos(x)$. That's great. So $f'(x) = \cos(x)$.
- 6.12** But I can say, "I give you the function $\cos(x)$ and take its derivative." You say, "That's $-\sin(x)$ " from the rules that we've discussed. So what is that? Is that the first derivative of f' or is that the second derivative of the original f ? Yes, it is. It's both. You can take higher derivatives— the second derivative, the third derivative, the fourth derivative—you can keep taking those derivatives as long as they exist. That's completely fine. But what do they actually tell us? That's what we care about. What do they tell us about the function? What do they tell us about its behavior and things like that?
- 6.13** If $f''(x) > 0$ in some interval, then f is—I'm going to state this, in my personal opinion, the proper way. But I'll tell you both languages that are used here because you'll see it in both contexts. But I'm going to say: if $f''(x) > 0$ in some interval, then f is convex. I think that's technically, in my opinion, that's the appropriate language to use with this. But you'll also see people say that it is what's called the concave up. I understand why

people say that, but, in my personal opinion, I don't think it's proper. What this is getting at is $f''(x) > 0$ in some interval, then f is convex. It means the graph looks like this. It looks like this. It's like a bowl that's faced upwards. That's what it means.

- 6.14** I understand why people call it concave up because it's saying up is something like this. It cues you in to say this is what it looks like. That's fine. But I disagree with the language that's used. It should be called convex, ultimately. When you get into higher order optimization problems and things like that, convex is the language that is actually used. I'm going to stick with convex.
- 6.15** The last one is if $f''(x) < 0$ in some interval, then f is concave. You can imagine— what do you think the other version is? If it looks like this, if it looks like it's down, you're probably going to call it concave down, realistically. That's what the language you'll see is used. But I don't necessarily think that that's proper, personally. Convex is up, concave is down. That's the idea. This gives us a way of trying to classify this type of behavior.
- 6.16** This is getting at what I was talking about here. If these are graphs of functions, they're both increasing, but they're increasing very differently. Do we agree? This one is increasing, but it seems to be going up at a faster rate, whereas this one is increasing and it seems to be tapering to some extent. How can I classify that? What do I notice? This one looks like it's up—we would say that the second derivative is positive, whereas this one looks like it's down, and I would say that the second derivative is negative. The second derivative allows me to classify: how am I increasing or how am I decreasing in that way? That's what we're getting at.
- 6.17** Why is this useful? It all boils down to sign lines. What do I mean by a sign line in this case? Let's say I give you the following scenario. I'm not even going to give you a function. I'm not going to tell you anything about it, aside from just giving you a visual representation as what its derivative is doing. That's all I'm going to give you. We're going to see if that thing actually is of use. Is that visual representation of the derivative of a function useful or not in terms of its optimization? This is what I'm going to give you.
- 6.18** Suppose I gave you the following thing. I'm just going to represent my x -axis as this line, and I'm going to write f' off to the side of it to indicate that I'm looking at f' values associated to this. Now, let's suppose that I tell you that at $x = a$, $f'(a) = 0$, so I put a zero above there. Then I'm going to give you another value that's bigger than a —let's say $x = b$, and I'm going to tell you that the derivative doesn't exist there. I'm just going to say DNE means "does not exist."
- 6.19** If I take f' and I evaluate it at b , I get something that doesn't make sense. As an example of that—what if I had something that was like $f'(x) = 1/x$? Then $f'(0)$ does not exist because I can't divide by zero. It doesn't exist. Does that make sense? These types of functions do actually exist where derivatives don't at certain points. Then I'll say at $x =$

c, again, I get zero again. Then I'm going to tell you the following thing. Everywhere else, f' exists, and it is never zero or does not exist. I found all of the "points of interest" to some extent. I found all of the points where $f' = 0$, so at a and c , those are the only points at which $f' = 0$, and b is the only point where f' does not exist. It exists everywhere else.

- 6.20** So think about this for a second. The values of a function are real numbers. How can you classify a real number? One way to classify it would be to say that a real number is either positive or zero or negative. There's only three possibilities for any real number. You give me any real number whatsoever, I can say it's either positive, zero, or negative. Do we agree? Okay. That means that for everything less than a , f' is either positive or negative. Do we agree? It can't be zero because after all, $x = a$ and $x = c$ —those are the only places where $f' = 0$. Everything to the left of a is either positive or negative.
- 6.21** Let's say they're positive. I'm going to indicate that with a plus sign here. Let's say to the right of a , everything is negative. In between a and b , I'm negative. Maybe in between b and c , I'm negative again. Maybe in between c and infinity, I'm positive. If I gave you that, that's called the sign line associated to the first derivative, f' . If I give you this, what can I say about the behavior of the original function, f ?
- 6.22** Let's go back to our terminology. If $f' > 0$ in some region, then I know that f is increasing. In particular, up from negative infinity—as far left as I can possibly imagine—all the way up to a , it says that my derivative is positive, which means that my function is increasing from there. If I'm increasing, I'm doing something like this. Now, in between a and b , my function is decreasing. So what's happening? My function is now coming down. In between b and c , my function is coming down. In between c and as far as I can think of, my function is going up. It's telling me that my function is doing what? Around $x = a$, what is my function getting at? It's a peak. It's increasing, zero, decreasing. That's a peak. This would be a peak.
- 6.23** This one's confusing at $x = b$. It's saying something like: I'm going down, and then my derivative doesn't exist at a point, and then I keep going down. What does that look like? That doesn't look like a peak for sure, because I'm going downhill, and then I don't exist, and then I'm going downhill again. Well, that's not a valley either. That's not a peak or a valley. I can't really say anything about what $x = b$ is going to be doing.
- 6.24** But then what happens at $x = c$? It's saying: you're going downhill, and then you're temporarily horizontal where you're zero, and then you're going uphill. What is that saying? That's saying I have a valley. I have a valley here.
- 6.25** Simply based on this information alone, this sign line, I've been able to determine what? I've been able to determine where I have a peak, where I have a valley, and where I have some really strange behavior that I'm not quite sure how to classify. That's okay. But if we're looking for optimization—optimization is trying to find those

peaks and the valleys, the first derivative alone is telling me: that's how you can find them.

- 6.26** Let's put some more terminology to these particular things. Where your derivative is equal to zero or does not exist are called critical points of the function. We're going to get that terminology out of the way as well. Let's do that right here. Where $f'(x) = 0$ or $f'(x)$ does not exist, are called critical points. Critical points of your function. So "pts" is points. In this example, we would say that $x = a$, $x = b$, and $x = c$ are critical points of the function.
- 6.27** Now, you want to make sure that these things are actually in the domain of the function. You want to make sure that your function is actually defined at these particular points as well. If you have an asymptote or something like that—like $1/x$ has $x = 0$ as an asymptote—my function is not defined at $x = 0$. Its derivative is not defined at $x = 0$. Of course not. It's not going to be defined where the function isn't defined. If I can't evaluate the function, it doesn't make sense to talk about tangent lines or anything like that. The critical points we're going to be talking about are things that are in the domain of my function to begin with.
- 6.28** So that's telling me the first derivative. What the first derivative tells me: increasing, decreasing, peaks, valleys, things like that. When I have the behavior that looks like this—this is a peak, then we call a —it's a value at which you're getting a local peak. You're getting a local maximum type behavior. The local maximum—this thing is going to be called the argmax, the argument maximum.
- 6.29** Basically what it's saying is: the argument typically refers to the domain value. The domain value. The fact that this is giving me a peak, a maximum type behavior, I would call $x = a$ the $\text{argmax}(f)$. It's where my function is maximized. Whereas this one at $x = c$, I go down and then I come back up. That's a valley type behavior. I would call that an argmin associated to it, a minimum type behavior. So this one would be called my argmin associated to that.
- 6.30** That's the language that we have. We have critical points—that's where my derivative is equal to zero or does not exist, and they're in my domain. If my sign line has a positive behavior, a negative behavior—so it's a peak type behavior—then I call $x = a$ an argmax. If it has a down and an upward trajectory—so I'm getting a valley—then I would call $x = c$ an argmin associated to it.
- 6.31** Similarly, I could draw the same type of graph for f'' , the second derivative of my function. I could have things where I'm positive, negative, and don't exist. But then what am I going to be measuring there? I'm not measuring increasing or decreasing anymore because when $f'' > 0$, I say that my function is convex. When $f'' < 0$, I say that my function is concave.
- 6.32** I'm not determining whether or not my function is increasing or decreasing. I'm determining the shape of it. Do I look like this or do I look like this? Am I convex or am I

concave associated to that? The second derivative is going to help classify what type of shape I have—am I facing downward or am I facing upward?

- 6.33** I want to illustrate this with a toy example to some extent. A little toy example. This is what's going to help get us to our third way of determining where a function is equal to zero. This one is a very big buzzword that people throw around all over the place. You look at headlines and all of this, you look at "Nature" magazine or whatever, big scientific science publications and things like that. You look at these big journal articles and you'll see things use this word, this phrase, all the time. They refer to it. It's called gradient descent. It's this big buzzword that's talking about optimization and machine learning and all that stuff. What is it actually saying? What does it actually mean?
- 6.34** You have people who will I say things like, "I need an advanced degree in mathematics to actually understand gradient descent. I need to understand this." It's like, "No, if you've ever gone hiking, you understand gradient descent very well." Because if I want to get off of the mountain, what direction do I go in? I go downhill. That's literally what gradient descent is. You don't need an advanced degree to get through that.
- 6.35** I'll go through gradient descent later regarding that. But that's going to be our third method for trying to find points at which the derivative is equal to zero. We're trying to find valleys or we're trying to find peaks. Now, gradient descent is referring to finding valleys because you're looking for minimizers. That's because in machine learning and optimization, you're typically looking at a loss function for the model that you're considering, and you want to minimize the error associated to it. So you want to find a valley associated to it rather than a maximum. You don't want to maximize the error of your model. That's a pretty terrible model. You want minimize the error of your model associated to that.
- 6.36** Let's run through a little toy example where I'm going to ask you the following questions. I'm going to give you the function, and then I'm going to say, "I want to know where it's increasing or decreasing. I want to know any of the local behavior"—this is a local maximum, this is a local minimum. In a small window around $x = c$, I have a valley. In a small window around $x = a$, I have a peak. It's a local type behavior. Then I want to understand, "What does the shape of the graph look like? Is it convex or is it concave? What can I say about that?" Let's run through an example to tie all of this terminology and vocabulary together for optimization. Let's walk through that and see what this looks like.
- 6.37** Now, don't get me wrong—gradient descent is complicated, and it can be difficult and intimidating and all of that. But I don't think it's appropriate to say you need an advanced degree or you need advanced knowledge or things like that to actually get at it. I think you just need to teach the principles correctly and concisely to cut through everything to actually get to what you need. That's what we're trying to do. That's what we're trying to do.

- 6.38** So here we go. Here's my example: Find where the function $f(x) = x^4 - 4x^3$ is: increasing, decreasing, convex, concave, and any local behavior associated to it. I want you to find the argmaxes, the argmins—where do the local maxes and the local mins occur? I want to know where they occur, because that's my optimization strategy at the end of the day. If this function, for example— this wouldn't be a good example of a function that occurs in real life—but if this function were representing the revenue or the profit of my business or things like that, and x is representing whatever quantity that I'm selling, whether it's, I don't know, computers or watches, books, or anything like that— whatever x is, the product that you're selling.
- 6.39** I would like to know, number one, "How many of x should I sell to maximize my profit? How many of x should I sell to minimize my cost, so to speak?" Finding the argmaxes and the argmins is extremely important. Finding where they occur is important. So I'm going to say any argmaxes, argmins, and things like that. Basically, I want you to take this function and I want you to dissect it, everything that you can tell me about it. What's amazing is that the derivative tells me so much about it. The second derivative tells me even more.
- 6.40** Once we get there, I'll be able to give you what's called the second derivative test, which allows you to classify your maximizers or your minimizers. Sometimes they're called your optimizers or your extremizers, sometimes, to classify them as argmaxes or argmins. Let's get into this. This is a step by step. You can take this as an algorithm to some extent as well. Every problem, you can run this sequence of steps, and it will give you the information that you're looking for.
- 6.41** Step one is to find the derivative because after all, what am I trying to do? Let's step back for a second and say, "What am I actually trying to do?" I'm trying to figure out where am I increasing and where am I decreasing at the end of the day. I have to step back and say, "Do I have a tool that tells me how I increase or decrease?" The answer is, "Yes, the first derivative tells me how to do that." Let's look at that.
- 6.42** Let's take $f'(x)$. I step back and I say: my function $f(x) = x^4 - 4x^3$. It's a difference of functions so I can differentiate each one individually, and each one is simply a power rule associated to it. I bring the exponent down and I decrease the exponent by 1. This is going to be $f'(x) = 4x^3$ — what am I going to get here? The 3 comes down, it multiplies the 4, so that's going to give me a 12, and then I decrease the exponent by 1—so x^2 . So we have that. That's fine.
- 6.43** Now, in order to get my sign line, I want to understand where my derivative is positive and where it is negative. So think back to how we classified real numbers. We said a real number is classified by being either positive, zero, or negative. If I want to know where am I positive, I may as well find out where am I zero first. Think about that. You have three categories.

- 6.44 Think about it in another context, for example. You want to categorize the color of, let's say, a box full of ping pong balls or something like that. The ping pong balls are three colors: red, green, and blue, something along those lines. Those are the only possible colors they can be. You want to say, "I want to know when are they red or blue? When are they red or blue?" So what do you do? You find all the green ones first, you take all the green ones out, and then everything else has to be either red or blue. Do you agree?
- 6.45 If I'm trying to figure out where my derivative is positive or negative, what am I going to do? It's either positive, negative, or zero. So I'm going to find out where is it zero. I'm going to take those out, and then everything else has to be positive or negative. Does that make sense, how that works?
- 6.46 So I set this equal to zero. That might not be easy—we've seen that multiple times. I might need to run Newton's method associated to that. I may need to run the bisection method in order to find these things, potentially. But in this case, I can simply algebraically do this. This one has a 4 and an x^2 . This one has a 4 and an x^2 . What do I need to get my x^3 ? I need $1x$. What do I need to get a $-12x^2$? I need a -3 , and so that's equal to zero associated to that.
- 6.47 Now, that's equal to zero where? I have the product of two real numbers is equal to zero, and one of the properties of the real number says that that means that either this is 0, so $x = 3$, or this $x = 0$. So $x = 3$, or this is zero, so $x = 0$. This is saying that $x = 0$ or $x = 3$. If I wanted to give a name to that, that's where my derivative is equal to zero so what do I call them? Those are my critical points. These are critical points. $x = 0$ and $x = 3$ are the critical points associated to the function $f(x) = x^4 - 4x^3$. Notice that they're both in the domain. I can take this function and evaluate it at zero. I can take this function and evaluate it at three. They're in the domain of the function. They're valid inputs that you're allowed to put in.
- 6.48 I know where the derivative is equal to zero. Now, what is my sign line in this in this case? Here's f' . I have $x = 0$ —that's where my derivative vanished at zero—and $x = 3$ —that's where my derivative vanishes. Now, to figure out when $f' > 0$ or $f' < 0$, what you do is you simply test points to the left or to the right.
- 6.49 If I choose a point to the left of $x = 0$, a great choice would be -1 , for example. If I take $x = -1$ and I plug it into my derivative, what do I get? I can do it off to the side: I can say $f'(-1) = -$ if I look at it from here, what am I going to get? I'm going to get $f'(-1) = 4 * (-1)^2 * (-1 - 3)$. Do I care what the value is? No, I just care if I'm positive or negative. That's all I'm trying to figure out in the first place. Do I really need to write this out? Not really. Because what's $(-1)^2$? $(-1)^2 = 1$. So $4 * 1 > 0$, so this is a positive number. I'm going to multiply that by $(-1 - 3)$, which is -4 —that's negative. So I'm getting positive times negative. If I take a positive number times a negative number, am I positive or am I negative? Negative in that case. So this is going to be a negative here.

- 6.50** You might argue, you might say, "Robbie, you just plugged in $x = -1$. What about $x = -2$? It doesn't matter. It doesn't matter because real numbers are classified as positive, negative, or zero, and I took away the ones that are zero, and so my only possibilities are positive or negative. If I had another one where I was positive over here—let's say I took $x = -3$ and I got positive, but at $x = -1$, I get negative. That means I would have had to increase and then decrease, which means there'd have to be another point over here at which you were zero or don't exist. But I already found all of those. You can choose any point to the left of zero. It doesn't matter: $x = -1$, $x = -5$, $x = -50$ —I don't care.
- 6.51** You could choose any point in between zero and three. Say $x = 1$. I'm not going to evaluate the function at the function $f'(1)$. I'm simply going to come up here and say, "If I plug in $x = 1$, then $1^2 > 0$, so this quantity is positive. If I plug in $x = 1$ here, $1 - 3 < 0$, so I'm getting positive times negative. I'm negative. I'm negative here. What about x is equal to something bigger than three? Say $x = 4$. This is $4 * 4^2$ —that's positive. $4 - 3$ is also positive. Positive times positive is positive. I'm done. You don't have to care about what the values are. You just care about if they're positive or negative. That's it.
- 6.52** So that's my sign line. That's my sign line for f' . What is that telling me? When $f' < 0$ —I have to go back to my terminology—when $f' < 0$, I'm decreasing. When $f' > 0$, I'm increasing. This is saying that my function is decreasing here, temporarily horizontal—zero—decreasing here, temporarily horizontal—zero— and increasing here. What parts of this question can I answer? They wanted to know, "Where am I increasing? Where am I decreasing? Where am I convex? Where am I concave? What are my argmaxes, my argmins—all of that stuff."
- 6.53** This line alone tells me where I'm increasing—I'm increasing from $x = 3$ to ∞ . It tells me where I'm decreasing—it tells me I'm decreasing from $-\infty$ to $x = 0$. I'm temporarily stuck here, and then from $x = 0$ to $x = 3$. So I've answered increasing and decreasing already simply by this one visual tool. This is incredibly powerful. Moreover, it's telling me that I'm going down here and then up here. It's telling me that $x = 3$ is an argmin. I'm going downhill, then uphill, so it's a valley. I now have argmin. Do I have any argmaxes? If I had an argmax, then I would have to go positive and then negative. The only place where I'm going positive is to the right of $x = 3$. I'm never positive over here, so I don't have any argmaxes associated to it.
- 6.54** The only questions that I have left to answer are convex or concave. But those deal with my second derivative. I find my second derivative next. My second derivative. I think it's miraculous, actually, that all of this information is literally encapsulated in this one visual diagram. I think that's powerful to some extent. This very simple thing is just a line with some values on it and pluses and minuses. That's it. And it somehow tells me almost everything I need to know about my function. It's pretty amazing.
- 6.55** Now, computing your second derivative—here's the strategy. When evaluating values of f' , it's better when it's factored out because I can simply—think about it. When I was

evaluating here—when $x = 1$ —all I was taking advantage of was positive times negative is negative. That was it. If I took advantage of this way of writing f' , I actually need to know what the value is. Because I can't say anything about positive times negative because this is a subtraction. This isn't a multiplication.

- 6.56** If I plug in $x = 1$, I get $f'(1) = 4 - 12 = -8$. It's negative. It's better to evaluate f' in factored form when looking at its positives and negatives, but you do not differentiate it when it's in factored form because then you have to say, "I have to use the product rule. It's a product of two functions—I have to multiply." Go back to this. Differentiating this is far easier any day of the week. So I'm going to look at this form of f' . What do I get? $f'(x) = 4 * x^3$ —the 3 comes down, I get a 12, and I decrease the exponent by 1. The 2 comes down, and I decrease the exponent by 1. I get that. I want to figure out where is the second derivative positive or negative. What do I do? I take away all the zeros. It's the ping-pong ball analogy again. If I want to know what ping-pong balls are red and blue, I take out all the green ones. Then what's left is the red or blue.
- 6.57** What can I take out? Each one has a 12 and an x , so I can take out a $12x$. What do I need to get an x^2 ? I need a $1x$ here. What do I need to get a $-24x$? I don't need any more x s, but I need a 2, and I can set that equal to zero. When is that equal to zero? These are real numbers after all. The product of two real numbers is zero precisely when one of them is zero. That means that $x = 0$ or $x = 2$ in this case.
- 6.58** Now, there's a very subtle difference here. Critical points are where my first derivative — my first derivative is either zero or does not exist. I call them critical points. I can't call these things critical points for the original function f , because they come from the second derivative, not the first. $x = 0$ and $x = 2$ —they're coming as zeros of the first derivative. So $x = 0$ and $x = 2$ are critical points of f' , but not f . There's a very subtle difference here. There's a very subtle difference here.
- 6.59** If you're anything like me, you're looking at this saying, "Wait, wait, wait, wait—you're saying that $x = 0$ is not a critical point." I'm saying technically $x = 0$ is a critical point because it's where $f'(x) = 0$. The fact that the second derivative is zero at $x = 0$ is a fluke to some extent. So because this $x = 0$ and this $x = 0$ match—only because this $x = 0$ satisfies $f'(x) = 0$ —can I designate it as a critical point? The fact that it coincides that $f''(0) = 0$ is a fluke. But because $x = 0$ and $x = 2$ make my second derivative equal to zero, it doesn't mean I'm allowed to call them critical points of the original f . I can only call them critical points of the f' function. There's some subtleties here that are a little tough to flush out. But at the end of the day, we now know that $f''(x) = 0$ at $x = 0$ and $x = 2$.
- 6.60** So let's do a sign line, but let's do it for f'' now. Let's do it for f'' . I draw my line. I have f'' . I know that at $x = 0$, I'm here. Let's see, if that's three, then let's say here's $x = 2$. My second derivative vanishes at those two points. My second derivative—notice that my second derivative is defined for every other x value. If I had something where I was dividing by x or something along those lines or dividing by $(x - 1)$ or something, then I

would have to worry about where my second derivative doesn't exist either, just like I did up here. But in this case, this function—the first and second derivatives—are always defined.

- 6.61** Now, let's do the same thing. Let's choose a point to the left of zero, so $x = -1$. If I plug it into the second derivative, I want to plug it into the factored form so I can take advantage of this positive-negative product. If I plug in -1 , $12 * (-1)$ is negative— $(-1 - 2)$ is negative. This is negative times negative. If I take a negative number times a negative number, I get a positive number. So I'm going to put a plus here.
- 6.62** In between $x = 0$ and $x = 2$, let's choose $x = 1$. If I choose $x = 1$ and I plug it in, $12 * 1$ is positive, $(1 - 2)$ is negative. So this is positive times negative. That's negative. Very good. If $x = 3$, $12 * 3$ is positive, $(3 - 2)$ is positive. So positive, positive, I get positive again. Perfect. Now, I'm looking at the second derivative, so I'm not going to put increasing or decreasing. I'm looking at the second derivative. I'm looking at convex or concave. I'm looking at those two things. Remember, where the $f'' > 0$, I am convex, smiley face. Where I'm positive, I'm convex, convex. Where I'm down, so $f'' < 0$, I am concave, frowny face.
- 6.63** Now, can I answer the remaining parts of the question—"Am I convex? Am I concave? Can I find anything about argmaxes?" We still know there are no argmaxes associated to this. But now I can answer every single one of these questions. Once I have all of this data, I can say, number one, I am increasing, so INC—INC is I'm increasing. Where am I increasing? I go to the first derivative and I say, "I'm increasing from, not here, not here, from $x = 3$ to ∞ . I'm going to write it in interval notation. I'm going to say $(3, \infty)$. That's where I'm increasing.
- 6.64** Number two, where am I decreasing? So DEC is going to represent decreasing. Well, I'm decreasing from as far left as I can imagine— $(-\infty$ up to $0)$. I'm temporarily stuck at 0 , and then I go from 0 to 3 . I'm going to denote that as $(-\infty, 0) \cup (0, 3)$. This little $\cup$ just means this interval, and then include this one as well. From $-\infty$ to 0 and then 0 to 3 . That's where I'm decreasing.
- 6.65** Some people say, "If your derivative is equal to zero, you're just decreasing from $-\infty$ to 3 ." I don't agree with that because decreasing means my derivative is strictly negative and I'm zero at $x = 0$. I want to split the interval to say that I'm decreasing, decreasing—I'm actually sticking to the correct definition.
- 6.66** Okay, where am I convex? Convex is where I'm happy face. That's up, right? That's $f''(x) > 0$. I'm going to have—let's do convex. Where am I convex? From $-\infty$ to 0 and from 2 to ∞ . That's $(-\infty, 0) \cup (2, \infty)$. Where am I concave? Everywhere else, from 0 to 2 . So I'm concave from $(0, 2)$. What about argmaxes and argmins? I have no argmaxes. So five—no argmax. And six—where was my argmin? Argmins are my valleys. It's where my valleys occur. So that's where I'm going down and up. So that's going to be at $x = 3$. I have $\text{argminf}(x) = 3$.

- 6.67** All of that information now allows me, if I wanted to, I could graph this. If I wanted to, I could graph this. What is it telling me? My function is decreasing until I get to zero, then I have a horizontal tangent line where I'm temporarily stopped. I then continue to decrease. I then increase from then on out. What does that look like? I'm going to decrease— temporarily horizontal—continue to decrease, and then I increase. My behavior—my function has to look something like that. All simply from knowing what the first derivative is in that case.
- 6.68** Now, let's check that against the second derivative. I've got to be convex here, so I've got to be happy face up to the left of 0. To the left of 0—this looks like it's up. It looks like it's convex. Then it tells me in between 0 and 2, I need to be concave—that looks like down a little bit here. Then from $x = 2$ on, I need to be convex again. I need to be up. It looks like this is actually what the function $f(x) = x^4 - 4x^3$ looks like. Do we agree? Simply based on this alone. I don't need a graphing utility. I don't need anything like that. Now I know exactly what this guy looks like.
- 6.69** Now, do I know what the values are? No, I haven't plugged in $x = 0$, $x = 3$, $x = 2$, or $x = 0$. I haven't plugged those values in. I don't know how this is oriented on some axis or anything like that. I don't know if it's really far below the x-axis or high above the x-axis. I don't know, but that's fine. I at least know what its general shape looks like overall. This is incredibly powerful, and it allows you to evaluate minimums and maximums accordingly.
- 6.70** But again, It's a toy example to illustrate the principles that we're getting at. In the wild, when you're dealing with this in a function and you have 10,000 terms to this particular thing, you're not going to take the derivative by hand, number one. You're going to plug it into a machine. Trying to figure out where it's equal to zero is a futile effort. You're not going to get anywhere with that. If you have 10,000 terms and you take its derivative, goodness knows how many terms you're going to have, and you're not going to be able to set that equal to zero.
- 6.71** You're going to have to invoke Newton's method, or bisection method, or something along those lines—or gradient descent, which we'll talk about later—to try and figure out where I actually equal zero. But once I know where I'm equal to zero, I can take advantage of writing out a sign line for both f' and f'' to determine the behavior of the function. Do I look like this? Am I convex? Do I look like this? Am I concave? Things like that. And classify my local behavior associated to that. Does that make sense overall, how that's working?
- 6.72** Now, one thing that we didn't take into consideration is this function. I'm assuming that this function is defined for all x values. It is. It's a polynomial. I can take any x value whatsoever, and it's valid. Any real number whatsoever, I can plug in, and it's completely valid. But if this is modeling a particular real-world problem—if x is the number of things that I produce from a business standpoint: x is the number of books that I sell, x is the number of computers that I sell, x is the number of whatever— then

it doesn't make sense that x is negative. It doesn't make sense that x could be infinite because I'm not going to sell an infinite number of books, things like that.

- 6.73** There has to be constraints associated to it. Do we agree? There has to be some constraints. I can't have infinite cost. I can't have infinite revenue. I can't have infinite things like that. I have to have constraints associated to it. How do I handle this? The question that I want to throw into this is—let's look.
- 6.74** We found a minimum, a minimizer, an $\operatorname{argmin} f(x) = 3$. Notice that that is literally the bottom most part of my curve of my function. That is literally my smallest value that my function can attain—when $x = 3$. So whatever $f(3)$ is, that's the smallest value that f can attain. So even though this is really only getting at local behavior—it's saying it's a local valley—it's technically a global thing, too. It is literally the smallest that my function can attain over the entirety of the domain. It's the smallest value this function can attain. That's a global type behavior.
- 6.75** To illustrate local versus global, think about it like this. If you have a business, for example, at the end of each quarter, you have to talk about how your business performed overall. Every three months, you have to talk about how your business performed. That's a local phenomenon: how did my business perform in January, February, and March?
- 6.76** But if my business has been open for 20 years, how my business performed in January, February, and March of this year, that's a local type behavior: what's my local minimum? What's my local maximum associated to that? Whereas if I'm considering the entire lifespan of the company, that's a global behavior. If I'm including the entire domain of the function: what is the largest revenue that I've ever attained in my business ever? What's the lowest revenue that I've attained in my business ever?—that's a global behavior. That's including the entirety of the domain.
- 6.77** Now, local and global can coincide, and this is an example of one. This is saying that locally, I have a minimum. But it just so turns out that that's also the global minimum. It is the lowest possible value that my function has ever attained across its entire domain.
- 6.78** Let's talk about constraints. I'm going to give you the same function, but now I'm going to tell you: what if I want to constrain it? What if I want to constrain it in a closed interval? Actually, before I do that, I want to mention the following thing. I want to mention how the second derivative—I want to give you the second derivative test. This is called the second derivative test, and this is so important. Second derivative test. It says the following.
- 6.79** If I hand you a critical point of the function—so someplace where the derivative is equal to zero—I hand you a critical point. I say $x = 4$. So I know that, let's say, my derivative vanishes there. $f'(4) = 0$. I want to understand: is $x = 4$? Is it a minimizer or is it a maximizer? If $f'(a) = 0$ —so a is a critical point of the function—if a is in my domain and $f''(a) > 0$, what can I say? Let's interpret this.

- 6.80** If $x = a$ is where my derivative is equal to zero—so it's a critical point if it's in my domain — and my second derivative is positive, that means I'm convex. That means I'm convex. If I have a horizontal line— $f' = 0$, horizontal tangent line—and my function looks like this, am I a local min or am I a local max? I have to be a local min because I'm concave. Then, $a = \operatorname{argmin}f(x)$. It has to be a minimizer. Again, if I look like this at the bottom—so $x = a$ is where my hands come together, $x = a$ —I have to be a minimizer. I have to be a minimizer.
- 6.81** Conversely, if f' —let's choose another letter— $f'(b) = 0$, and $f''(b) < 0$, let's think about that for a second. $f'(b) = 0$. It means I have a horizontal tangent line. I have zero slope there. It's a critical point, $f'(b)$. Now $f''(b) < 0$. That means I'm concave. I look like this. I've got a horizontal line, and I look like this at that point. What do I have to be? I have to be a maximizer. So then $b = \operatorname{argmax}f(x)$. It's a maximizer of $f(x)$.
- 6.82** Now, what can I say if $f'' = 0$ at the same point? I don't know. It doesn't tell me anything. If $f'' = 0$, I don't know if I look like this. I don't know if I look like this. I don't know if I look like this, where I come down and then I continue going down. This was an example where f' and f'' were both zero at $x = 0$. And what did it tell me? It told me absolutely nothing. It tells me this isn't a valley. Because if I'm approaching from the left-hand side, it's like a valley, I'm going downhill. But if I'm approaching from the right-hand side, I'm going uphill. What the heck is it?
- 6.83** It's what's called a saddle point. It's what's called the saddle point. Now, it's hard to see it in this sense, but a saddle point is effectively the following. A saddle point is this. Let me give you more of a visual for this. It's better if you have a function with multiple variables. But a saddle point is basically the following. It looks like this. That right there is a saddle point associated to it.
- 6.84** Why? Why can I not classify this? Well, if I start trying to approach this point by walking this way or walking from the backside of this—so you think about this as a three-dimensional object. It's like a piece of paper that I've bent this way and then folded in on the corners a little bit—if I approach this point along this, what am I really getting? That looks like a maximum to me. I'm going up and then down. Do we agree? However, if I approach it this way, if I approach this way and then this way, now it's a minimum because I'm going down and then up the other side. Does that make sense?
- 6.85** It's directionally dependent, classifying this as a minimizer or a maximizer. It's called a saddle. It's called a saddle point. That's exactly what this is doing over here in one variable. Here I'm decreasing, so I would think, "Oh, it's a minimizer." But here I'm increasing, so I'd say, "Oh, it's a maximizer." No, it's neither. It's neither. It's what's called the saddle point. The second derivative allows me to diagnose that.
- 6.86** What if I change the problem and I say: this function, I'm now going to restrict it. I'm going to restrict it. I'm going to say I'm only allowed to consider x -values in the interval, let's say, $[1, 4]$. I'm only allowed to consider x values in between 1 and 4. If I restrict

this, what happens now? Restrict f to the new domain, which is $[1, 4]$. Again, hypothetically, let's suppose that $f(x)$ is telling me x is the number of units that I sell, whatever my business is. Maybe in a particular week, month, day, year, whatever, I only sell between one and four things, whatever that is. I want to understand how is my business being governed in that particular interval, if I'm only allowed to sell one to four things? Does that make sense, what I'm talking about? Okay.

- 6.87** If I restrict to my domain, how do I determine what my maximum and minimum is? Number one, you do the exact same analysis we just did: you find out where is the derivative equal to zero? Where is the second derivative equal to zero? You look at the sign lines associated to those things. But now you restrict it to only involve $[1, 4]$.
- 6.88** What were my critical points for my first derivative being zero? That was $x = 0$ and $x = 3$. If I'm not allowed to consider anything outside of this interval, then I have to throw away $x = 0$. I'm not allowed to consider it because it's not even in my interval anymore. I'm only allowed to consider anything that lives in between 1 and 4. So $x = 3$ is my only critical point because I'm restricted. Does that make sense? Similarly here, these were the values at which my second derivative was equal to zero. So I'm going to throw away this one because I'm not allowed to consider $x = 0$ in there. I'm only allowed to consider $x = 2$.
- 6.89** Then when I want to determine what are my global max and mins, all I do is the following. I go and I say, "Where was my derivative equal to zero?" My derivative was equal to zero at $x = 0$ and $x = 3$. I'm not allowed to include 0, so I throw that one out. So $x = 3$. Does $x = 3$ live in my interval? Yes, it does. So I consider $x = 3$ as a value. Now, because I'm on my endpoints as well, I have a closed interval $[1, 4]$, I also include my endpoints. I include $x = 1$ and $x = 4$ as potential candidates that I have to evaluate my function at to find my global max or global min from that behavior.
- 6.90** Why is that the case? Think about it. If I have a function that only exists— let's say it looks like this—and it only exists in between 1 and 4. I'm not saying that this is the function here. I'm just drawing a squivel to try and get this principle through. I'm not allowed to know what happens to the left at $x = 1$. So I can't say whether or not 1 is a minimum because I don't know if my function does something like this. And I'm not allowed to say that it's not a minimum because I don't know if my function continues down. I don't know. I'm not allowed to know what's to the left of $x = 1$. So I can't say anything about it.
- 6.91** Similarly for $x = 4$. Is this a maximum because my function goes like this afterwards? Or is it nothing because my function goes up like this? I don't know. All I can do is I can say, "I can evaluate the original function at $x = 1, 4$, and 3, and that's going to give me three values that I can compare. Well, three I can compare. One's going to be the biggest, one's going to be the smallest, and then I'm done."

6.92 If I take what is $f(1)$ versus $f(3)$ versus $f(4)$, and I plug these in—if I plug in 1 to here, I get what? $f(1) = 1^4 - 4 * 1^3 = 1 - 4 = -3$. If I plug in 3, I get big numbers. So $f(3) = 3^4 - 4 * 3^3 = 81 - 4 * 27 = 81 - 108 = -27$. Got to keep me honest. I'm doing this on the fly. I think that's right. Then $f(4)$, if I plug 4 in, this one's beautiful. I get $f(4) = 4^4 - 4 * 4^3 = 4^4 - 4^4 = 0$. One of these is the smallest: -27 —that's my global min. One of these is the largest: 0 —that's my global max.

6.93 When you are on a constrained interval like this, a closed interval, and your function is continuous, you are always guaranteed a global max and a global min. The way that you do it is you find your critical points by setting the derivative equal to zero, and then you check your endpoints as well and compare those values. That's a finite set. One of them is the biggest, one of them is the smallest. You might have multiple of them are the biggest, and multiple of them are the smallest. They're the same values, but you're guaranteed that those exist.

Lecture 7: Gradient Descent

At a glance — Gradient descent — the third algorithm of the course.

- 7.1 All right, so today we get our third algorithm, which is gradient descent. Gradient descent is awesome because you see it a lot in the headlines when people are talking about machine learning algorithms and all this, they're like, "Gradient descent, gradient descent, gradient descent"—stochastic gradient descent, which is the industry standard at this point.
- 7.2 But what actually is it? Do I need some advanced computer science or math degree to actually understand it? Or can I actually cut through everything and actually understand it from a fundamental level in a single variable course? The answer is yes, you can. That's what we're going to do today. We're going to talk about gradient descent and how I can use it for finding where my derivative vanishes. Remember where the derivative is equal to zero is where we were finding our critical points, and that was giving us our peaks and our valleys associated to that. That's where $f' = 0$. We're going to use gradient descent today to talk about how can I actually find where $f' = 0$. Then we're going to work out a bunch of examples. Today is all about gradient descent.
- 7.3 What's the underlying principle with all of this? The question that we want to ask ourselves is, "What does f' tell us about values of f ?" We saw this last time when we were talking about if $f' > 0$, then my function is increasing, and if $f' < 0$, then my function is decreasing. But what's interesting about this is that f' , if I have a single variable—so here's some answers to this question.
- 7.4 We know that $f' > 0$ means I'm increasing. We know that $f' < 0$ means I'm decreasing. Let's look at a quick example. "What direction should I go to actually increase or decrease?" Knowing that $f' > 0$ tells me my function is increasing, but it doesn't tell me which direction I should travel to continue increasing. It just says you're increasing. But how do I know what direction I should go in in this case?
- 7.5 Let me illustrate this with a quick example. By the way, for single variable functions, f' is equal to your gradient. I've been hiding it this entire time, but we've been talking about the gradient the entire time. $f' = \nabla f$. If it's a single variable function—if f contains more variables like x, y, z , however many variables you want—the gradient is a different animal. It doesn't make sense to say what f' is anymore. The gradient is a little bit different. But for single variable functions, which is everything that we've been talking about, $f' = \nabla f$.
- 7.6 Here's an example. Let's start off very simple. If I take $f(x) = x^2$. I can go to any graphing utility, figure out what that looks like, or if I know what it looks like, I could graph it out and see what happens. It's a parabola—I'm going to do my best here. It looks like this, roughly. $y = x^2$. I can choose an x value and evaluate my derivative at that point. We know how to compute the derivative. The derivative is given by $f'(x) = 2x$. That's our

power rule. The exponent comes down. We decrease the exponent by 1, so that gives me a $2 * x^1 = 2x$.

- 7.7** I can say, "Let's choose a point." Let's say $x = 1$. I'm going to take $f'(1) = 2 * 1 = 2$. But notice that that's positive. That's meaning that my function is increasing. Okay, great. At $x = 1$, I look at the slope. That slope is equal to 2, and it's positive. It's going from bottom left to top right. What do I notice? It's positive. So that's going to tell me if you move in the direction of the gradient—the gradient here is $f' = 2$, positive meaning I move to the right, negative meaning I move to the left—if I move in the direction of the gradient, it's going to say that my function is increasing in that case. Does that make sense?
- 7.8** If I say 2 is positive—at this point, it says: if I move in that direction, my function should increase. Because $f'(1) = 2$, that's positive. The positive direction we say is moving to the right, it's saying my function is going to increase. That's the underlying principle. The direction of greatest increase—direction of greatest increase at $x = 1$ —is $f'(1)$. It's in the gradient, i.e. the gradient.
- 7.9** That's the biggest takeaway: if I want to go in the direction of greatest increase, I go in the direction of the gradient. Therefore, if I want to decrease my values, I want to find a valley, I want to find the minimizers, the argmins, what do I need to do? Well, if going in the direction of the gradient goes in the direction of greatest increase, then going in the direction of minus the gradient should be the direction of greatest decrease. That makes sense.
- 7.10** If I'm starting here at $x = 1$, the gradient is 2—so it says go to the right, which means if I negate the gradient, which is -2, it's saying go in the left direction. What I notice is that if I start moving in the left direction, the function values decrease. That's the underlying principle: move in the $-f'(1)$ direction to decrease values of f . After all, it's gradient descent, not gradient ascent. We're not trying to find peaks, we're trying to find valleys. We're trying to descend to the bottom of the valley. That's what we're trying to do.
- 7.11** That's all fine and great. Now we know that we're going to move in the direction of $-\nabla f$ or $-f'$. That's the idea. By the way, that generalizes to functions of a thousand variables. You still go in the direction of minus the gradient every single time. The same principles apply for multiple variables. But I need to understand, "How do I write where I'm located once I start moving?"
- 7.12** Let me reiterate that. If I start standing at $x = 1$, and I say I want to move to the right, and I want to move to the right at the rate of the gradient, or I want to move to the left at the rate of minus the gradient—how do I write where I'm located? What is my position on the x -axis at any given time? How do I do that? Let's see how we would do something like that.
- 7.13** Effectively, what I'm asking you, mathematically, is I'm saying: give me a function which moves at the rate of whatever I tell you, either the gradient or minus the gradient. It

doesn't matter. It doesn't even have to be a gradient. It could just be abstractly, move at the rate of 5 whatever units per time or move in the direction of -5 units per time. How do you actually give me a function which represents my position on that line?

- 7.14** Here's how you do it. It's very simple, actually. You start at a point, whatever that is. Let's call that $x = a$. So $x = a$ is fixed. Now, if I want to move to the right at the rate of whatever. Let's say I want to move to the right. To move to the right at rate $b > 0$ in this case because I'm directionally saying: if I move to the right, that's positive; if I move to the left, that's negative. If I want to move to the right at the rate of $b > 0$, what is my position out here? Give me my position right there. That's the game. That's what I'm saying.
- 7.15** Here's how you do it. Any point along this line can be written in the following way. I want you to think about this intuitively rather than just memorizing a formula. I want you to think about this intuitively. For example, if I say, "I want you to start at the point a ." What do I do? I walk over to the point a . Great. Then I give you further information. I say, "I want you to either move to your right or to your left at a certain rate." I say, "I want you to move to the right at the rate of b ." Great. What do I do? I turn myself to the right. I'm going to move at the rate of b , and then I start walking that direction. Do we agree? That's literally all I'm doing. Or if I wanted to move to the left, I would say, "Okay, I move to the left," and then I simply go in that direction as well.
- 7.16** At the end of the day, that's all we're asking. What am I going to do? I'm going to say that my function— let's call the function, $P(t)$ for position at time t . It's going to be, "Start at a ." And if I think of t as a time parameter— I imagine having a stopwatch or something like that—and I say, "I start at a " and I click the stopwatch and now I start moving, whatever the direction may be. Then I would say, "My rate is going to be b ."
- 7.17** Look at the units on this thing: a would be a position. Let's say a is given by meters or something along those lines. That's my location. My rate with respect to time is going to be meters per second or meters per minute or meters per hour or whatever the case may be. If I take meters per second and I multiply by a time variable of seconds, the second units cancel and I'm left with meters. This is simple dimensional analysis from any chemistry class or something like that. If I have two quantities, this one is meters per second and this one is seconds, the units cancel and I'm left with meters in total. Does that make sense?
- 7.18** If my rate is b meters per second, and I multiply that by t seconds, I'm getting b meters when I take their product, or $b * t$ meters when I take their product— $P(t) = a + bt$. But I need $t \geq 0$. Why do I need $t \geq 0$? Think about that. If $t < 0$, if $t = -1$ or $t = -2$, which direction am I moving? I'm going $a - b$ —I'm going that way. If I want to go to the right, I need $t \geq 0$. That's a key concept for gradient descent.
- 7.19** In higher dimensions, it's going to say the following: start at some point in, say, three-dimensional space, whatever that is. Let's say this is my point in three-dimensional

space. Then it's going to tell you: go in the direction of $-\nabla f$. Now you're looking all over three-dimensional space and you say, "Okay, the gradient is that direction." I point my body in that direction, and I multiply by t , and I walk that way. But it's the exact same formula, except a and b now would simply be vectors. They have multiple components instead of just being scalars, real numbers. But it's the exact same underlying principle. That's what we have.

- 7.20** Let's apply this to gradient descent. What do we want to do? Gradient descent says the following thing: my line of motion will be what? It'll be $P(t)$, again. I start at my point. The point was $x = 1$, going back to our previous example. I start at $x = 1$, and I add on to that my gradient: $f'(1) * t$, where $t \geq 0$. This would be my position if I'm moving in the direction of the gradient, which means that I'm increasing, though. Remember, we want to go in the direction of $-\nabla f$ because we want to decrease. How do I modify that? I don't change t . t is ≥ 0 . I change the plus sign to say I go in $-\nabla f$. So this will be my position on the x -axis, if you will, moving in the direction of minus my gradient, $-f'(1)$. Therefore, the values of f are decreasing.
- 7.21** Now, let me give you the method of gradient descent. Very much so like Newton's method, or the bisection method, this is an iterative method. Meaning, think back to Newton's method—what did we do? We chose an initial point, I plugged it into a formula, and that gave me a new point x_1 . I take that point, I plug it into the formula, it gave me a new point x_2 . The hope was that $x_0, x_1, x_2, x_3, x_4, x_5$ converged to the thing that I was interested in. That's what we were looking at.
- 7.22** The method of gradient descent is no different. You start with an initial guess, x_0 . That's your initial guess. Then you're going to generate new iterates, let's call them—new iterates by the following recursive formula. You generate new iterates as follows. This is going to be a little complicated, but bear with me. The formula is going to say: my next one, x_{n+1} — is equal to what? This is equal to my previous iterate, x_n . Then I want to move in the direction of $-\nabla f$.
- 7.23** What am I going to do? I'm going to take $-\alpha_n * f'(x_n)$, where $\alpha_n > 0$. It's playing the role of this t , like here. I take my previous iterate, I move in the direction of minus the gradient times some positive parameter, potentially zero, but more often than not, it's not. So $\alpha_n \geq 0$. More often than not, it's positive.
- 7.24** But now I have to tell you where α_n is coming from. Here's the trick. Where α_n is equal to the following thing. Here's what I'm going to do. Let's go back to this example of x^2 . I have this, I have that. Over here at $x = 1$, we found that the $f'(1) = 2$. It says I need to move in the direction minus at the rate of -2 . I start at 1 , and I move in the backwards direction. If I move in that particular direction, which is minus the gradient, I'm looking at this thing as being: start at 1 and move at $-2 * t$. Do we agree? That's going to govern my position along the x -axis.

- 7.25 Then here's what happens. If my position at any point along the x-axis is given by this particular formula where $t \geq 0$, what I'm going to do is I'm going to take that position and I'm going to plug it into the function itself. If I am standing at this point here and I plug it into the function, I get this point. If I'm standing here, I get this point. If I'm standing here, I get this point, here, this point, and so on. It's going to show me where I am on my function. But only the values of the function on this particular line. That's all it's giving me.
- 7.26 It's giving me the values of my function when I'm on that line: $1 - 2t$. I'm moving in the direction of minus the gradient. What I'm going to do is I'm going to say the following. I'm going to say: take the function and evaluate it on this specific line. I'm starting at x_0 or x_n , and I'm moving in the direction of minus the gradient. That's all I'm doing. I'm going to say: evaluate $f(x_n - \alpha \cdot f'(x_n))$, where $\alpha \geq 0$, and minimize that function with respect to the variable α . Here's how we denote that: $\operatorname{argmin}_{\{\alpha \geq 0\}} f(x_n - \alpha \cdot f'(x_n))$.
- 7.27 What am I actually saying? Mathematically, this is very heavy. The notation is heavy, the ideas are very heavy. But let's get the intuition behind it. The intuition is saying the following: start at x_n . Ask yourself: what is the gradient? The gradient is $f'(x_n)$. Then turn yourself in the opposite direction, $-f'(x_n)$ —the negative gradient. I'm going to decrease. I want to say: how far do I need to go along that path? How far do I need to move from 1? How far do I need to move until I get to the bottom? How far I need to go? What is that really saying?
- 7.28 It's saying: along this line, which is given by the $1 - 2 \cdot t$ —it's saying there's a specific t value which will give me the bottom trough. Do we agree? There's a certain value of t which will give me the bottom of the trough. In this case, if $t = \frac{1}{2}$, I get $1 - 2 \cdot \frac{1}{2} = 1 - 1 = 0$. That would tell me, right there at the bottom, $x = 0$. That's where it is.
- 7.29 But how do I actually find that? What do I do? I take my function, I evaluate it on the line—so I'm getting these dots. I'm getting those dots all the way around, and I say: there has to be a parameter α which minimizes that quantity, which finds the bottom of the function. What do I do? I take the minimum, the minimizer. I take the argmin over all $\alpha \geq 0$. That is the specific α_n that you're looking for. Then that gets funneled into this formula to compute your next iterate. It's complicated, but the idea is very simple. The idea is: start at a point, move in a certain direction and walk in that direction. Walk downhill until you feel like you've hit the bottom. Once you feel like you hit the bottom, start again and do the next thing.
- 7.30 That's all fine and great, but how do we do it in practice? Let's go through some examples. These examples—the first one's fairly nice; the next one's a little heavy. But we can do hard things. I'm going to keep this up, the method of gradient descent. Here's the example. Apply gradient descent to the function $f(x) = x^2 - 4x$ with $x_0 = 1$. Let's see what happens.

- 7.31** Here's what I need to do. Procedurally, this is like a scavenger hunt to some extent. Procedurally, they tell me what I need to do. I need to compute x_1 , and I have a formula for knowing what x_1 is equal to. They tell me what the formula is. From the formula, we know that x_1 is going to be equal to what? Remember, this is a recursive formula saying: if you want x_1 , you need to ask yourself the following question: what does n need to be in order to give me x_1 on the left-hand side? In this case, since this is x_{n+1} , I'm going to take $n = 0$. If $n = 0$, I get x_1 on the left. Fine. Then everywhere I see an n , I replace it with $n = 0$. I'm going to get $x_1 = x_0 - \alpha_0 f'(x_0)$. You're with me so far? We're just plugging into the formula at the end of the day. That's step one of the scavenger hunt.
- 7.32** The next thing says: what do I need to compute? What do I know and what do I not know? They gave me x_0 . They told me $x_0 = 1$. That's my starting guess. I know this and I know this—that's equal to 1. What I don't know is, I don't know f' yet, so I'm going to have to go and compute f' . But that's not a barrier because we actually know how to compute that. That's the power rule. We've done that quite a bit. So I don't know f' yet, but I can compute it. Then α_0 —that's the tricky bit. α_0 is like, "I need to run this particular procedure and figure out what's going on."
- 7.33** Step one, let's find f' . Let's find f' and see how that goes. So f' , just from basic principles—I come back up to my function, which is $f(x) = x^2 - 4x$, and I recognize that this is a difference of functions. The derivative can be computed term by term like we've done. The derivative of a difference is the difference of the derivatives. That's fine. The derivative of x^2 —power rule: the two comes down—so I'm going to get a $2 * x^1 = 2x$. Then the derivative of $-4x$ is -4 . I get $f'(x) = 2x - 4$. Great. What does the method tell me? You need to evaluate that at x_0 . So $f'(x_0)$ —I'm going to simply substitute in for my variable—so $2x_0 - 4$. But I know what x_0 is. They told me $x_0 = 1$. So this is $2 * 1 - 4 = 2 - 4 = -2$. That's telling me that's my gradient. Now, if I know what this thing looks like, I can confirm that the sign of the gradient in this case, which is -2 , either makes sense or it doesn't in this case. Let's see that. The graph of this thing looks like what? It goes through the origin and it goes through $x = 4$. If I plug in 4, I get $4^2 - 4^2 = 0$. So it goes through these two points: $(4, 0)$ and $(0, 0)$. It goes through those two. It's a quadratic. It's an x^2 . So I know it either looks like this—it's convex—or it looks like this—it's concave—depending on the coefficient. The coefficient is positive, so it's convex. It's going to look like this. At the end of the day, it looks something like that.
- 7.34** Now, $x = 1$ —that's pretty close to zero. $x = 1$ is here, let's say. If I go down to my graph, I notice what should the slope be—positive or negative? It should be negative because I've got this behavior here. It's going top left to bottom right. The slope should be negative, and that checks out because what was my derivative at $x = 1$? It was -2 . That would say that since the derivative or the gradient is negative, it says: start at 1 and move to the left. If you move to the left, the values of your function will increase. Does that pan out with what we see from the graph? If I start moving to the left, my function's values get bigger and bigger and bigger. If I want my function values to

decrease, I need to move in the negative direction of the gradient. I need to move in the $-(-2) = +2$ direction—the positive direction. I need to move to the right to decrease.

7.35 So far, what do we have? We have $x_1 = x_0 - \alpha_0 \cdot f'(x_0) = 1 - \alpha_0 \cdot (-2) = 1 + 2\alpha_0$. Do we agree? I get the +2 because I'm moving in the right direction because I want my function values to decrease. Let's let that sit in for a second. $x_1 = 1 + 2\alpha_0$. But now I have to solve this optimization problem. I need to figure out what is α_0 . Here, $\alpha_0 = \operatorname{argmin}_{\{\alpha \geq 0\}} f(x_0 - \alpha f'(x_0))$. That's what I need to solve. We have all the ingredients of gradient descent on the board. I'm going to erase this particular part so we can have some room to work because now we have to solve this particular problem. Now we've got to solve that problem. Let's see what's going on.

7.36 How do I minimize this? The first thing I'm going to do is I'm simply going to give this function a name. Notice—that function is a function of what variable? x_0 is a number—that's 1. $f'(1)$ is a number. We already know that. The only thing that's changing here is α . α is my variable of interest. I'm going to say: let $g(\alpha) = f(x_0 - \alpha \cdot f'(x_0))$. $x_0 = 1$. $f'(1) = -2$. So $g(\alpha) = f(1 + 2\alpha)$. We want to understand how to minimize that particular function with respect to α . How do I minimize a function? I need to find out: where does it bottom out? Where are its valleys? We found that its valleys occur where its derivative is equal to zero. I need to understand what is the derivative of g with respect to α . Let's figure that out.

7.37 Do I have a formula for f ? Yes, I do. It's right here. What is g ? g is the function, which is: take that as an input, plug it in, and square it. So I'm going to get $g(\alpha) = (1 + 2\alpha)^2 - 4 \cdot (1 + 2\alpha)$. That's what my function is. I could work it out in all of its gory detail to some extent and expand this out and make it as big as possible, and I could do all of that. I don't need to because the chain rule says that it's actually better if you don't. Because at the end of the day, what am I trying to compute? $g'(\alpha)$, and I'm trying to set $g' = 0$. Let's see how that works.

7.38 How would I differentiate this function right here? How would I differentiate this thing right there? That's a chain rule. It's a composition of functions. I have the function $1 + 2\alpha$, but it's trapped inside of the squaring function, something squared. So think back to the chain rule when we went through that. What did the chain rule say? This was the onion analogy. You work your way from the outside to the inside.

7.39 I step back for a second and I say: what is the outermost function here? It's something squared. I go back and I say, "How do I differentiate something squared?" That's a power rule. If I have variable squared, I bring the two down in front, I rewrite the variable, and I decrease the exponent by 1. I'm going to do that. The 2 comes down front, so I get $g'(\alpha) = 2 \cdot (1 + 2\alpha)$. I decrease the 2 by 1, so it becomes a 1. Now I've worked my way through that outer layer in my derivative.

7.40 But the chain rule says: now that you're through the furthest out layer, you move to the next layer and differentiate that, and then the next layer, and then the next layer

until you're all the way done. Once I'm done with the outer square layer, I'm looking at the function $(1 + 2\alpha)$, and I say: what is its derivative with respect to α ? The derivative of a constant—the derivative of 1—is 0. That's fine. But what's the derivative of 2α in that case? It's like $2x$. What's the derivative of $2x$? It's 2. I have to multiply this by 2. That's coming from the chain rule. That's my chain rule.

- 7.41** Then I keep moving forward. Then what do I get? -4 , which is my constant, times the derivative of $(1 + 2\alpha)$. We just did that. The derivative of $(1 + 2\alpha) = 2$. I'm done. That's my derivative: $g'(\alpha) = 2 * (1 + 2\alpha) * (2) - 4 * (2)$.
- 7.42** Notice that I had to sacrifice one of two things. If I expanded this—I distributed, I collected all the terms and all of that—I had to sacrifice the size of the function by expanding it for ease of differentiation. Because then I could go term by term, power rule, and I'm done. Versus—rather than expanding, which could be potentially terrible, I could rather say, "Why don't I use the chain rule and deal with the fact that the chain rule is hard, but I can do hard things?" Then I can differentiate it simply by keeping everything compact. I don't have to expand everything. It's this trade-off. What do you choose to do?
- 7.43** If this thing had been $(1 + 2\alpha)^{50}$, you're never going to expand that any day of the week. So the chain rule is going to be your go-to. That's the point I'm trying to make. The chain rule—yes, it's hard. However, it's the way to go 9 times out of 10.
- 7.44** Here's $g'(\alpha)$. Let's see what we get: $2 * 2 = 4$. So I get a $4 * (1 + 2\alpha)$, and then $-4 * 2 = -8$. This is simple enough. I could expand it. It's not that bad. What is this? $g'(\alpha) = 4 + 8\alpha - 8 = 8\alpha - 4$. Remember, at the end of the day, what was I trying to set this equal to? I was trying to set it equal to zero to find the value of α that's going to give me the minimizer for g . What do I get? I add the 4 over, divide by 8: $\alpha = \frac{4}{8} = \frac{1}{2}$.
- 7.45** That happens to be bigger than zero. It's a valid point. We have a constraint. We said that $\alpha \geq 0$. If I had solved for it and I got $\alpha = -5$, that's not valid. Again, remember, why does α need to be positive? Because I want to do what? I want to move in the direction of $-\nabla f$. If α were, say, -4 , then this would say: start at x_0 and move it four times the gradient. I'm moving in the wrong direction. α has to be positive.
- 7.46** Now, I need to check: is this actually a minimizer? Does that actually minimize g ? A natural question would be to say, "Does $\alpha = \frac{1}{2}$ minimize g ?" How could I check that? I could run a sign line on g' . I could run a sign line. We saw that last time that the sign line on the first derivative tells me behavior about g . Let's run a sign line on that. If I have g' , we saw that at $\alpha = \frac{1}{2}$, I get 0. Now I have to pick points to the left and to the right of $\frac{1}{2}$.
- 7.47** But remember, you have this bound on the left where $\alpha = 0$. You want $\alpha > 0$. You have to be a little bit careful. What's a point in between 0 and $\frac{1}{2}$? This is 0.5, 0.25 would work. So a quarter, $\frac{1}{4}$. If I take $\frac{1}{4}$ and I plug it into my derivative—what was my

derivative? It was this guy. If I take $8 * \frac{1}{4} = 2$. $8 / 4 = 2$. $2 - 4 < 0$. So I'm going to get a negative. That means I'm decreasing here.

7.48 What do I hope is going to happen on the right-hand side? If I want to find a valley, what do I hope is going to be on the right-hand side? Am I going to positive or negative? If I want a valley, I need to decrease, so my derivative has to be negative, and then I'd have to increase on this side. So I hope that it's going to be positive over here. Let's choose a value of α to the right. Say $\alpha = 1$. If I plug in $\alpha = 1$ to my derivative, what happens? $8 * 1 = 8 - 4 = 4$. That's positive. I'm positive here. I'm increasing. So this is actually a minimizer. That's a minimizer. Check.

7.49 All right, so we're almost done. Since that's a minimizer, I can say: what is α_0 ? $\alpha_0 = \frac{1}{2}$. It is precisely the thing that minimized g . Now I've collected all the ingredients via our scavenger hunt for gradient descent. I can finish the problem. I can say: what is x_1 ? $x_1 = 1 + 2 * \alpha_0 = 1 + 2(\frac{1}{2}) = 2$. This entire thing is 2.

7.50 How did gradient descent do? Did it find the right point? Remember, it's trying to find the bottom of the valley. That's what it's trying to find. What is f' at my new value? What is $f'(x_1)$? Let's check and see what happens. I got to erase. Let's see what happens. What is $f'(x_1)$? That's $f'(2)$. Where was my derivative? We found it—my derivative is right here. What happens when I evaluate at 2? This gives me $f'(2) = 2 * 2 - 4 = 0$. We found the bottom.

7.51 In one step with gradient descent, we found the bottom. In that picture that I drew earlier, the function that came down like this, there was $x = 2$. That was the bottom of the valley. Gradient descent found it in one step. It's a very powerful method. It's a very powerful method for finding the bottoms of functions. However, I want to give you an example that should be cause for concern at the end of the day. It should be cause for concern.

7.52 Because what if I gave you a function that did something like this? I'm going to give you a function that does— say it does something like that. Let's put some axes on this. Let's say it does something like that. I don't know what it's modeling. I literally just drew a scribble on the board. That's fine. I'm trying to illustrate a principle. I'm not saying that this is—I mean, it is a function, but I don't know what it is. I'm not going to write out its formula necessarily. But I want to illustrate the principle.

7.53 Gradient descent told me what? If I start here at x_0 and I compute my gradient—my gradient is saying it's negative there. I need to move in $-\nabla f$. If my gradient is negative at x_0 , that would say I need to move to the left to increase my values. That makes sense. If I move this way, the values of my function are increasing. So it says: move in the right direction, so your values decrease, so I'm going to move in this direction. But what is gradient descent going to find? Gradient descent is going to find the bottom of that valley.

- 7.54** So if my question at the end of the day is, "What is my overall global minimizer?" What I'm trying to find is this guy. That's the smallest value of my function. Gradient descent isn't going to find that if I start here. Gradient descent is going to find a local minimum. It's going to find that bottom of the valley right there. It's not going to find the very, very most bottom of it. That's concerning. That says: then is gradient descent even worth doing? Absolutely, it's worth doing. But it means that you should probably run gradient descent with multiple initial values and see what happens.
- 7.55** If I chose this to be x_0 , I get that bottom. If I chose this to be x_0 , I would get that bottom. If I chose this to be x_0 , I would get that one. There's a systematic way to say: choose different values of x_0 , run the method for gradient descent, see what gets outputted, and whichever one is the smallest, that's what you go for with your minimum. There's an art form to this a little bit, much like Newton's method, where Newton's method was: where do I guess my initial method? Because I could potentially guess a point for which the method fails. Gradient descent—you might pick a method where you get a local as opposed to a global minimizer.
- 7.56** I want to give you an example of this. This is going to be heavy. This is going to be very heavy. Apply gradient descent to $f(x) = x^4 - 2x^3 - 9x^2$. That's my initial function. Now, remember, at the end of the day, I'm trying to figure out where the derivative equals zero. That's the game. As a first approximation to this, I could say that this is a polynomial. I can compute its derivative and hope that I can actually just solve it analytically by hand, and then I'm done. Let's see what actually happens.
- 7.57** If I take $f'(x)$, the power rule tells me this is $4x^3$ — what happens here? Remember, the 2 stays here—the -2, technically. The 3 comes out front and multiplies it. That's going to give me a -6, and then I decrease the exponent by 1. The -9 here, the 2 comes down, multiplies that, gives me a -18 times x : $f'(x) = 4x^3 - 6x^2 - 18x$. I want to set that equal to zero.
- 7.58** This is one of the most beautiful things about mathematics in general is mathematics is incredibly complicated, and it's very deep and there's a lot of results and things like that. But at the end of the day, solving equations—even if I made this a 40, for goodness' sake. If I said x^{40} —something like that, and I try and write it down by hand—no chance am I going to solve that by hand. Something seemingly so simple—it's got three terms, but I increase its exponent by a certain ridiculous factor, and suddenly I've got no shot of being able to do it by hand. I have to have methods like gradient descent or Newton or bisection to be able to solve these things.
- 7.59** How do I do that? Let's see. What do they have in common? They all have a 2 in common. That's fine. They all have x in common, and that's the largest one I can take. Then what's left? In order to get a $4x^3$, I need a 2 and an x^2 . In order to get a -6 from a 2, I need a -3, and I need one more x . Then in order to get a -18, I need a -9, I think. Is that right? You got to keep me honest. You got to spot check my work here. Can I factor this? It's a quadratic. It's a degree 2. Worst case scenario, I could throw the quadratic

formula at it and be like, "Here's my answer." That's fine. But can I actually factor it? Let's see.

- 7.60** The $2x$ goes along for the ride. If I want to factor this out— this would be what? A $2x$ and an x for sure, because $2x * x = 2x^2$. But then whatever the product of this number and this number is, it has to be -9 . That means it's either $+9$ and -1 or -9 and $+1$ or $+3$ and -3 or something along those lines. Let's see. If I plug in a -3 here and a $+3$ here— let's spot check it and see if it works.
- 7.61** $2x * x = 2x^2$. Great. $2x * -3 = -6x$. $3 * x = 3x$. $3 * -3 = -9$. Those two combine to give me $-3x$, and then my -9 , so that is my factorization. This actually works, which allows me to say—what do I have? Either $2x = 0$, $2x + 3 = 0$, or $x - 3 = 0$. If $2x = 0$, that tells me that $x = 0$. If $2x + 3 = 0$, that tells me that $x = -\frac{3}{2}$. If $x - 3 = 0$, that says $x = 3$. Those are my critical points. Those are my candidates for my minimizers.
- 7.62** How do I know which one's which? Just knowing that the derivative is equal to zero doesn't tell me that my function is actually minimized there. It could be a peak. It could be a maximizer. So how do I diagnose that? The sign line is how you diagnose it. You look at your sign line. You put these in order. Let's say $x = -\frac{3}{2}$. That's a zero. If $x = 0$, I get zero. If $x = 3$, I get zero.
- 7.63** Now you wiggle around each critical point and determine the sign of f' . Is it positive or is it negative? Let's see. $-\frac{3}{2} = -1.5$. If I chose something that's like -2 —that'd be on this side— and I plug it into my derivative— $f'(x) = 2x(2x+3)(x-3)$ —what do I get? $2 * -2$ —this is negative. $2 * -2 + 3 = -4 + 3$ —this is negative, and $-2 - 3$ is negative. So I get negative, negative, negative. If I have three numbers, all of which are negative, how do I get those?
- 7.64** I'm sorry, I think you might have forgotten to factor out 2.
- 7.65** Factor out 2 from where?
- 7.66** $2x + 3$. You could have taken out $2x$.
- 7.67** $2x$ from here?
- 7.68** Yeah, but now there's still a 2 as a scale—
- 7.69** For this thing here? That's because of this guy right here. So this term here, this thing right here, this is a great question. So this thing right here is this term times this term. I'm trying to factor out this quadratic. This guy stays out by himself. This guy's out here. If I cover that up and I say, "What is the factorization of that thing?" I need some number times some number gives me this thing. It could be all sorts of things, but the $2x$ and the x would give me a good candidate at the very least.
- 7.70** Then I need two numbers here that multiply to -9 . That's $+3$ and -3 . That's going to give me -9 . I have to check their ordering to figure out which one's which. If I work out the

distribution—this times this, this times this, this times this, this times this—I get this thing, which simplifying $-6x + 3x = -3$. I think that works. All right.

7.71 Let's see. What happens now? When I plug in $x = -2$ to the left, we said that that was negative, negative, negative. If I have three negatives: -2 , -3 , -4 , multiply together, is it positive or is it negative? It's negative. So I get this. If I plug in something in between -1.5 and 0 —so -1 would be a great choice. This is -2 . This is positive, and this is negative. Negative, positive, negative: two negatives and a positive is going to give me positive. That's a positive. If I plug in something in between 0 and 3 , like $x = 1$ — $2 * 1$ is positive, $2 * 2 + 3$ is positive, $1 - 3$ is negative, so I get positive, positive, negative. If I have positive, positive, negative, that's negative. If $x = 4$ —positive, positive, positive, I'm going to get positive. So I'm increasing, decreasing, increasing, decreasing. This one is going to give me a minimizer. This one is going to give me a minimizer. This one is going to give me a maximizer. So I don't care about that one. I'm trying to minimize things. But the question becomes: which one is best? Which one gives me the bottom of the valley?

7.72 That's what gradient descent is going to do for us. We're going to apply gradient descent to this. Now look, if I'm close to this one, then I would—if x_0 is close to this one, then that probably suggests that you're going to find that one— $x = -3/2$. If I start with $x_0 = -1$, I'm probably going to find this one. If I start with $x_0 = 1$, I'm probably going to find that one. Let's see if that actually pans out.

7.73 Here's the solution. Let $x_0 = -1$ in this case; that's my initial guess. Then what does gradient descent say? It says that $x_1 = x_0 - \alpha_0 f'(x_0)$. We're on the scavenger hunt. We need to find f' , we need to find α_0 , we need to plug all that stuff in. You'll see very quickly why you do this with a machine; you don't do this by hand. This example is meant to illustrate that point.

7.74 What is my derivative up here? Well, we already computed it. $f'(x)$ was equal to what? We computed that thing. It's right here. This thing right there. That was my $f'(x) = 2x * (2x + 3) * (x - 3)$. Okay, so what is $f'(-1)$? When I plug in -1 , I get $-2 * -1 = -2$, $(-2 + 3) = 1$, $(-1 - 3) = -4$. So I'm left with $-2 * 1 * -4 = 8$. I'm left with $+8$. Checking my sign line, it should be positive because I plugged in $x_0 = -1$, and this told me you're positive there anyway. That's my gradient. It says move to the right. That means my function values are going to increase. That should make sense based on this.

7.75 $x_0 = -1$. It says: you're in the increasing range. It says: move to the right to go uphill. What do I need to do? I need to move to the left to go downhill to decrease. I'm going to look at $x_1 = x_0 - \alpha_0 f'$ —so $-8\alpha_0 \rightarrow x_1 = -1 - 8\alpha_0$. Great. Where what? Where $\alpha_0 = \operatorname{argmin}_{\{\alpha \geq 0\}} f(-1 - 8\alpha)$.

7.76 Now I need to minimize the function $f(-1 - 8\alpha)$ with respect to α . Just like we did in the previous exercise. So I'm going to define a new function, $g(\alpha)$, and I'm going to minimize that. Let $g(\alpha) = f(-1 - 8\alpha)$. This is where you see the beast emerges. This is

where the beast emerges. Because look— what is this saying? It's saying: go to f and plug in $(-1 - 8\alpha)$. Fine. Go to f . It's got three terms, one of which is raise it to the fourth power, plug that in, and do something with it. Good luck with that. That's a monster. I'm not going to do that. That's incredibly difficult. I'm not going to expand it out— $(-1 - 8\alpha)^4$. It's going to take me another hour just to figure out what the thing even looks like. What am I going to resort to?

7.77 What did we say last time? You don't expand it out. You use the chain rule to simplify the computation. Great. So what is this going to look like? This is equal to $g(\alpha) = (-1 - 8\alpha)^4 - 2 * (-1 - 8\alpha)^3 - 9 * (-1 - 8\alpha)^2$. Now I take its derivative. Let's see what we get. If I take derivative of $g'(\alpha)$ —the chain rule, so the 4 comes down, I get $(-1 - 8\alpha)$, I decrease the power by 1, and then I multiply by the derivative of the inside, which is -8 . The next one: I get $(-2 * 3) * (-1 - 8\alpha)^2 * (-8)$. I get that. Then my last term, let's get this example out of the way. We don't need this anymore. My last one is: $-18 * (-1 - 8\alpha) * -8$. I set that whole thing equal to zero.

7.78 Here's the beautiful thing, though. Every single one of those terms has a -8 . So I can divide out everything by -8 . Those cancel. What am I left with? I'm left with $4 * (\text{variable})^3 - 6 * (\text{variable})^2 - 18 * (\text{variable})$. That was precisely how we factored this guy. I already know how to factor now—replace x here with $(-1 - 8\alpha)$. It's a bit of a fortunate happenstance that that actually works for us. This is equal to what? $2 * (-1 - 8\alpha) * 2 * (-1 - 8\alpha + 3) * (-1 - 8\alpha - 3) = 0$. I'm just plugging in $(-1 - 8\alpha)$ into there, and that's what I get. Let's see—what does that mean? That means that either this is zero, so either $(-1 - 8\alpha) = 0$, or this thing is zero, so I get—what do I get here? $-2 - 16\alpha + 3$. That's what? $1 - 16\alpha = 0$. Or this one is zero, so that's what? $-4 - 8\alpha$. One of those three things has to be true.

7.79 But notice, look—this one tells me that $\alpha = -1/8$. That's not allowed because α has to be positive. That one's no good. This one tells me $\alpha = 1/16$. That's potentially valid. That's a positive number. But this one tells me that $\alpha = -1/2$. That's no good because it's negative again. So I can throw it out. That means that $\alpha = 1/16$.

7.80 Now, I would have to go back and check that it actually minimizes g' by going into a sign line and checking that value, which we should do. But you can check that $\alpha = 1/16$ minimizes g . It does. You can check that. But what that means is that my α_0 should be $1/16$. Then what does gradient descent say? $x_1 = -1 - 8 * 1/16 = -1 - 1/2$, otherwise known as $-3/2$. That's exactly what we said it should have been based on our sign line associated to that.

7.81 I'll leave it to you as an exercise—for you to take if $x_0 = 1$, you run the exact same process for gradient descent, and you'll find out that you run into this one. If you look at the graph of what this thing it looks like, it looks like this. It comes down, up, down, up. If I want to find the bottom of the function, the very lowest of my function, I have to be careful where x_0 is starting, because if I start over here, I'll find this one. That's what we just showed. I leave it as an exercise to you to start over here with $x_0 = 1$, run

the same process. You'll run into $x = 3$, which gives me the bottom of my function. It gives me the lowest peak.

- 7.82** You don't need big fancy machinery and all this stuff to actually understand gradient descent. It boils down to, "Do you know what a first derivative is? Can you think through the process of saying, 'I want to be able to learn how to walk downhill using this formula.'" Now you know how.
- 7.83** Could you go back and show how you factored that the derivative of the $g'(x) = 4x^3 -$
- 7.84** Derivative of $g(x)$ —this one right here.
- 7.85** When you see $f'(x)$ up there with the $2x$, $2x+3$.
- 7.86** $f'(x) = 2x * (2x + 3) * (x - 3)$?
- 7.87** Yeah, because I was getting a $2x * (x - 3)^2$. I did that a couple of times. It seems like the right thing, but I don't know what I'm missing.
- 7.88** I see. Yeah. So let's go through that factorization. So if I understand correctly, the question is really, "How does f' factor?" How did I get this particular thing right here? Okay. Good. All right, so what was the original f' ? The original $f'(x)$ was what? $4x^3$. And then it was what? $-2x^3$ —so this was like a $-6x^2$. And what was it? $-18x$. That was the original, right? I think that's right— $f'(x) = 4x^3 - 6x^2 - 18x$. Okay.
- 7.89** When I look at something like this, I say— the, first thing is I notice that this is a cubic, and so factoring that is going to be brutal, typically. Now, in this one, I can bring that down because I notice that everything has an x associated to it. Because I have an x in every single term, I can pull that x out. But I also notice that everything is even: there's a 4, a 6, and an 18. Those are all divisible by 2. I can factor out a $2x$ associated to that first.
- 7.90** That's where this thing—I can say $f'(x) = 2x *$ —what is going to be here? In order to get a $4x^3$, I need this to give me another 2. So $2 * 2 = 4$, but I need $2x$ s to jump that up to a cube. Does that make sense? Then over here, I need $-2x *$ what gives me 3? I need -3 . $2 * -3 = -6$. Then I need an additional x to jump that up to x^2 . The last one is—I need what? I need -9 here. $2 * -9 = -18$, and I already have an x , so that's what I have. That's step number one.
- 7.91** Now, if I want to factor out the rest, factoring out this term, let's take that off to the side. If I say: what is $2x^2 - 3x - 9$? If I want to try and factor that into two terms that are multiplied together, I have to ask myself a couple of questions. The first question is: whatever goes here and whatever goes here, they have to multiply together to give me a $2x^2$. A natural choice would be a $2x$ and an x , for example. That would be a natural choice as to what would go there. It may not work, but we're trying to figure it out. Let's try it and see what happens.

- 7.92 The next thing that I notice is that whatever goes here and whatever goes here—their product has to be a -9. What are my choices there? It's either 9 and -1, or I could reverse the order: -9 and 1. Or 3 and -3—I don't really need to reverse the order because if I have 3 and -3, it's the same thing. So now I just exhaust over all of my choices. It turns out this is the right one to use. But if I throw this in as 3, -3, or something like that, or maybe I did it in reverse order. Maybe I said this one was the minus and this one was the plus. I'm just checking to see which one goes where.
- 7.93 Now I work it out. What do I get? I would get $2x * x = 2x^2$ —that would be this guy times this guy. This guy times this guy gives me a $+6x$. This one gives me a $-3x$, and this one gives me a -9 . But notice that the middle term is wrong. The middle term—this gives me a $2x^2 + 3x - 9$ and that sign is wrong because I want it to be $-3x - 9$. I say, "Maybe my signs were wrong. Maybe this is the -3 and this is the +3. If I make that positive and I make this negative, what does that do? $2x * x = 2x^2$, but now $2x * -3 = -6x$, so I get a minus. $3 * x$ is > 0 . Then when I work that out, I get $-3x - 9$. That's exactly what I have here. This whole factorization becomes: $2x * (2x + 3) * (x - 3)$. That's where that's coming from. Does that make more sense?
- 7.94 Recently, I saw that data science is starting to be offered in high school. I was wondering—how would you go about teaching some of this to high school students or even middle school?
- 7.95 Yeah, this is a great question. Absolutely. Data science is being taught much earlier now. I think we've seen that as a consequence of this big boom of people realizing that data is so valuable and being able to understand data and run these different algorithms on it to get predictive power on the back-end side of things, that it's running the world to some extent. Every business, every market, everything is trying to take advantage of analyzing data. It's not surprising that it's being taught earlier and earlier and earlier because it's such an important thing.
- 7.96 Now, how do you do that? Ultimately, I think teaching any principle whatsoever, you need to model correct principles, number one, which, in my opinion, means you need to modify your curriculum accordingly to actually teach principles that matter, not just teach because this is how we've done it for the past 100 years. Because there's a lot of stuff in a lot of classes that we teach that ultimately don't matter that much. That's a harsh thing to say, but it's true.
- 7.97 You learn a lot of principles, but it's like, "When am I ever going to use this?" It's the time-old question that every math student ever has asked, "When am I going to use this in my daily life?" More often than not, the answer is, "The way it's being taught? Probably not." Because they're not motivating it with a question that matters. That's how you teach it. You give the students questions and you say, "What matters to you?"
- 7.98 If I have a group of high school students—let's say I've got a class full of 15 to 20 high school students, those students are going to range drastically across all sorts of

measures. I might have a student who really loves sports. They really love European football or things like that, or they love tennis, or they love golf, or they love anything like that. I would tell the students, I would say, "Find something that's of interest to you, and let's try and find a data set about that." If I have a student who loves sports analytics, I would say— let's say they love golf, fantastic, or they love tennis—I would say, "Let's go find some data on golf or tennis, and then let's see if I can ask a question that I'm interested in knowing."

- 7.99** Let's say it's or something along those lines. I say, "The past five major champions have all had this characteristic. Is that characteristic indicative of major championship success?" Something along those lines. Or, "All of them trained in this facility," or something along those lines. Or, "All of them had this coach," or something. "Can I extract some predictive power based on this that data." But the student will not care if it does not motivate them.
- 7.100** So step one: you give them something to care about, and you give them the power. You say, "Fine. What do you like?" "I like sports." Fantastic. "I'm interested in data associated to the stock market," or, "I'm interested in data associated to medical device sales," or whatever the case may be—hopefully, a high school student is not interested in that, in medical sales, but it's possible. But whatever they're interested in, you have them find that particular data set, whatever it is.
- 7.101** Then you craft your curriculum accordingly so that you say, "What do they need to be able to analyze that data? Do they need computer skills? Do they need Python programming?" Most definitely they would. "What Python programming?" I don't necessarily need to have them hard code various data structures. I just need them to understand, "Can you program something or use a package, or something along those lines, to actually apply it to your data set?" Then I figure out how I work that into my curriculum.
- 7.102** "What's the underlying mathematical principles that I need them to understand?" Well, optimization, for sure. So then I craft the course based on optimization. "What do I need for optimization?" I need derivatives. So I teach them how to compute derivatives. But I cut out all the fluff. I cut out all the stuff that they're not going to use. I give them everything that they need so that they can be successful. I don't couch it in a way so that you need some advanced degree or you need to take this, this, this, this, and this course. That's nonsense. That's what needs to be modified in order for that course, that high school analogy that you mentioned— that's what needs to be done in order for those students to be taught the correct principles.
- 7.103** Give them something that matters to them, not some problem that's—I don't know—that's been around for hundreds of years where somebody wrote this question and it was a good question, it illustrates good principles, but nobody cares about it at the end of the day. Give them something that matters to them. It'll give them the motivation

necessary to pursue that and keep them learning because they're hungry for it now, because it matters.

- 7.104** With all this data being available and the means to actually make something useful out of it, how big of a threat could weaponized applied mathematics be—fused with ill intent, or just a means for doing something bad?
- 7.105** Yeah, for nefarious purposes? Absolutely. This is a serious problem. It's something that—there's a lot of thought going on about this right now. For example, you read headlines in various news sources, talking about talks in country x about what to do about AI, or talks in country y about, "AI is causing issues in this particular thing. What do we do? Do we pass legislation that says that we can control this? Do we pass legislation that says that this isn't allowed?" Things like that. So people are thinking about this actively, and it's a very important question.
- 7.106** What do you do with things like that? One thing I think that is an important thing to consider is that at the end of the day, somebody coded this. Somebody coded something into an algorithm, and that was applied in whatever software, or whatever recommendation engine, or whatever the case may be. There has to be a discussion about the ethics of coding. There has to be a discussion about that.
- 7.107** What that looks like? I don't know because that's a very hot topic and a very difficult topic to navigate. What's ethical to one person is not ethical to another person. To say that there's some universal standard—that gets a little concerning. I think there need to be guidelines. I think there need to be trainings built into not only do you understand how to manipulate the data or use the data, but there have to be trainings that specifically say, "You're a human being, your environment, your circumstances, whatever, naturally endow you with personal biases." So if you're going to be coding something up, you need to be hyper vigilant and aware of those personal biases associated to that. That's an important thing.
- 7.108** For example, one of the biggest things with AI—this is actually fairly old, but it's incredibly sophisticated now and terrifying, to be honest. For me, I think it's terrifying—is what are called deep fakes. You've seen these things all over the place where somebody takes a deep neural net effectively—which you can think about as machine learning, effectively. They train it on speeches and mannerisms and gesticulations and all that stuff of a particular person, and then they train a machine to actually reproduce that thing.
- 7.109** For most people—like if somebody did a deep fake on me, for example—is that really going to have that much significance? Probably not unless they're saying horrible things or slandering me and all that stuff. That would be horrible. But at the end of the day, probably not. But if you take somebody like the president of a country and you start making a deep fake, and suddenly that person is allegedly saying x, y, and z, that's a serious problem. You can see how somebody could manipulate this potentially for

really nefarious purposes. You say, "Oh, this president said this, this, and this," or, "That ambassador said that, that, and that," and so on. It's like, "Whoa, did they actually say it?" No. So then it begs the question, "What's reality? What's real?" And that can be incredibly dangerous.

- 7.110** I think as a starting point, we need to have open dialogs, we need to have open discussions, and we need to be able to be open-minded about what actually ethical data analysis is. What is it? How do we—I hesitate to say "regulate it" because that has a negative connotation to it, but I think there needs to be some oversight associated to it as to what is appropriate and what is not appropriate. What that domain looks like in the coming decades, and things like that? I don't know, but it is something that we need to be hyper-vigilant, in my personal opinion, and realize that, yes, data analytics, and the more data we have, have beautiful consequences for improving lives and all of that. But everything that has light has darkness, and we need to be careful that whatever tool we create, somebody is going to use it for negative purposes. We need to try and hedge against that in whatever way that looks in the most ethical and loving, in my opinion, way possible, where nobody is excluded whatsoever to open voices and things like that.
- 7.111** With AI getting bigger and bigger and being able to solve basically any problem, wouldn't it be more efficient to teach kids how to come up with the right questions instead of just teaching them how to solve random questions?
- 7.112** That's a good question. That's a really good question. I like that. You're right. AI is proving to be more and more sophisticated at solving all sorts of problems and things like that. I hesitate to say that it can solve all problems because, for example, the autonomous vehicle type thing—we're not there yet. Autonomous vehicles, as a parent, I would be terrified if suddenly everything went autonomous vehicles and it wasn't 100% accurate. Because if my son decides to bolt into the road last minute and there's an autonomous vehicle coming down, I would hope and pray that that autonomous vehicle can recognize that my son is in the road and stop. That would need to be 100% accurate, but we're not there in the reality of the situation.
- 7.113** I do think there is value in asking the right questions and allowing the technology to help answer those questions. I think there is value in teaching kids, especially at a young age, to be able to understand, "What are good questions? And how can you use those tools of the AI?" Stuff like that. "How can you use those tools to solve those questions?" But I want to make sure that we don't become dependent upon the AI for our answers. I want to make sure that we're still controlling the scenes. We're still controlling the flow of data, the flow of how these things work. In order to do that, we have to understand what's under the hood, and we have to understand how these algorithms are working.
- 7.114** I'm not saying that your question is saying this, but I'm saying my worry is that if we put 100% of blind faith into an AI, that's a very bleak future if we don't understand the

underlying principles because of it, because of the things that I've mentioned, like biases are built into these things and all of that. I think it's important. I think your question is very valid in that we need to teach how to ask the right questions and then how to ethically use the correct principles with AIs or whatever it may be to actually solve the problems.

7.115 For example, we have AIs now that can simply generate answers, as you mentioned, to pretty much anything we can think of to some extent. So if I wanted to use that in a negative way, I could use it for plagiarism. I could use it for copying somebody else's work, or taking somebody's idea, or something along those lines, simply because the AI is giving me that answer. It's giving me that information. If I'm not aware of the ethical underpinnings of this and where that information is coming from, then that gets into a lot of issues. Then what is intellectual property now? Who owns what? All of those types of things. It gets to be a very sticky issue. But I think starting with asking the right questions and teaching that is very key. Thank you.